Software Design Document

Project: **Project-3**

Date: March 30, 2026

Version: 1.0 | Standard: IEEE 1016 | Classification: Internal

Total Design Components	134
Total Modules	53
Component Types	Module, Class, Interface, Service, Database, Algorithm, API Endpoint

Modules	ControlModule, SensorModule, CommModule, ConfigModule, SafetyModule, ComplianceModule, HALModule, SignalProcessingModule, SchedulerModule, UartModule, AdcModule, SensingModule, ControlLoopModule, HardwareAbstractionLayer, VoltageMonitorModule, FaultManagementModule, DiagnosticsModule, PowerSafetyModule, UartCommModule, CommandModule, ActuatorModule, LoggingModule, StartupModule, ADCModule, BoardSupportModule, ActuatorControlModule, HAL, Driver, Middleware, Application, StateManagementModule, WatchdogModule, FaultHandlingModule, SafetyControlModule, CommsModule, FaultMonitoringModule, PowerMonitoringModule, ReliabilityModule, StateMachineModule, SensorProcessingModule, DataPersistenceModule, SystemConstraintsModule, AcquisitionModule, CommunicationModule, FaultModule, DataLayer, StateModule, InterfaceModule, ArtifactGenerationModule, DocRepositoryModule, VCSModule, ReleasePackagingModule, TestReportingModule
Standard	IEEE 1016 — Software Design Description
Prepared By	SDD Agent (AI-Generated)

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Acronyms & Definitions

Acronym / Term	Definition
SDD	Software Design Document
SRS	Software Requirements Specification
CRS	Customer Requirements Specification
IEEE	Institute of Electrical and Electronics Engineers
DD-COMP	Design Document Component Identifier
SW-REQ	Software Requirement Identifier
API	Application Programming Interface
MVC	Model-View-Controller (design pattern)
ISR	Interrupt Service Routine
FSM	Finite State Machine
UART	Universal Asynchronous Receiver-Transmitter
I/O	Input / Output
DB	Database
JSON	JavaScript Object Notation
PDF	Portable Document Format
TOC	Table of Contents
LOT	List of Tables
N/A	Not Applicable

1. Introduction

This Software Design Document (SDD) describes the architectural and detailed design of the **Project-3** software system, derived from the Software Requirement Specification (SRS) following the IEEE 1016 standard. It covers **134 design components** across **53 modules**, providing full traceability from design decisions back to software requirements.

2. System Architecture Overview

The following table summarises all design components grouped by type, providing a high-level view of the system architecture.

Type	Count	Components
Module	1	MISRA Compliance Process
Class	38	CommandParser, UARTDriverConfig, Supply Voltage Assumption, Control Output Calculator, Feedback Fault Monitor, ThresholdConfigRepository, ActuatorStateMachine, ActuatorStateMachine_OFF, AdcDriverConfig, ADC Sample Buffer, ActuatorSafetyMonitor, ActuatorTargetStopController, OvervoltageDetector, OvervoltageDriveDisabler, ActuatorCommandGate, FaultTransitionHandler, FaultStateRetentionLogic, UART Command Parser, Voltage Range Configuration, StartupDelayManager, UART Peripheral Initializer, System State Machine, FaultStateManager, ThresholdStopEvaluator, HysteresisBandManager, RawAdcDataPolicy, UartStatusMessageBuilder, UartTransmitIntervalValidator, ShortCircuitDetector, UndervoltageMonitor, FaultStateTransitionHandler, HumidityIndependencePolicy, ADCReader, ShippingLimitGuard, ActuatorEnableController, SystemStateMachine, FaultStatusTracker, DocumentationRepositoryAdapter
Interface	2	Layered Architecture Enforcement Interface, FaultInputProvider

Service	74	ControlLoopTask, ADCSampler, UARTCommandHandler, UARTDiagnosticsService, ConfigService, ADCValidationService, UARTFaultHandler, FaultManager, SafeStateController, ADC Channel Configuration, ADC Read Service, Control Loop Scheduler, Fault Reporter, Actuator Drive Output, ThresholdConfigService, UartCommandParser, UartCommandParser_OFF, UartAckService, AdcSamplingScheduler, AdcContinuousAcquisition, Flap Control Computation, Open-Loop Inhibit Gate, ADC Validity Monitor, Target ADC Validator, SupplyVoltageADCChannelConfig, SupplyVoltageSampler, OvervoltageThresholdStore, OvervoltageUartNotifier, FaultExitVoltageChecker, OvervoltageMonitor, UartActuatorCommandInterface, Command Dispatcher, Actuator Controller, Invalid Command Handler, Invalid Command Logger, Startup Safety Manager, ADCResolutionConfigurator, ADCReferenceConfigurator, ADCSamplingScheduler, ADCPinInitializer, ActuatorGPIOInitializer, PWMTimerInitializer, ADC DMA Configuration Service, Independent Watchdog Service, Clock and Timer Configuration, Reset Initialization Handler, Nominal Transition Controller, HardwareWatchdogInitializer, WatchdogServicer, FatalMalfunctionResetHandler, ControlLoopScheduler, UartStatusScheduler, UartConfigService, OvercurrentMonitor, ThermalMonitor, StallDetectionService, WatchdogService, NvmConfigStorage, AdcAcquisitionService, ControlLoopScheduler, UARTCommandHandler, FaultMonitor, FaultLogger, FaultToStateTransitionService, RecoveryCommandHandler, StateStatusProvider, ArchitectureDiagramExporter, SchematicPdfExporter, PcbGerberExporter, FirmwareReleaseTagger, CalibrationDocIncluder, TestReportGenerator, DemonstrationVideoIncluder, ReleasePackageBuilder
Database	4	ConfigStore, CalibrationStore, Voltage Configuration Store, FaultLogStore
Algorithm	14	PIDControlAlgorithm, Position Scaling Algorithm, ShippingPositionStopLogic, Actuation Range Guard, Real-Time ADC Freshness Check, Drive Permission Check, SupplyVoltageScaler, ActuatorOnCommandParser, ADCSaturationDetector, ADCOnlyControlAlgorithm, UartBaudRateConfig, UartFrameFormatConfig, TemperatureProcessing, CommandCalculator
API Endpoint	1	StateStatusAPI

3. Module Descriptions

3.1 ControlModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-001	ControlLoopTask	Service	Observer	SW-REQ-001
DD-COMP-002	PIDControlAlgorithm	Algorithm	N/A	SW-REQ-001
DD-COMP-016	Supply Voltage Assumption	Class	Singleton	SW-REQ-010
DD-COMP-020	Control Output Calculator	Class	Strategy	SW-REQ-014
DD-COMP-027	ShippingPositionStopLogic	Algorithm	Observer	SW-REQ-018
DD-COMP-030	ActuatorStateMachine	Class	State	SW-REQ-019
DD-COMP-031	ActuatorStateMachine_OFF	Class	State	SW-REQ-020
DD-COMP-037	Flap Control Computation	Service	Strategy	SW-REQ-026
DD-COMP-052	ActuatorCommandGate	Class	Guard	SW-REQ-041
DD-COMP-088	ADCOnlyControlAlgorithm	Algorithm	Strategy	SW-REQ-077
DD-COMP-089	ControlLoopScheduler	Service	Singleton	SW-REQ-078
DD-COMP-092	RawAdcDataPolicy	Class	Facade	SW-REQ-081
DD-COMP-110	ControlLoopScheduler	Service	Observer	SW-REQ-097
DD-COMP-112	CommandCalculator	Algorithm	Strategy	SW-REQ-097
DD-COMP-115	ActuatorEnableController	Class	State	SW-REQ-099

3.2 SensorModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-003	ADCSampler	Service	Factory	SW-REQ-002

DD-COMP-111	ADCReader	Class	Singleton	SW-REQ-097

3.3 CommModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-004	UARTCommandHandler	Service	Command	SW-REQ-003
DD-COMP-005	CommandParser	Class	Builder	SW-REQ-003
DD-COMP-006	UARTDiagnosticsService	Service	Facade	SW-REQ-004
DD-COMP-007	UARTDriverConfig	Class	Singleton	SW-REQ-005

3.4 ConfigModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-008	ConfigService	Service	Repository	SW-REQ-006
DD-COMP-009	ConfigStore	Database	Repository	SW-REQ-006
DD-COMP-025	ThresholdConfigRepository	Class	Repository	SW-REQ-017
DD-COMP-026	ThresholdConfigService	Service	Singleton	SW-REQ-017
DD-COMP-043	Target ADC Validator	Service	Factory	SW-REQ-032
DD-COMP-049	OvervoltageThresholdStore	Service	Repository	SW-REQ-038
DD-COMP-065	Voltage Configuration Store	Database	Singleton	SW-REQ-054
DD-COMP-066	Voltage Range Configuration	Class	N/A	SW-REQ-055

3.5 SafetyModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-010	ADCValidationService	Service	Chain of Responsibility	SW-REQ-007

DD-COMP-011	UARTFaultHandler	Service	Strategy	SW-REQ-007
DD-COMP-012	FaultManager	Service	Observer	SW-REQ-007
DD-COMP-013	SafeStateController	Service	State	SW-REQ-008
DD-COMP-014	CalibrationStore	Database	Repository	SW-REQ-008
DD-COMP-022	Feedback Fault Monitor	Class	State	SW-REQ-016
DD-COMP-023	Fault Reporter	Service	N/A	SW-REQ-016
DD-COMP-038	Actuation Range Guard	Algorithm	Decorator	SW-REQ-027
DD-COMP-040	Open-Loop Inhibit Gate	Service	Chain of Responsibility	SW-REQ-029
DD-COMP-042	Drive Permission Check	Algorithm	N/A	SW-REQ-031
DD-COMP-099	OvercurrentMonitor	Service	Observer	SW-REQ-086
DD-COMP-100	ThermalMonitor	Service	Observer	SW-REQ-087
DD-COMP-101	StallDetectionService	Service	State	SW-REQ-088
DD-COMP-113	ShippingLimitGuard	Class	Decorator	SW-REQ-098

3.6 ComplianceModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-015	MISRA Compliance Process	Module	N/A	SW-REQ-009

3.7 HALModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-017	ADC Channel Configuration	Service	Factory	SW-REQ-011
DD-COMP-018	ADC Read Service	Service	N/A	SW-REQ-012
DD-COMP-024	Actuator Drive Output	Service	N/A	SW-REQ-014

3.8 SignalProcessingModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-019	Position Scaling Algorithm	Algorithm	N/A	SW-REQ-013

3.9 SchedulerModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-021	Control Loop Scheduler	Service	Observer	SW-REQ-015

3.10 UartModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-028	UartCommandParser	Service	Observer	SW-REQ-019
DD-COMP-029	UartCommandParser_OFF	Service	Observer	SW-REQ-020
DD-COMP-032	UartAckService	Service	Observer	SW-REQ-021
DD-COMP-058	UartActuatorCommandInterface	Service	Facade	SW-REQ-047
DD-COMP-059	ActuatorOnCommandParser	Algorithm	Factory	SW-REQ-048

3.11 AdcModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-033	AdcDriverConfig	Class	Facade	SW-REQ-023
DD-COMP-034	AdcSamplingScheduler	Service	Producer-Consumer	SW-REQ-024
DD-COMP-035	AdcContinuousAcquisition	Service	Observer	SW-REQ-022

3.12 SensingModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-036	ADC Sample Buffer	Class	Singleton	SW-REQ-025
DD-COMP-039	Real-Time ADC Freshness Check	Algorithm	N/A	SW-REQ-028
DD-COMP-041	ADC Validity Monitor	Service	Observer	SW-REQ-030

3.13 ControlLoopModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-044	ActuatorSafetyMonitor	Class	Observer	SW-REQ-033
DD-COMP-045	ActuatorTargetStopController	Class	Strategy	SW-REQ-034
DD-COMP-051	OvervoltageDriveDisabler	Class	Observer	SW-REQ-040

3.14 HardwareAbstractionLayer

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-046	SupplyVoltageADCChannelConfig	Service	Facade	SW-REQ-035

3.15 VoltageMonitorModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-047	SupplyVoltageScaler	Algorithm	N/A	SW-REQ-036
DD-COMP-048	SupplyVoltageSampler	Service	Observer	SW-REQ-037
DD-COMP-050	OvervoltageDetector	Class	Observer	SW-REQ-039

3.16 FaultManagementModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ

DD-COMP-053	FaultTransitionHandler	Class	Observer	SW-REQ-042
DD-COMP-055	FaultStateRetentionLogic	Class	State	SW-REQ-044
DD-COMP-056	FaultExitVoltageChecker	Service	Scheduler	SW-REQ-045

3.17 DiagnosticsModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-054	OvervoltageUartNotifier	Service	Observer	SW-REQ-043

3.18 PowerSafetyModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-057	OvervoltageMonitor	Service	Observer	SW-REQ-046

3.19 UartCommModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-060	UART Command Parser	Class	N/A	SW-REQ-049

3.20 CommandModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-061	Command Dispatcher	Service	Command	SW-REQ-050
DD-COMP-063	Invalid Command Handler	Service	N/A	SW-REQ-052

3.21 ActuatorModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-062	Actuator Controller	Service	Facade	SW-REQ-051

3.22 LoggingModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-064	Invalid Command Logger	Service	Repository	SW-REQ-053

3.23 StartupModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-067	Startup Safety Manager	Service	State	SW-REQ-056
DD-COMP-068	StartupDelayManager	Class	State	SW-REQ-057

3.24 ADCModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-069	ADCResolutionConfigurator	Service	N/A	SW-REQ-058
DD-COMP-070	ADCReferenceConfigurator	Service	N/A	SW-REQ-059
DD-COMP-071	ADCSamplingScheduler	Service	Observer	SW-REQ-060
DD-COMP-072	ADCSaturationDetector	Algorithm	State	SW-REQ-061

3.25 BoardSupportModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-073	ADCPinInitializer	Service	N/A	SW-REQ-062
DD-COMP-074	ActuatorGPIOInitializer	Service	N/A	SW-REQ-063

3.26 ActuatorControlModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ

DD-COMP-075	PWMTimerInitializer	Service	Factory	SW-REQ-064

3.27 HAL

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-076	UART Peripheral Initializer	Class	Factory	SW-REQ-065
DD-COMP-079	Clock and Timer Configuration	Service	Factory	SW-REQ-068

3.28 Driver

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-077	ADC DMA Configuration Service	Service	Facade	SW-REQ-066

3.29 Middleware

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-078	Independent Watchdog Service	Service	Singleton	SW-REQ-067

3.30 Application

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-080	Layered Architecture Enforcement Interface	Interface	Adapter	SW-REQ-069
DD-COMP-081	System State Machine	Class	State	SW-REQ-070
DD-COMP-082	Reset Initialization Handler	Service	Observer	SW-REQ-071
DD-COMP-083	Nominal Transition Controller	Service	Command	SW-REQ-072

3.31 StateManagementModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-084	FaultStateManager	Class	State	SW-REQ-073

3.32 WatchdogModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-085	HardwareWatchdogInitializer	Service	Factory	SW-REQ-074
DD-COMP-086	WatchdogServicer	Service	Observer	SW-REQ-075

3.33 FaultHandlingModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-087	FatalMalfunctionResetHandler	Service	Command	SW-REQ-076

3.34 SafetyControlModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-090	ThresholdStopEvaluator	Class	N/A	SW-REQ-079
DD-COMP-091	HysteresisBandManager	Class	State	SW-REQ-080

3.35 CommsModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-093	UartStatusScheduler	Service	Observer	SW-REQ-082
DD-COMP-094	UartStatusMessageBuilder	Class	Builder	SW-REQ-082
DD-COMP-095	UartConfigService	Service	Singleton	SW-REQ-083
DD-COMP-096	UartBaudRateConfig	Algorithm	N/A	SW-REQ-083
DD-COMP-097	UartFrameFormatConfig	Algorithm	N/A	SW-REQ-084

DD-COMP-098	UartTransmitIntervalValidator	Class	N/A	SW-REQ-085

3.36 FaultMonitoringModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-102	ShortCircuitDetector	Class	Observer	SW-REQ-089

3.37 PowerMonitoringModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-103	UndervoltageMonitor	Class	Observer	SW-REQ-090

3.38 ReliabilityModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-104	WatchdogService	Service	Singleton	SW-REQ-091

3.39 StateMachineModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-105	FaultStateTransitionHandler	Class	State	SW-REQ-092

3.40 SensorProcessingModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-106	TemperatureProcessing	Algorithm	N/A	SW-REQ-093

3.41 DataPersistenceModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ

DD-COMP-107	NvmConfigStorage	Service	Repository	SW-REQ-094

3.42 SystemConstraintsModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-108	HumidityIndependencePolicy	Class	N/A	SW-REQ-095

3.43 AcquisitionModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-109	AdcAcquisitionService	Service	Producer-Consumer	SW-REQ-096

3.44 CommunicationModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-114	UARTCommandHandler	Service	Command	SW-REQ-099

3.45 FaultModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-116	FaultMonitor	Service	Observer	SW-REQ-100
DD-COMP-117	FaultInputProvider	Interface	Adapter	SW-REQ-100
DD-COMP-118	FaultLogger	Service	Observer	SW-REQ-101
DD-COMP-123	FaultStatusTracker	Class	Observer	SW-REQ-103

3.46 DataLayer

DD-COMP ID	Component	Type	Design Pattern	SW-REQ

DD-COMP-119	FaultLogStore	Database	Repository	SW-REQ-101

3.47 StateModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-120	SystemStateMachine	Class	State	SW-REQ-102
DD-COMP-121	FaultToStateTransitionService	Service	Observer	SW-REQ-102
DD-COMP-122	RecoveryCommandHandler	Service	Command	SW-REQ-103
DD-COMP-124	StateStatusProvider	Service	Observer	SW-REQ-104

3.48 InterfaceModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-125	StateStatusAPI	API Endpoint	Facade	SW-REQ-104

3.49 ArtifactGenerationModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-126	ArchitectureDiagramExporter	Service	Facade	SW-REQ-105
DD-COMP-128	SchematicPdfExporter	Service	Facade	SW-REQ-106
DD-COMP-129	PcbGerberExporter	Service	Facade	SW-REQ-107

3.50 DocRepositoryModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-127	DocumentationRepositoryAdapter	Class	Repository	SW-REQ-105

3.51 VCSModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-130	FirmwareReleaseTagger	Service	Command	SW-REQ-108

3.52 ReleasePackagingModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-131	CalibrationDocIncluder	Service	Facade	SW-REQ-109
DD-COMP-133	DemonstrationVideoIncluder	Service	Facade	SW-REQ-111
DD-COMP-134	ReleasePackageBuilder	Service	Builder	SW-REQ-106

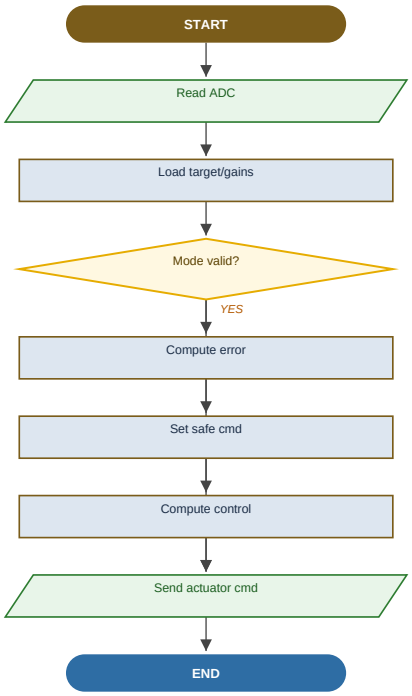
3.53 TestReportingModule

DD-COMP ID	Component	Type	Design Pattern	SW-REQ
DD-COMP-132	TestReportGenerator	Service	Builder	SW-REQ-110

4. Component Design Details

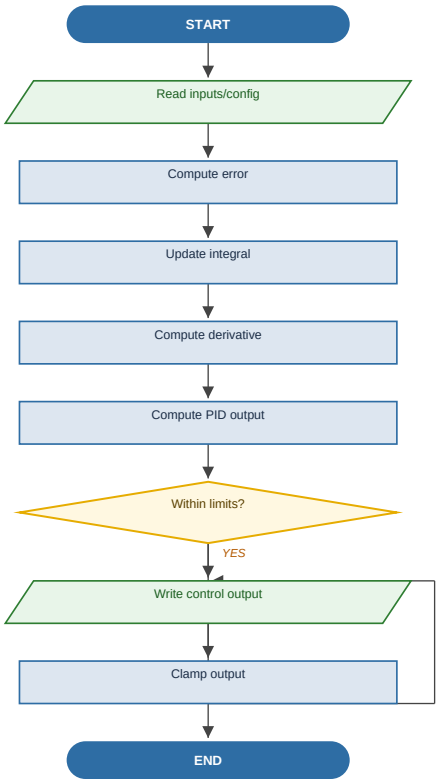
DD-COMP-001	**ControlLoopTask** *(Service)*
Description	Periodic task that executes the closed-loop control algorithm. Reads latest validated ADC feedback, compares to target position, computes control output using configured gains, and sends command to actuator driver.
Inputs	Validated ADC value, target position, control mode, control gains
Outputs	Actuator drive command (e.g., PWM duty/command value)
Dependencies	ADCValidationService, ControlConfigStore, ActuatorDriver
Design Pattern	Observer
Traces to	SW-REQ-001

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



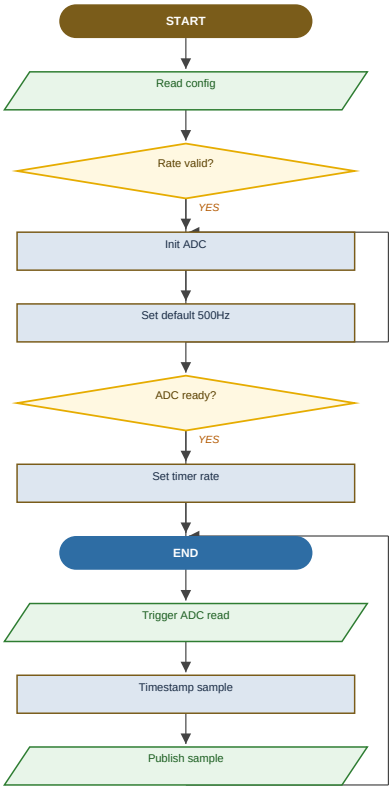
DD-COMP-002	**PIDControlAlgorithm** *(Algorithm)*
Description	Implements closed-loop control (e.g., PID) computing control output from error between target and feedback, with configurable gains and saturation handling.
Inputs	Target position, feedback position, Kp/Ki/Kd, previous error, integral state
Outputs	Control output value
Dependencies	ControlConfigStore
Design Pattern	N/A
Traces to	SW-REQ-001

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



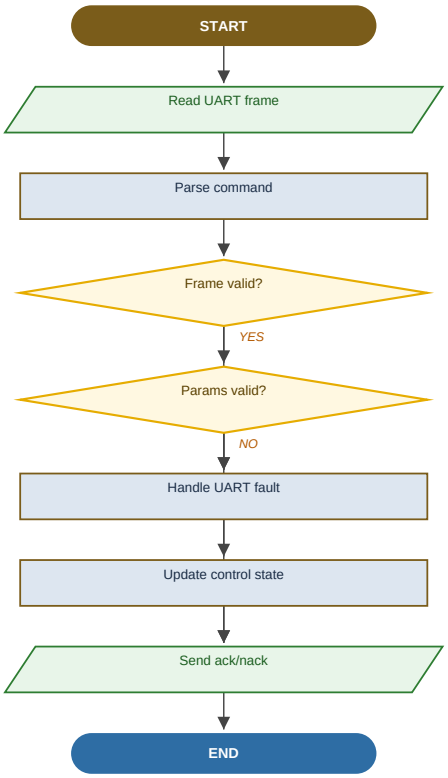
DD-COMP-003	**ADCSampler** *(Service)*
Description	Configures and triggers ADC sampling at a configurable rate between 100 Hz and 1 kHz, default 500 Hz, and publishes raw readings.
Inputs	Sampling rate configuration, ADC channel
Outputs	Raw ADC samples with timestamp
Dependencies	ConfigService, ADCDriver
Design Pattern	Factory
Traces to	SW-REQ-002

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



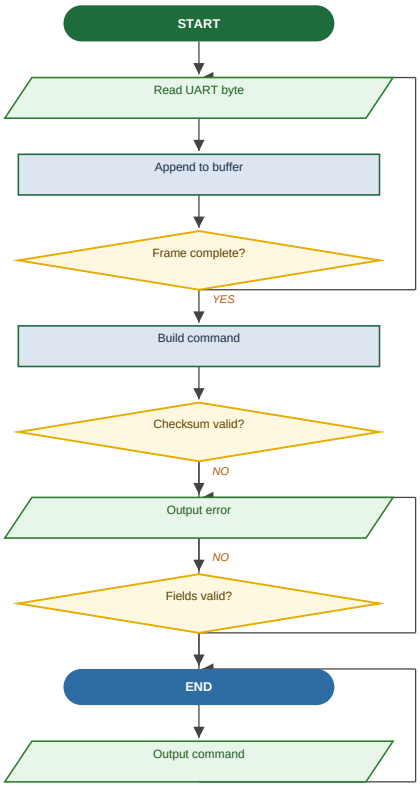
DD-COMP-004	**UARTCommandHandler** *(Service)*
Description	Parses UART command frames to set target position and control mode, validates inputs, and updates control state.
Inputs	UART RX bytes/frames
Outputs	Updated target position, control mode, command acknowledgements
Dependencies	UARTDriver, CommandParser, ControlStateStore, UARTFaultHandler
Design Pattern	Command
Traces to	SW-REQ-003

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



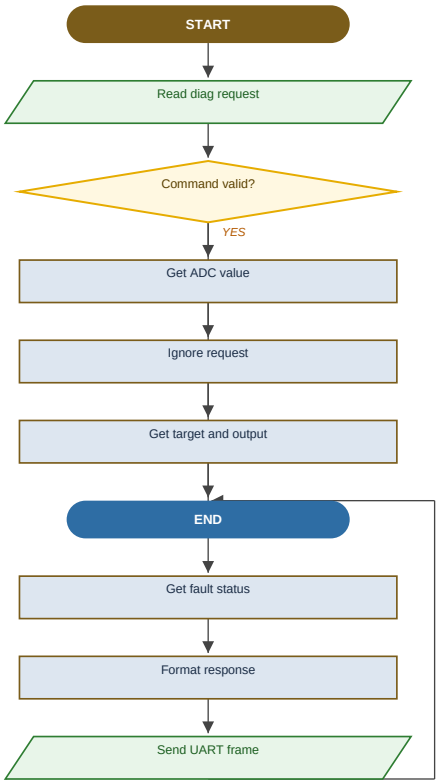
DD-COMP-005	**CommandParser** *(Class)*
Description	Parses incoming UART bytes into structured command messages with checksum and field validation.
Inputs	UART byte stream
Outputs	Command objects (set target, set mode) or parse errors
Dependencies	UARTProtocolConfig
Design Pattern	Builder
Traces to	SW-REQ-003

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



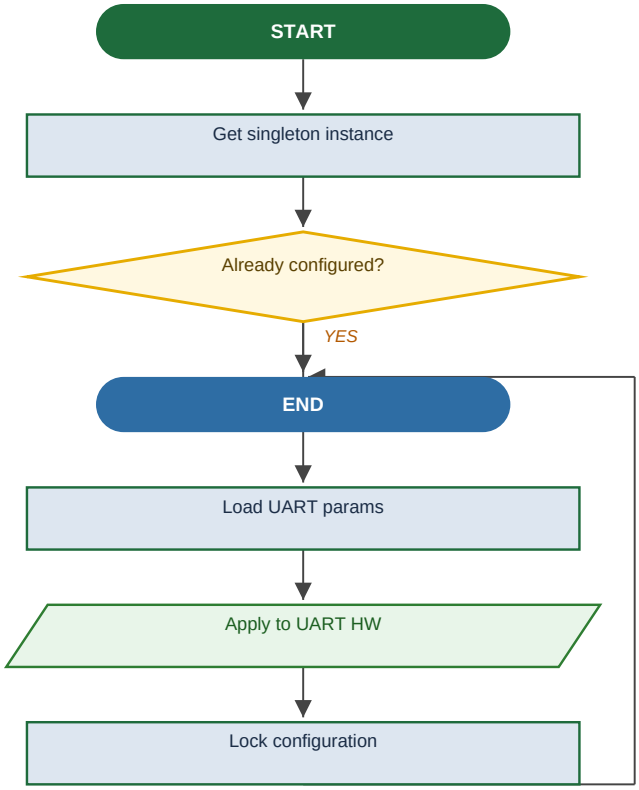
DD-COMP-006	**UARTDiagnosticsService** *(Service)*
Description	Responds to diagnostic request commands by gathering current ADC value, target position, control output, and fault status and sending a formatted UART response.
Inputs	Diagnostic request command, current system state
Outputs	UART diagnostic response frame
Dependencies	UARTDriver, SystemStateStore, FaultManager
Design Pattern	Facade
Traces to	SW-REQ-004

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



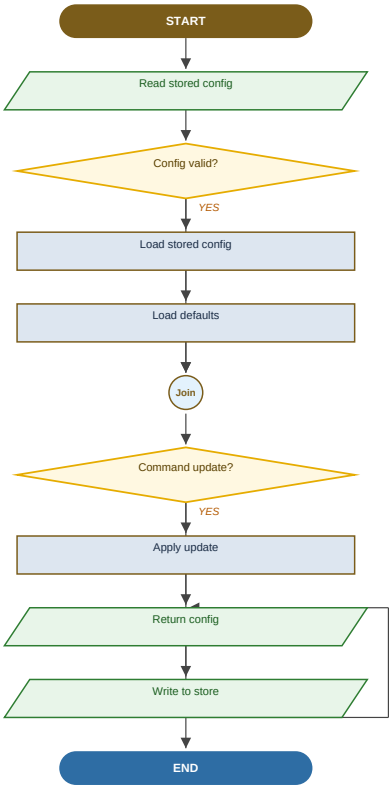
DD-COMP-007	**UARTDriverConfig** *(Class)*
Description	Initializes and enforces UART settings at 115200 baud, 8 data bits, no parity, 1 stop bit.
Inputs	Fixed UART configuration values
Outputs	Configured UART peripheral
Dependencies	HAL/UARTDriver
Design Pattern	Singleton
Traces to	SW-REQ-005

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



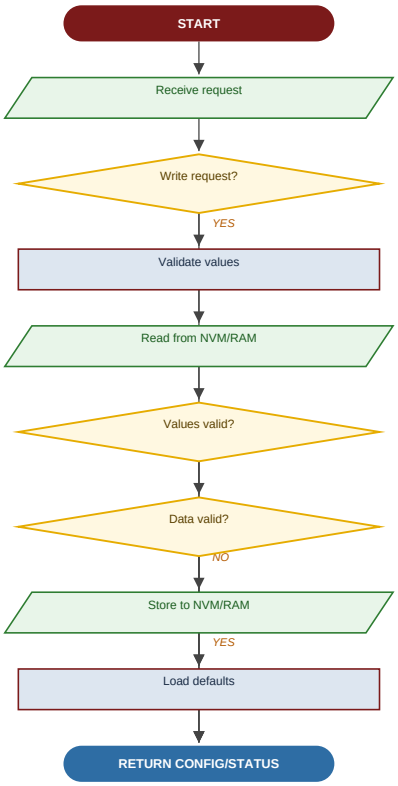
DD-COMP-008	**ConfigService** *(Service)*
Description	Provides runtime-configurable parameters for ADC sampling rate, control gains, and UART timeouts. Supports defaults and updates via commands or stored settings.
Inputs	Configuration requests, stored config values
Outputs	Current configuration values
Dependencies	ConfigStore, UARTCommandHandler
Design Pattern	Repository
Traces to	SW-REQ-006

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



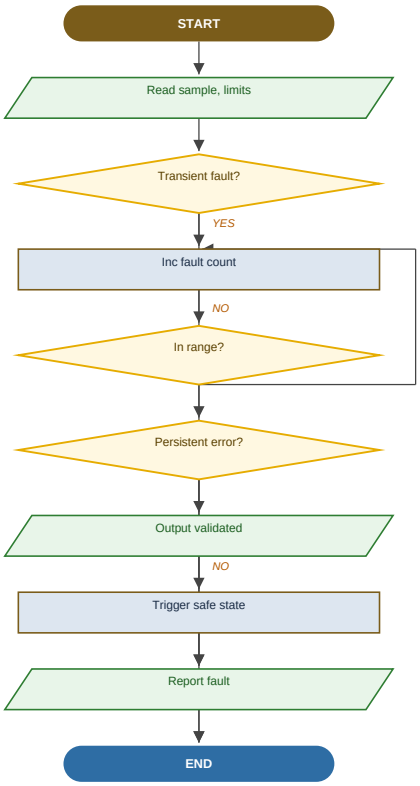
DD-COMP-009	**ConfigStore** *(Database)*
Description	Non-volatile and/or RAM storage for configuration parameters including sampling rate, gains, and timeouts with default values.
Inputs	Config write values
Outputs	Config read values
Dependencies	NVMDriver (if non-volatile)
Design Pattern	Repository
Traces to	SW-REQ-006

■ Flow Diagram *(data flow) — See Section 7 for full-page view*



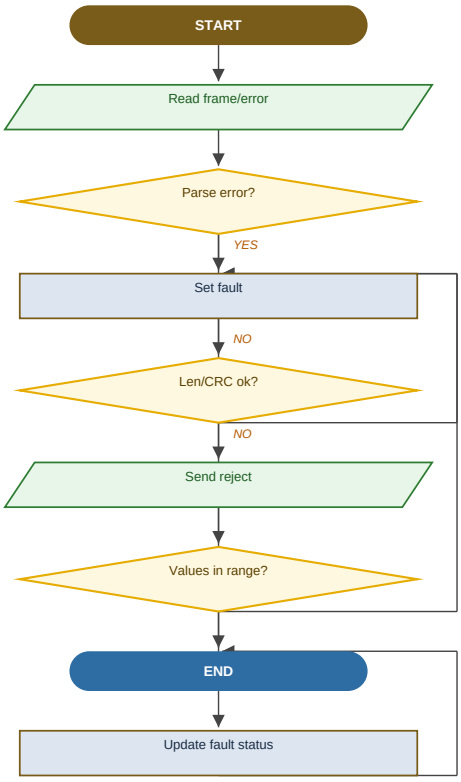
DD-COMP-010	**ADCValidationService** *(Service)*
Description	Validates ADC samples for transient faults and out-of-range values. Rejects invalid samples, increments fault counters, and triggers safe state on persistent errors.
Inputs	Raw ADC samples, calibrated min/max limits
Outputs	Validated ADC samples or fault indication
Dependencies	FaultManager, CalibrationStore
Design Pattern	Chain of Responsibility
Traces to	SW-REQ-007

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



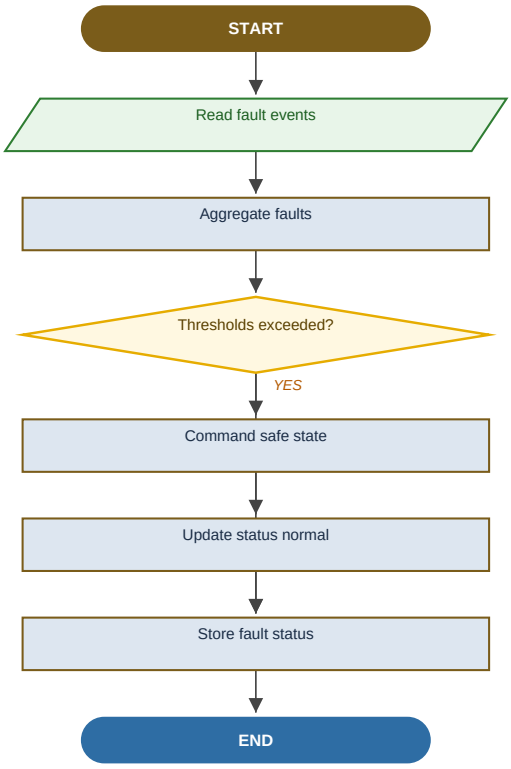
DD-COMP-011	**UARTFaultHandler** *(Service)*
Description	Validates UART command frames for integrity (length, checksum, value ranges) and manages fault state for invalid or malformed data.
Inputs	Parsed command frames, parse errors
Outputs	Fault status updates, rejected command responses
Dependencies	FaultManager
Design Pattern	Strategy
Traces to	SW-REQ-007

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



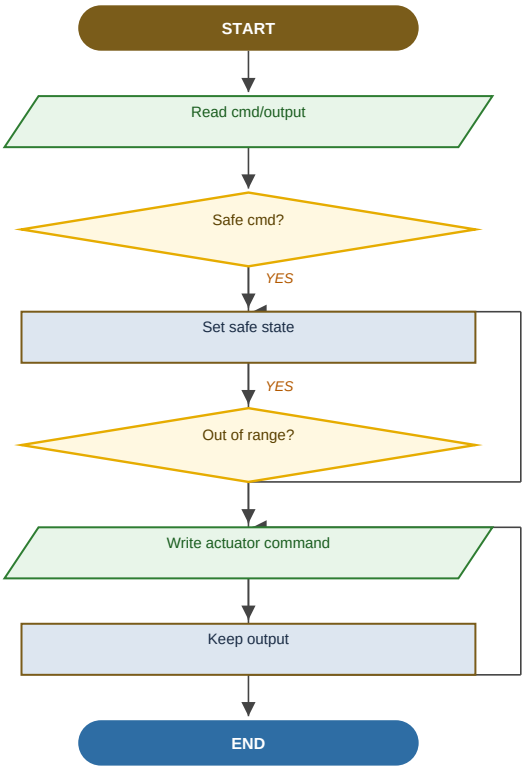
DD-COMP-012	**FaultManager** *(Service)*
Description	Centralized fault state handling. Aggregates ADC and UART fault conditions and commands entry to safe state when required.
Inputs	Fault events, thresholds
Outputs	Fault status, safe state command
Dependencies	SafeStateController, SystemStateStore
Design Pattern	Observer
Traces to	SW-REQ-007

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



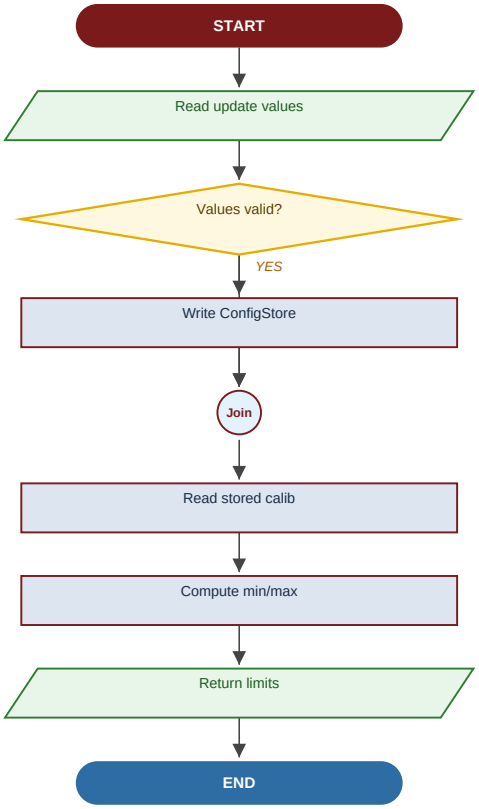
DD-COMP-013	**SafeStateController** *(Service)*
Description	Transitions actuator to a defined safe state (e.g., disable output or hold) when out-of-range ADC readings are detected.
Inputs	Safe state command, current control output
Outputs	Actuator disable/neutral command
Dependencies	ActuatorDriver
Design Pattern	State
Traces to	SW-REQ-008

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



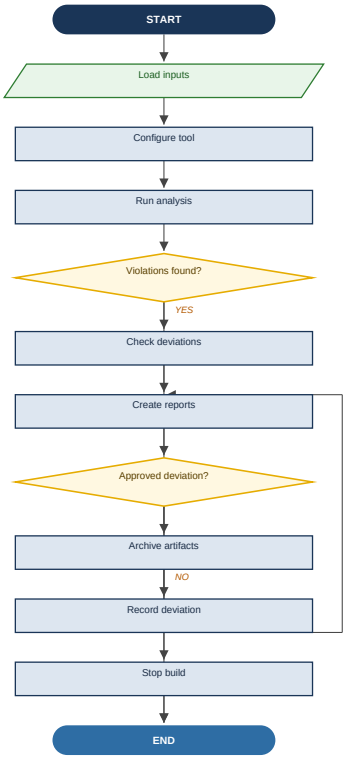
DD-COMP-014	**CalibrationStore** *(Database)*
Description	Stores calibrated potentiometer min/max limits used to validate ADC range.
Inputs	Calibration update values
Outputs	Min/max ADC limits
Dependencies	ConfigStore
Design Pattern	Repository
Traces to	SW-REQ-008

■ Flow Diagram *(data flow) — See Section 7 for full-page view*



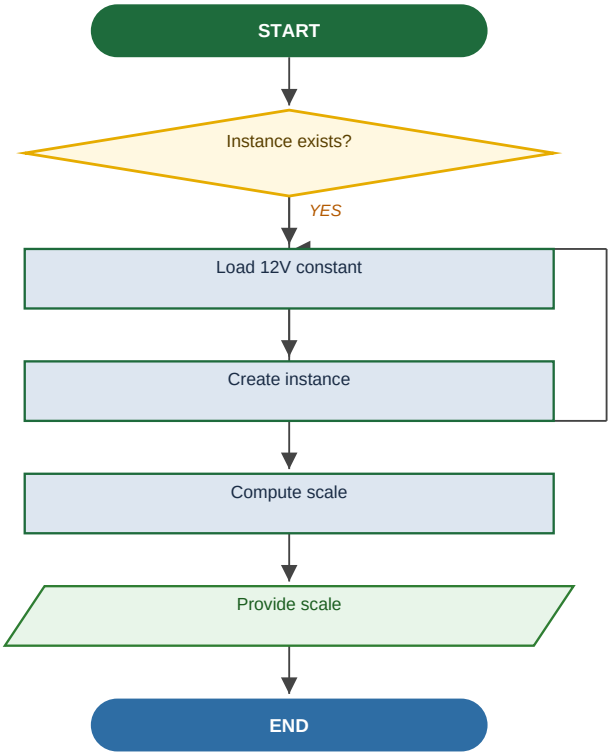
DD-COMP-015	**MISRA Compliance Process** *(Module)*
Description	Defines project-wide MISRA-C:2012 compliance workflow including static analysis configuration, rule enforcement, and deviation recording. Integrates tooling outputs into build artifacts and maintains a deviation register for approved exceptions.
Inputs	Source code files, MISRA rule set, approved deviation list
Outputs	Static analysis reports, deviation documentation artifacts
Dependencies	BuildSystem, StaticAnalysisToolchain, ConfigurationRepository
Design Pattern	N/A
Traces to	SW-REQ-009

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



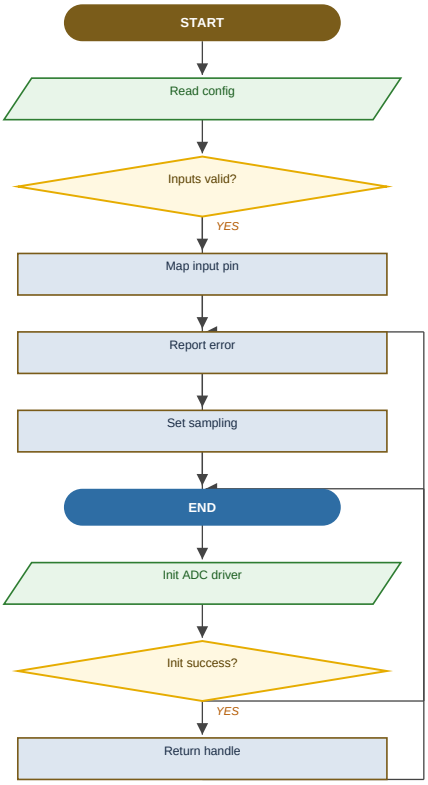
DD-COMP-016	**Supply Voltage Assumption** *(Class)*
Description	Holds constant for 12V actuator supply and provides scaling factor used by control output computation for duty cycle/drive limits.
Inputs	None (compile-time constant 12.0V)
Outputs	Scaling factor for control output duty/drive limit calculations
Dependencies	ControlOutputCalculator
Design Pattern	Singleton
Traces to	SW-REQ-010

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



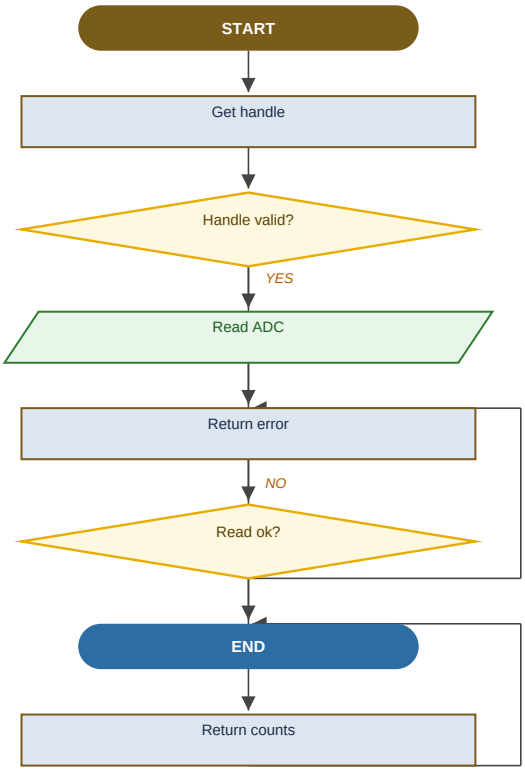
DD-COMP-017	**ADC Channel Configuration** *(Service)*
Description	Configures a single ADC channel for actuator potentiometer feedback with required sampling time, resolution, and input pin mapping.
Inputs	ADC channel ID, pin configuration, sampling settings
Outputs	Initialized ADC channel handle
Dependencies	MCU_ADC_Driver, ConfigurationRepository
Design Pattern	Factory
Traces to	SW-REQ-011

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



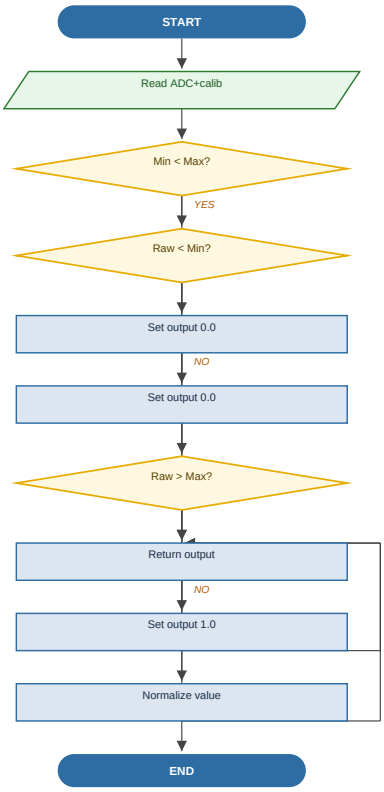
DD-COMP-018	**ADC Read Service** *(Service)*
Description	Reads the configured ADC channel for actuator feedback and provides raw ADC counts. Invoked by control loop each cycle.
Inputs	ADC channel handle
Outputs	Raw ADC value (integer counts)
Dependencies	MCU_ADC_Driver
Design Pattern	N/A
Traces to	SW-REQ-012

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



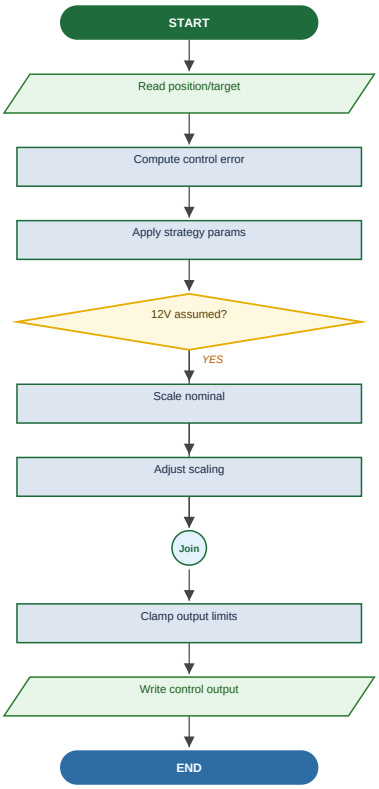
DD-COMP-019	**Position Scaling Algorithm** *(Algorithm)*
Description	Scales raw ADC counts to normalized position in [0.0, 1.0] using configurable min/max calibration values and clamps values outside range.
Inputs	Raw ADC value, calibration_min, calibration_max
Outputs	Normalized position (float 0.0 to 1.0)
Dependencies	CalibrationRepository
Design Pattern	N/A
Traces to	SW-REQ-013

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



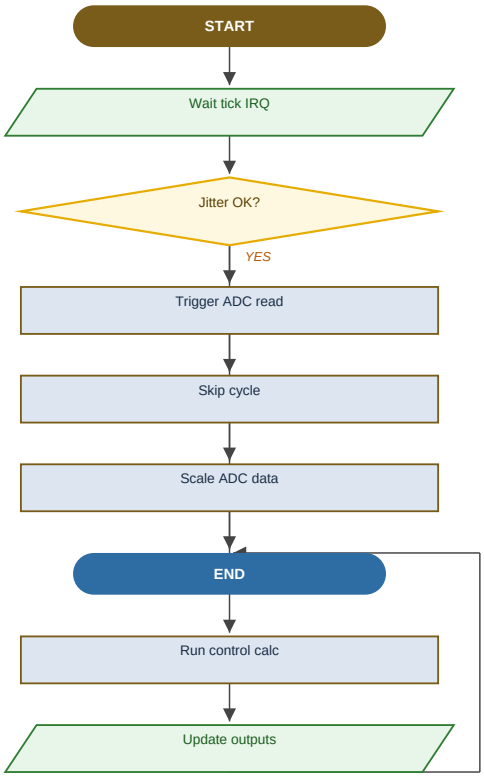
DD-COMP-020	**Control Output Calculator** *(Class)*
Description	Computes actuator control output each cycle based on measured position and commanded target. Applies scaling based on 12V assumption and output limits.
Inputs	Measured position, target position, control parameters
Outputs	Control output command (duty cycle/drive level)
Dependencies	SupplyVoltageAssumption, PositionScalingAlgorithm
Design Pattern	Strategy
Traces to	SW-REQ-014

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



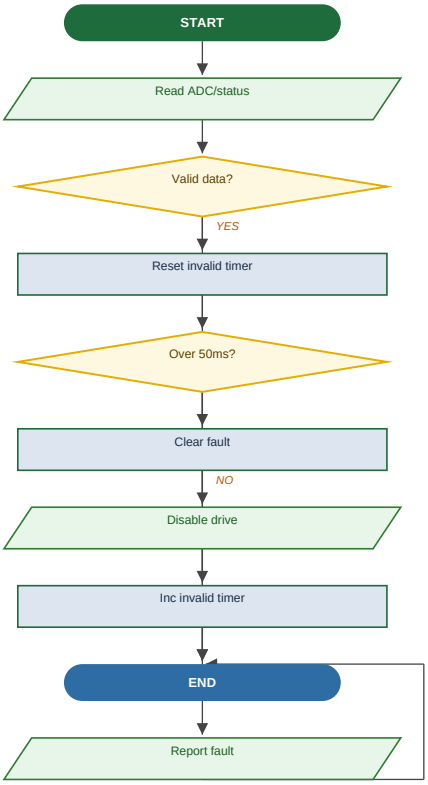
DD-COMP-021	**Control Loop Scheduler** *(Service)*
Description	Executes the control loop at a fixed period of 10 ms with allowable jitter of ±1 ms, triggering ADC read, scaling, control calculation, and output update.
Inputs	System tick timer interrupts
Outputs	Periodic control loop execution events
Dependencies	MCU_Timer_Driver, ControlOutputCalculator, ADCReadService
Design Pattern	Observer
Traces to	SW-REQ-015

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



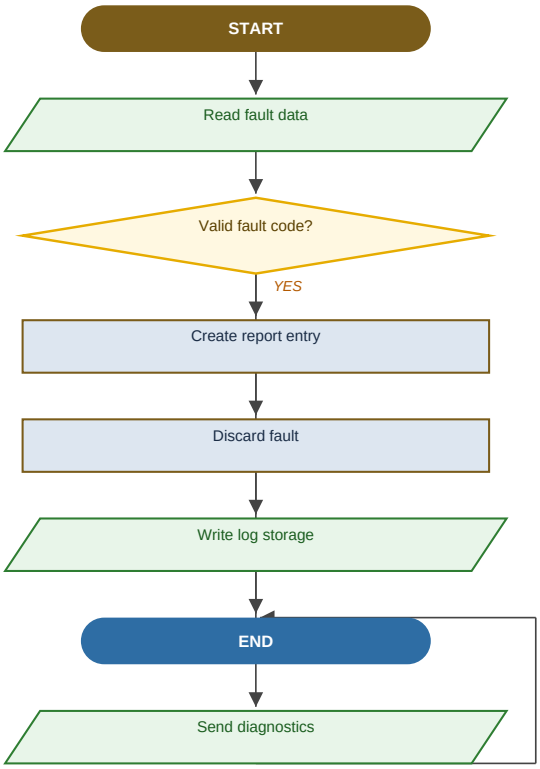
DD-COMP-022	**Feedback Fault Monitor** *(Class)*
Description	Monitors ADC feedback validity over time and disables actuator drive while reporting a fault if invalid data persists beyond 50 ms.
Inputs	Raw ADC value, validity status, time since last valid reading
Outputs	Fault status, drive disable command
Dependencies	ADCReadService, ControlOutputCalculator, FaultReporter
Design Pattern	State
Traces to	SW-REQ-016

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



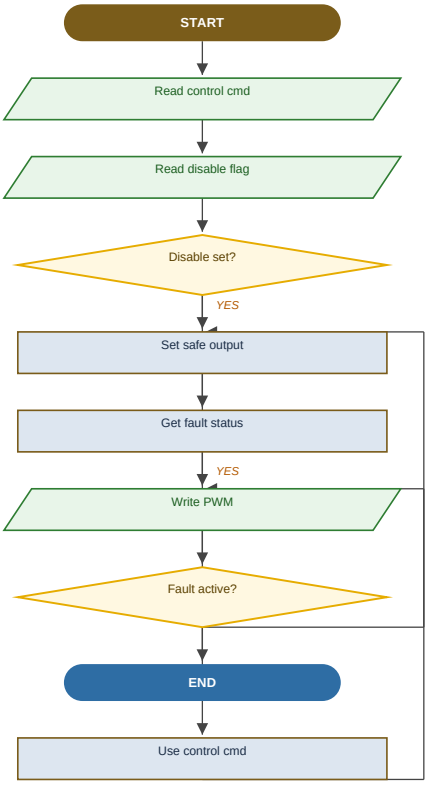
DD-COMP-023	**Fault Reporter** *(Service)*
Description	Registers and reports system faults to diagnostics interface and logs fault conditions for feedback invalidity.
Inputs	Fault code, timestamp
Outputs	Fault report entry
Dependencies	DiagnosticsInterface, LogStorage
Design Pattern	N/A
Traces to	SW-REQ-016

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



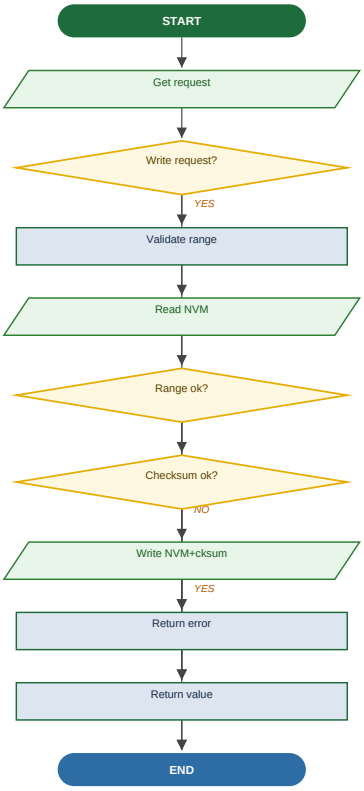
DD-COMP-024	**Actuator Drive Output** *(Service)*
Description	Applies the computed control output to the actuator driver hardware and supports disable command from safety module.
Inputs	Control output command, disable flag
Outputs	Updated actuator drive signal
Dependencies	MCU_PWM_Driver, ControlOutputCalculator, FeedbackFaultMonitor
Design Pattern	N/A
Traces to	SW-REQ-014

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



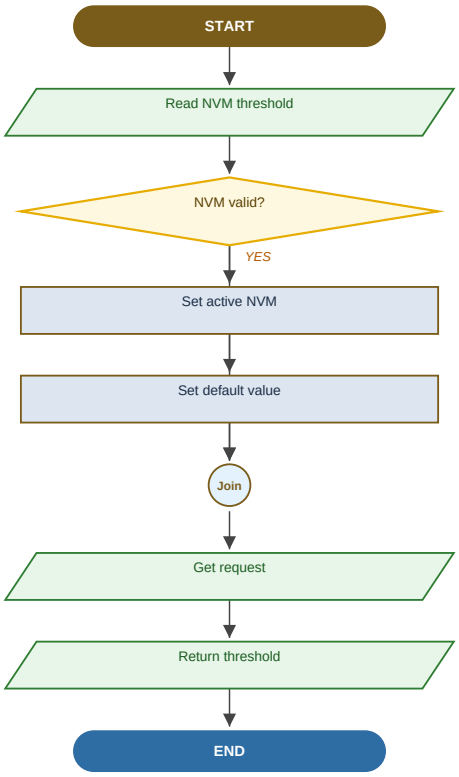
DD-COMP-025	**ThresholdConfigRepository** *(Class)*
Description	Persists and retrieves the shipping position ADC threshold from non-volatile memory. Provides read/write access with validation of range and checksum to ensure stored value integrity.
Inputs	Threshold value to persist; read request
Outputs	Stored threshold value; status codes for read/write success or failure
Dependencies	NvmDriver
Design Pattern	Repository
Traces to	SW-REQ-017

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



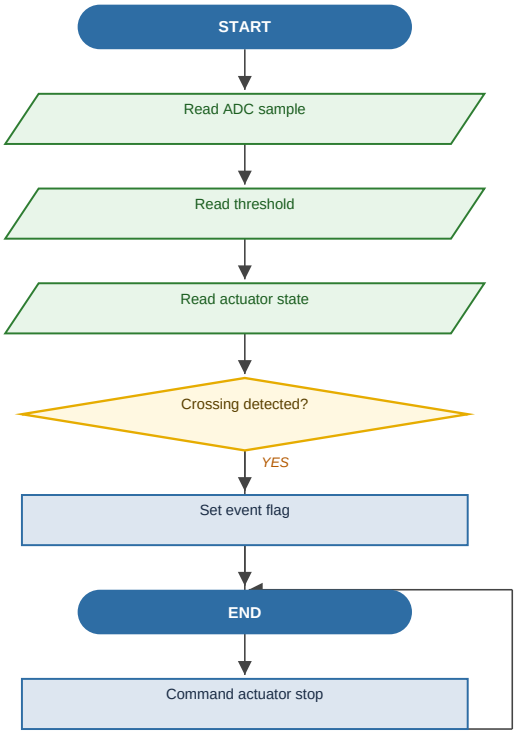
DD-COMP-026	**ThresholdConfigService** *(Service)*
Description	Loads the shipping position threshold during initialization and serves it to control logic at runtime. Falls back to default if NVM data is invalid.
Inputs	Initialization trigger; runtime get requests
Outputs	Active shipping threshold value
Dependencies	ThresholdConfigRepository
Design Pattern	Singleton
Traces to	SW-REQ-017

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



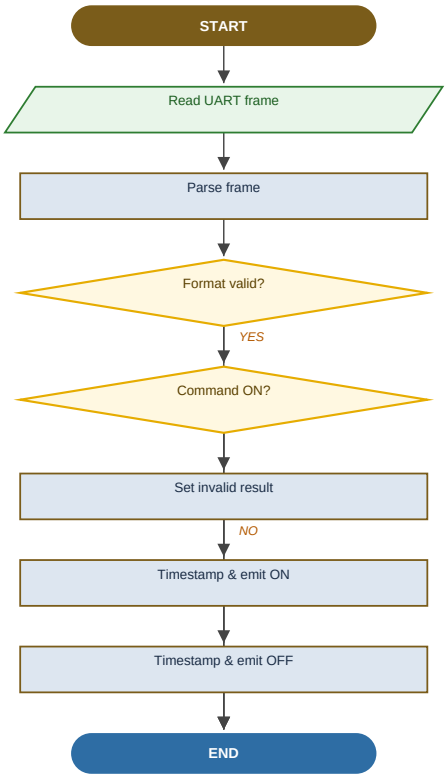
DD-COMP-027	**ShippingPositionStopLogic** *(Algorithm)*
Description	Compares each ADC sample to the configured shipping position threshold and commands actuator stop when crossing occurs. Logic executes within the ADC sampling ISR or high-priority task to meet 20 ms stop requirement.
Inputs	ADC sample stream; configured threshold; current actuator state
Outputs	Stop command to actuator driver; event flag for crossing detection
Dependencies	ThresholdConfigService, ActuatorDriver
Design Pattern	Observer
Traces to	SW-REQ-018

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



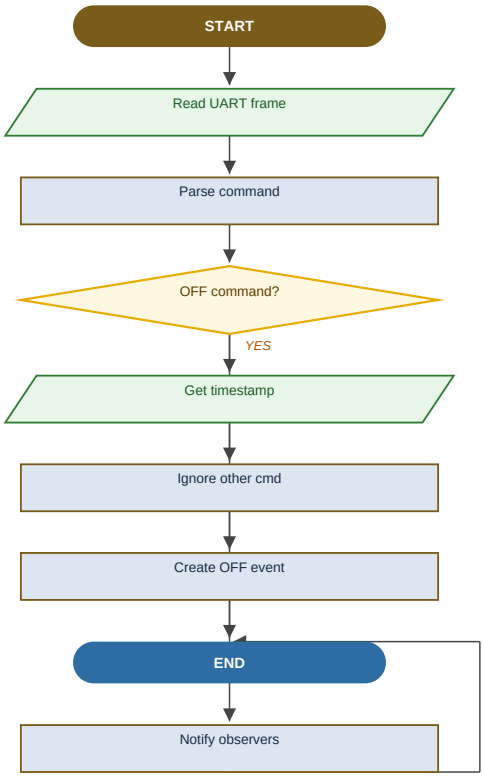
DD-COMP-028	**UartCommandParser** *(Service)*
Description	Parses incoming UART frames, validates command format, and emits ON/OFF command events. Ensures receipt-to-dispatch latency within 50 ms.
Inputs	UART RX bytes/frames
Outputs	Command event (ON/OFF) with timestamp; validation result
Dependencies	UartDriver, ActuatorStateMachine
Design Pattern	Observer
Traces to	SW-REQ-019

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



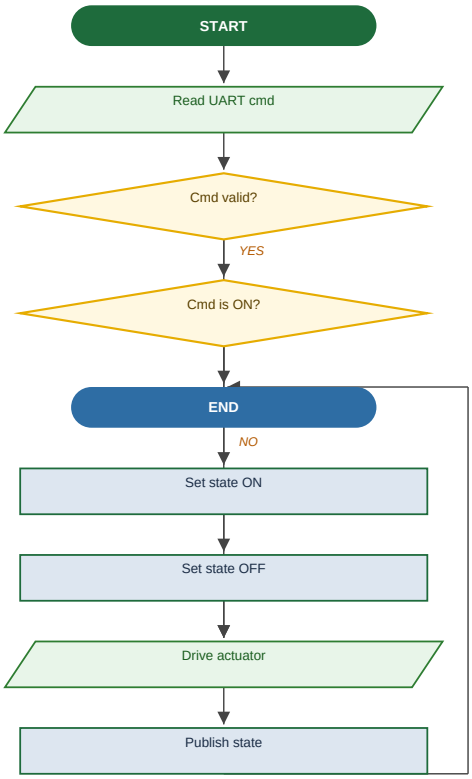
DD-COMP-029	**UartCommandParser_OFF** *(Service)*
Description	Handles OFF command parsing and dispatch in the same parsing service to trigger OFF transitions within 50 ms.
Inputs	UART RX bytes/frames
Outputs	OFF command event with timestamp
Dependencies	UartDriver, ActuatorStateMachine
Design Pattern	Observer
Traces to	SW-REQ-020

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



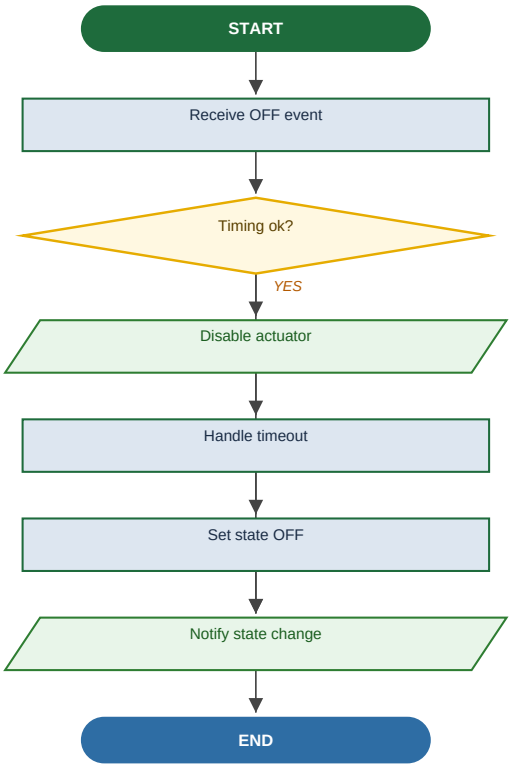
DD-COMP-030	**ActuatorStateMachine** *(Class)*
Description	Manages actuator ON/OFF states and enforces transitions triggered by UART commands. Publishes state changes to other services for acknowledgements.
Inputs	ON/OFF command events
Outputs	Actuator drive enable/disable signals; state change events
Dependencies	ActuatorDriver
Design Pattern	State
Traces to	SW-REQ-019

■ Flow Diagram *(state flow) — See Section 7 for full-page view*



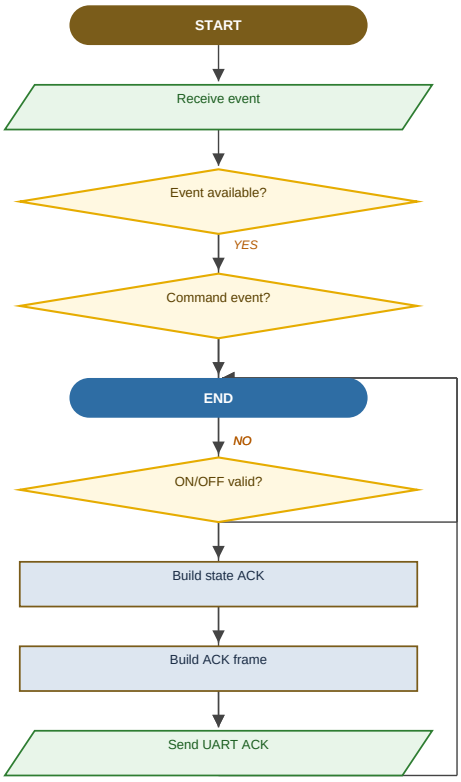
DD-COMP-031	**ActuatorStateMachine_OFF** *(Class)*
Description	Implements OFF transition path with de-energize command to actuator driver within timing constraints.
Inputs	OFF command event
Outputs	Actuator drive disable signal; state change event
Dependencies	ActuatorDriver
Design Pattern	State
Traces to	SW-REQ-020

■ Flow Diagram *(state flow) — See Section 7 for full-page view*



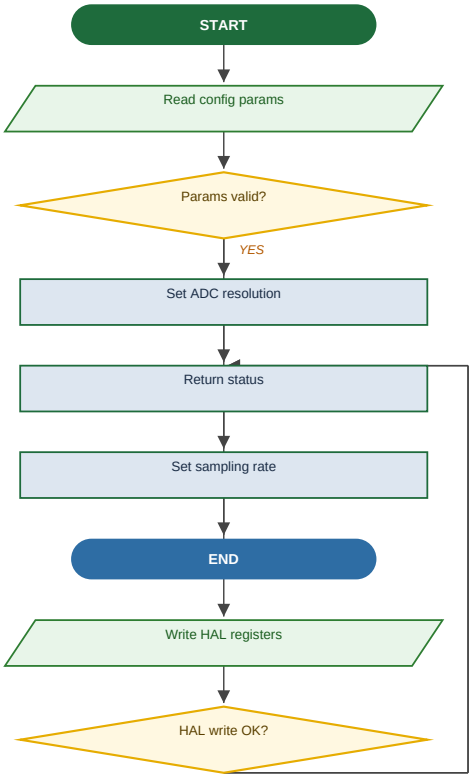
DD-COMP-032	**UartAckService** *(Service)*
Description	Sends UART acknowledgement frames for valid ON/OFF commands within 20 ms of command processing. Subscribes to state change or command validation events.
Inputs	Validated command event or state change event
Outputs	UART ACK message
Dependencies	UartDriver, UartCommandParser, ActuatorStateMachine
Design Pattern	Observer
Traces to	SW-REQ-021

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



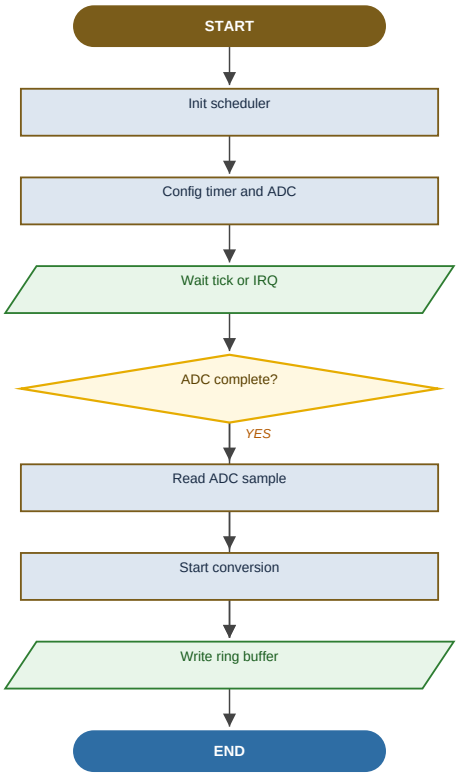
DD-COMP-033	**AdcDriverConfig** *(Class)*
Description	Configures the internal ADC hardware to 12-bit resolution or higher and applies sampling rate settings. Provides initialization API called at startup.
Inputs	Initialization trigger; configuration parameters
Outputs	Configured ADC peripheral; status code
Dependencies	HardwareAbstractionLayer
Design Pattern	Facade
Traces to	SW-REQ-023

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



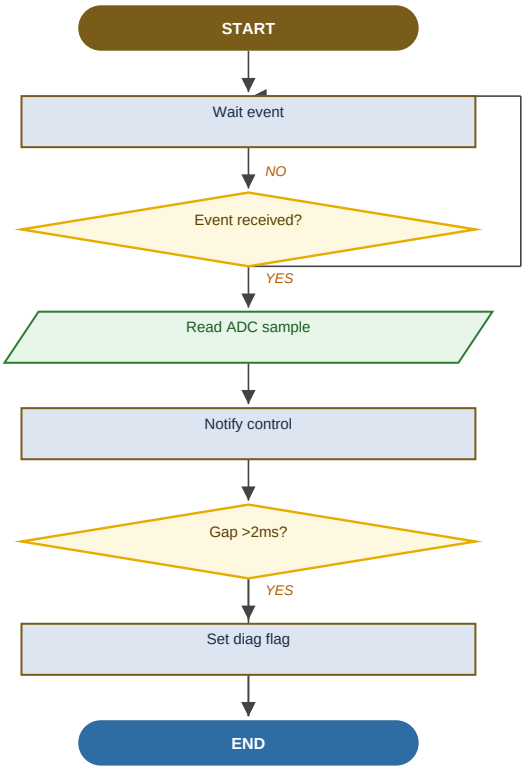
DD-COMP-034	**AdcSamplingScheduler** *(Service)*
Description	Schedules continuous ADC sampling at >=1000 samples per second using timer interrupt or DMA. Ensures no gaps longer than 2 ms by using a periodic trigger and buffering.
Inputs	Timer ticks; ADC conversion complete interrupts
Outputs	ADC sample stream into ring buffer
Dependencies	AdcDriverConfig, TimerDriver, RingBuffer
Design Pattern	Producer-Consumer
Traces to	SW-REQ-024

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



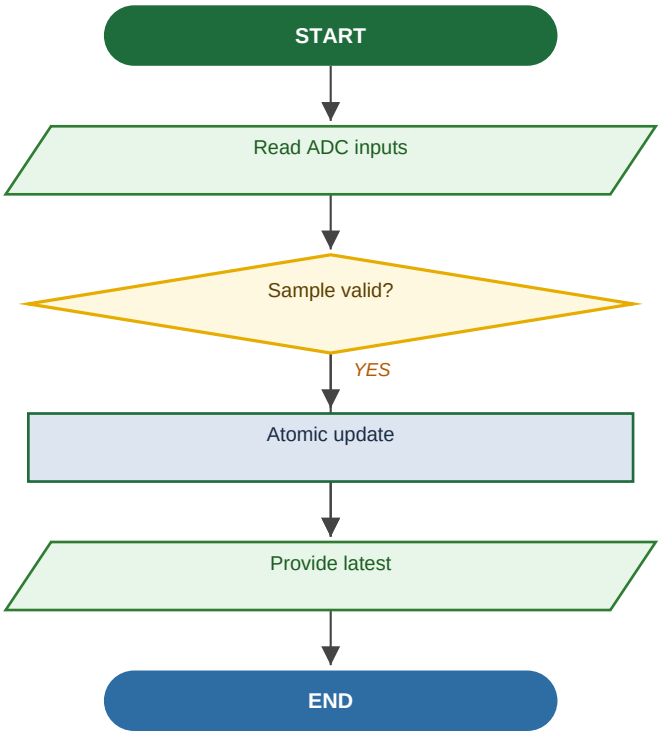
DD-COMP-035	**AdcContinuousAcquisition** *(Service)*
Description	Continuously acquires ADC samples during normal operation and feeds them to control logic. Monitors timing to ensure no gaps >2 ms and raises diagnostic flags if violated.
Inputs	ADC conversion complete events; system state
Outputs	Sample notifications to control logic; diagnostic status
Dependencies	AdcSamplingScheduler, ShippingPositionStopLogic
Design Pattern	Observer
Traces to	SW-REQ-022

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



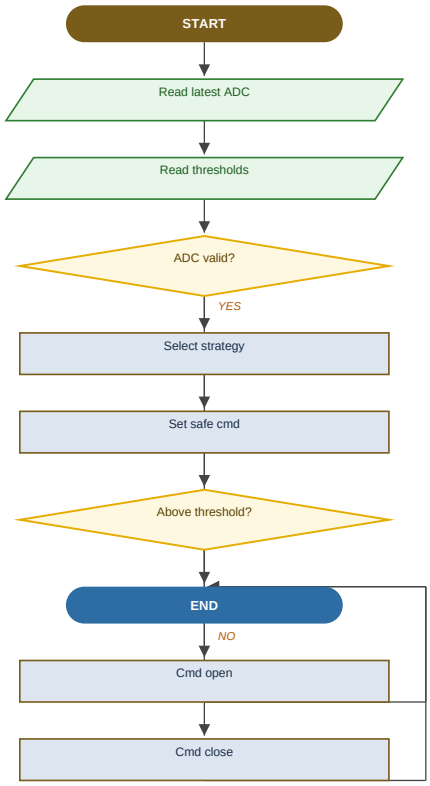
DD-COMP-036	**ADC Sample Buffer** *(Class)*
Description	Maintains the latest ADC sample and timestamp; provides atomic read access to the most recent sample for control computations.
Inputs	raw_adc_value (int), sample_timestamp_us (uint64)
Outputs	latest_adc_value (int), latest_timestamp_us (uint64)
Dependencies	ADCDriver
Design Pattern	Singleton
Traces to	SW-REQ-025

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



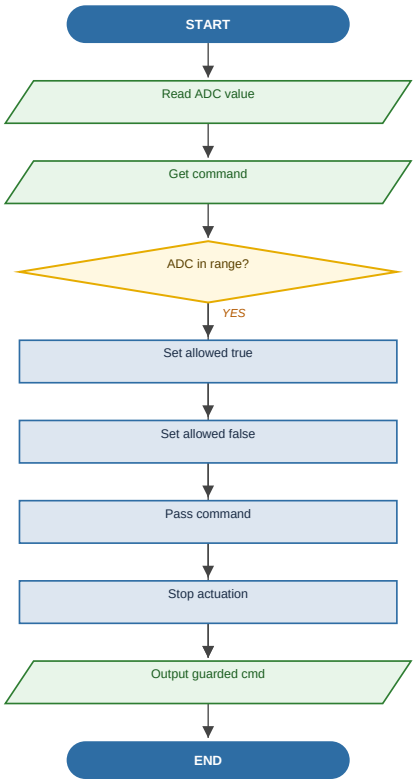
DD-COMP-037	**Flap Control Computation** *(Service)*
Description	Computes flap actuation commands using only the latest ADC value and configured thresholds; no other sensor inputs are consulted.
Inputs	latest_adc_value (int), threshold_config (struct)
Outputs	actuation_command (enum/struct)
Dependencies	ADCSampleBuffer, ThresholdConfigStore
Design Pattern	Strategy
Traces to	SW-REQ-026

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



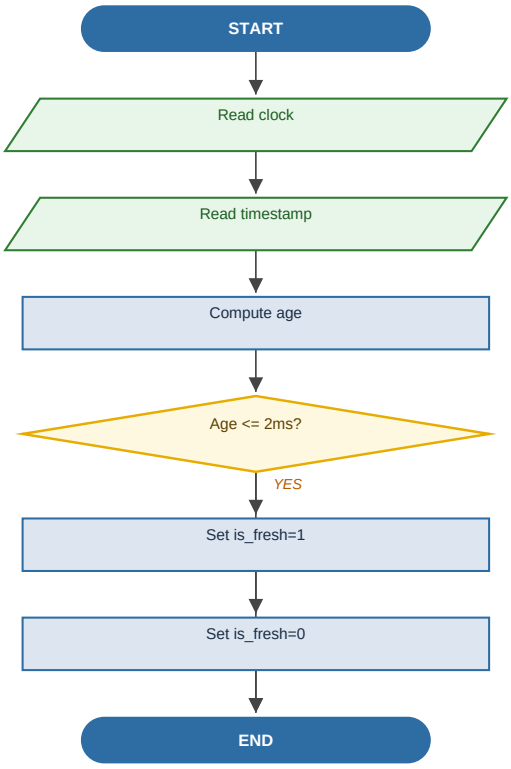
DD-COMP-038	**Actuation Range Guard** *(Algorithm)*
Description	Enforces actuation enablement only when ADC value is within 500–2600; disables or stops actuation when outside this range.
Inputs	latest_adc_value (int), requested_command (enum/struct)
Outputs	guarded_command (enum/struct), actuation_allowed (bool)
Dependencies	ADCSampleBuffer
Design Pattern	Decorator
Traces to	SW-REQ-027

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



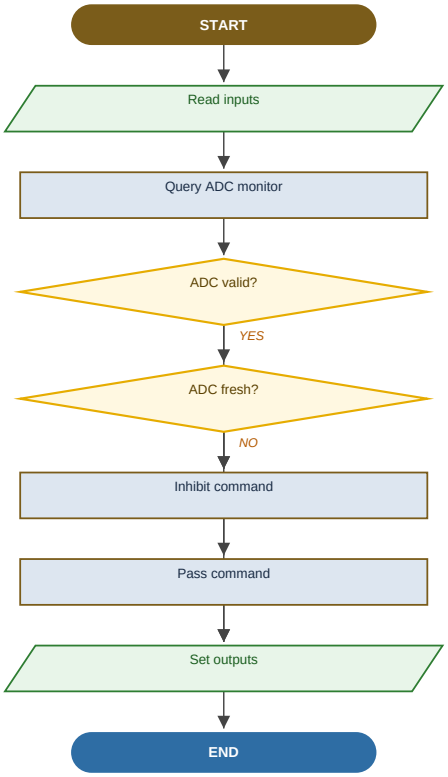
DD-COMP-039	**Real-Time ADC Freshness Check** *(Algorithm)*
Description	Validates that control decisions use a sample not older than 2 ms by comparing current time with latest sample timestamp before control execution.
Inputs	current_time_us (uint64), latest_timestamp_us (uint64)
Outputs	is_fresh (bool)
Dependencies	ADCSampleBuffer, SystemClock
Design Pattern	N/A
Traces to	SW-REQ-028

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



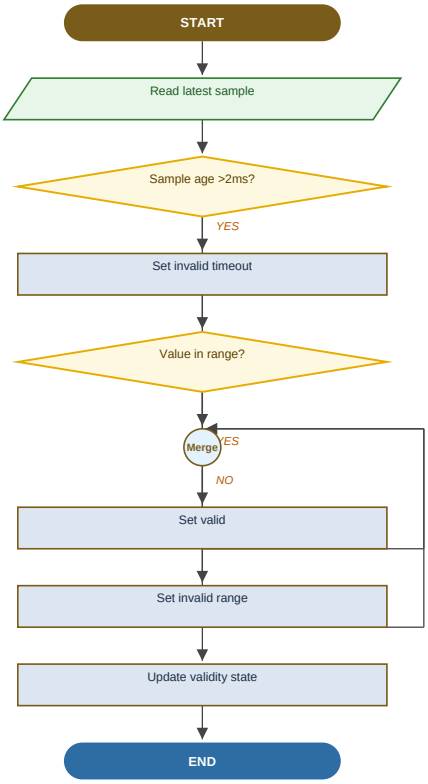
DD-COMP-040	**Open-Loop Inhibit Gate** *(Service)*
Description	Blocks actuator drive whenever valid ADC feedback is unavailable; allows drive only if ADC is valid and fresh.
Inputs	adc_valid (bool), requested_command (enum/struct)
Outputs	inhibited_command (enum/struct), drive_enabled (bool)
Dependencies	ADCValidityMonitor
Design Pattern	Chain of Responsibility
Traces to	SW-REQ-029

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



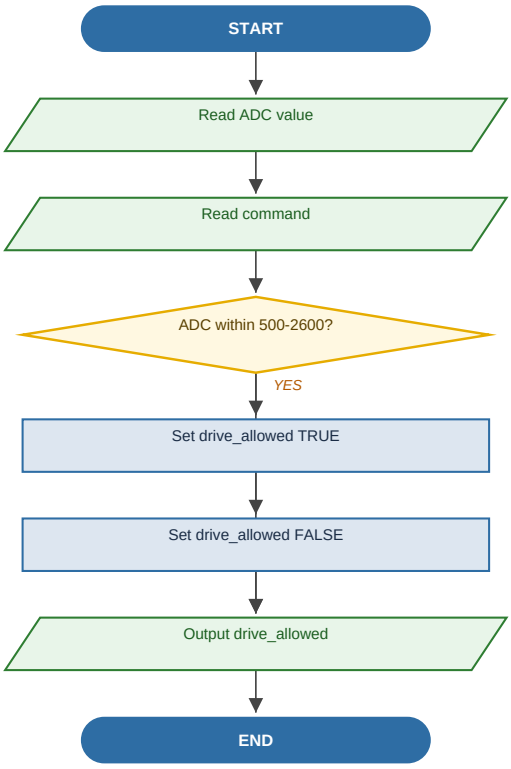
DD-COMP-041	**ADC Validity Monitor** *(Service)*
Description	Flags ADC feedback invalid if no new sample arrives within 2 ms or if value is outside 0–4095; updates validity state.
Inputs	latest_adc_value (int), latest_timestamp_us (uint64), current_time_us (uint64)
Outputs	adc_valid (bool), validity_reason (enum)
Dependencies	ADCSampleBuffer, SystemClock
Design Pattern	Observer
Traces to	SW-REQ-030

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



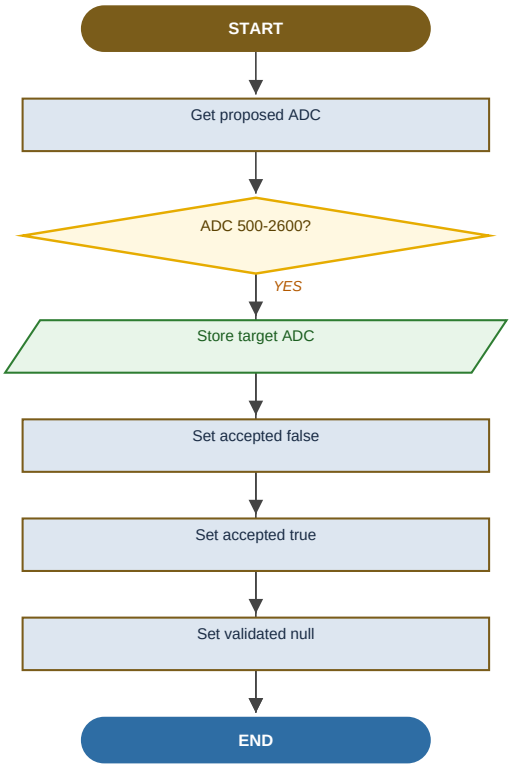
DD-COMP-042	**Drive Permission Check** *(Algorithm)*
Description	Permits actuator drive commands only when measured ADC value is within 500–2600 inclusive; otherwise denies drive.
Inputs	latest_adc_value (int), requested_command (enum/struct)
Outputs	drive_allowed (bool)
Dependencies	ADCSampleBuffer
Design Pattern	N/A
Traces to	SW-REQ-031

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



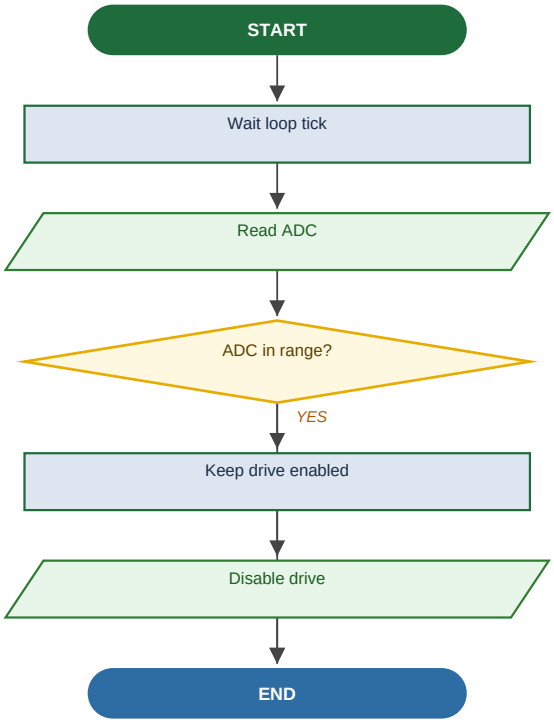
DD-COMP-043	**Target ADC Validator** *(Service)*
Description	Rejects configured target ADC values outside 500–2600 inclusive before storing or applying them to control logic.
Inputs	proposed_target_adc (int)
Outputs	accepted (bool), validated_target_adc (int|null)
Dependencies	ThresholdConfigStore
Design Pattern	Factory
Traces to	SW-REQ-032

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



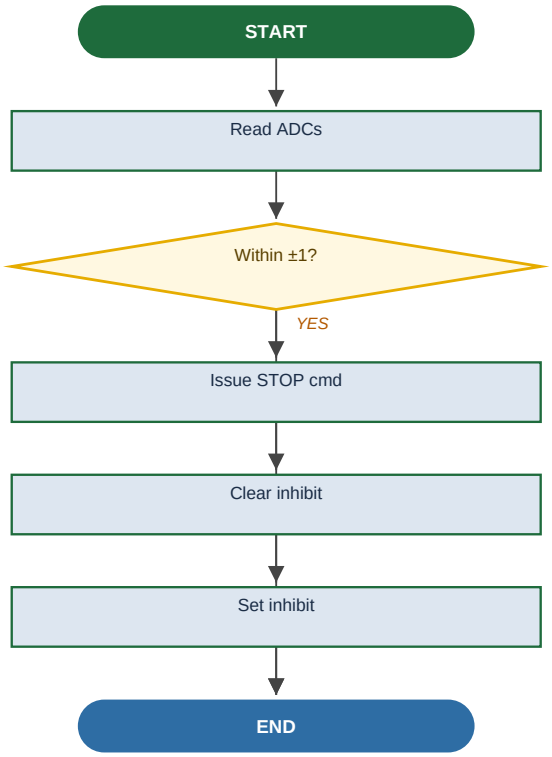
DD-COMP-044	**ActuatorSafetyMonitor** *(Class)*
Description	Evaluates ADC bounds each control loop cycle and disables actuator drive when the measured ADC value is <500 or >2600. Executes within the loop timing budget to ensure action within one cycle.
Inputs	measured_adc_value, control_loop_tick
Outputs	drive_enable=false when out of range
Dependencies	ActuatorDriveService, ADCService
Design Pattern	Observer
Traces to	SW-REQ-033

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



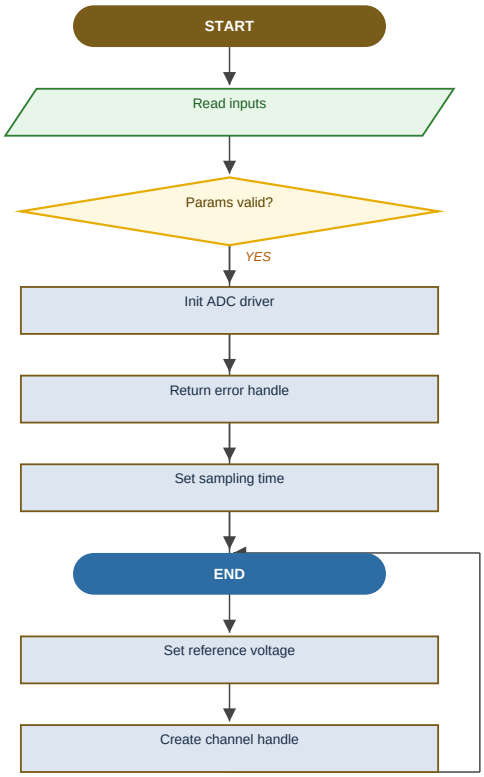
DD-COMP-045	**ActuatorTargetStopController** *(Class)*
Description	Compares measured ADC to configured target and stops actuator movement when within ±1 count. Issues stop command and inhibits further drive until outside tolerance or new command.
Inputs	measured_adc_value, target_adc_value
Outputs	drive_command=STOP
Dependencies	ActuatorDriveService, TargetConfigService
Design Pattern	Strategy
Traces to	SW-REQ-034

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



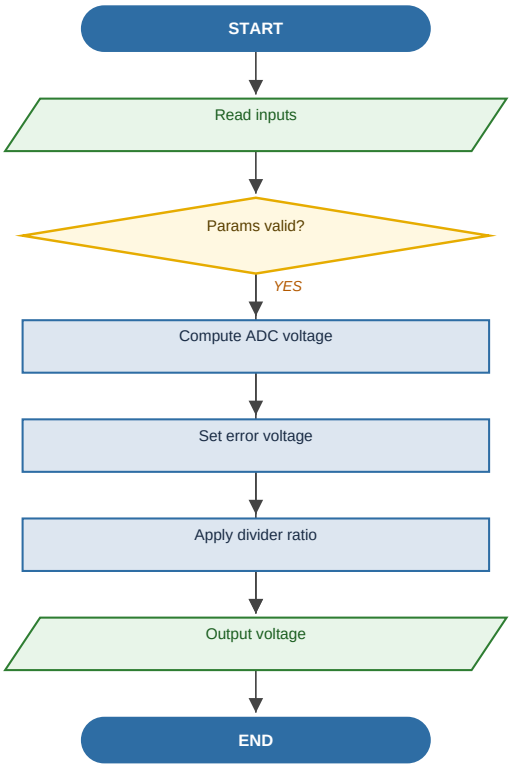
DD-COMP-046	**SupplyVoltageADCChannelConfig** *(Service)*
Description	Initializes and configures the dedicated ADC channel connected to the external supply voltage divider with proper sampling time and reference settings.
Inputs	adc_channel_id, sampling_time, reference_voltage
Outputs	configured_adc_channel_handle
Dependencies	ADCDriver
Design Pattern	Facade
Traces to	SW-REQ-035

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



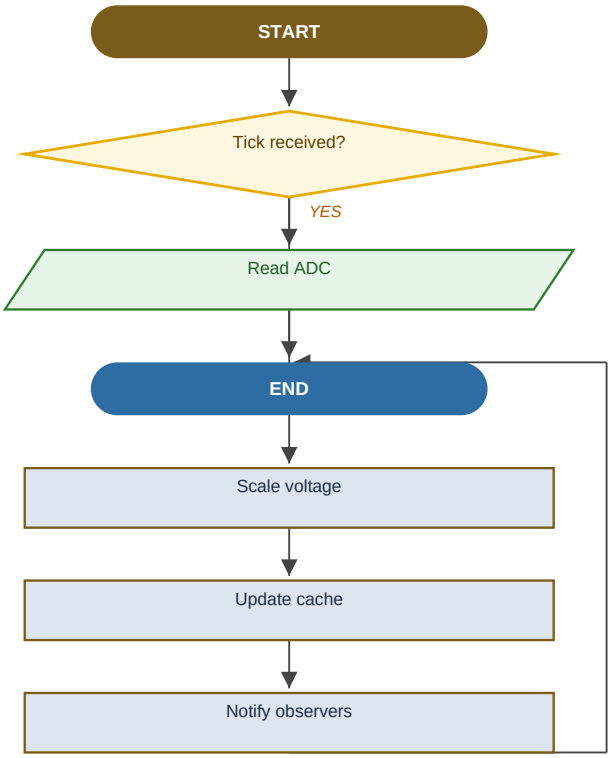
DD-COMP-047	**SupplyVoltageScaler** *(Algorithm)*
Description	Computes external supply voltage by applying the configured divider ratio to the raw ADC value and ADC reference scaling.
Inputs	raw_adc_value, divider_ratio, adc_ref_voltage, adc_resolution
Outputs	computed_supply_voltage
Dependencies	SupplyVoltageADCChannelConfig
Design Pattern	N/A
Traces to	SW-REQ-036

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



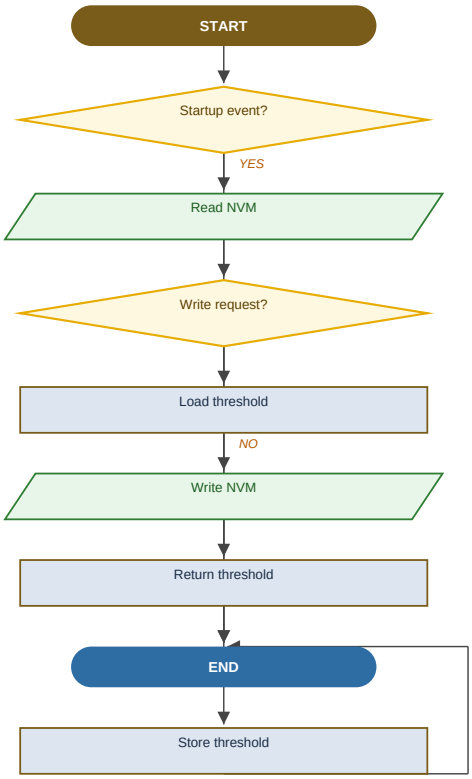
DD-COMP-048	**SupplyVoltageSampler** *(Service)*
Description	Samples the supply ADC channel at least once per control loop cycle and updates the computed supply voltage cache.
Inputs	control_loop_tick
Outputs	updated_supply_voltage_cache
Dependencies	ADCService, SupplyVoltageScaler
Design Pattern	Observer
Traces to	SW-REQ-037

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



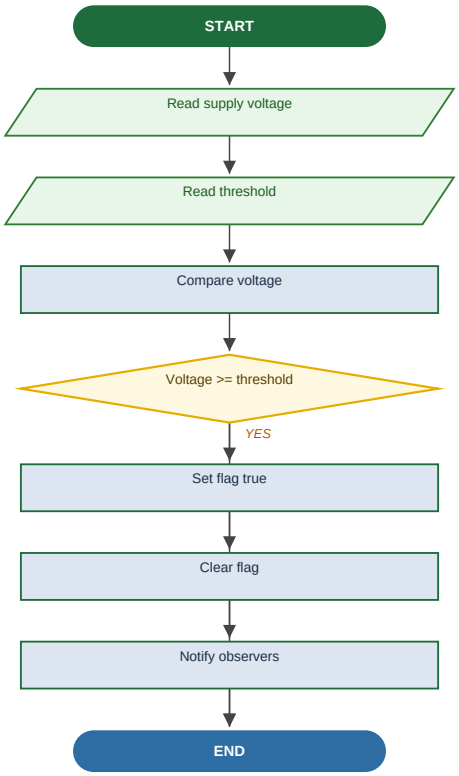
DD-COMP-049	**OvervoltageThresholdStore** *(Service)*
Description	Stores and retrieves Overvoltage_Threshold from non-volatile memory and loads it during startup for use by monitoring logic.
Inputs	threshold_value (on write), startup_event
Outputs	threshold_value (on read)
Dependencies	NVMDriver
Design Pattern	Repository
Traces to	SW-REQ-038

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



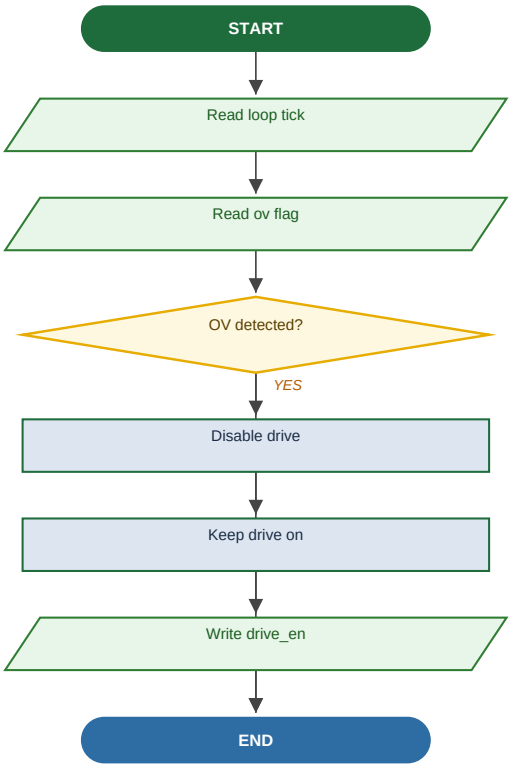
DD-COMP-050	**OvervoltageDetector** *(Class)*
Description	Compares computed supply voltage against Overvoltage_Threshold and raises an overvoltage condition when voltage >= threshold.
Inputs	computed_supply_voltage, overvoltage_threshold
Outputs	overvoltage_flag
Dependencies	SupplyVoltageSampler, OvervoltageThresholdStore
Design Pattern	Observer
Traces to	SW-REQ-039

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



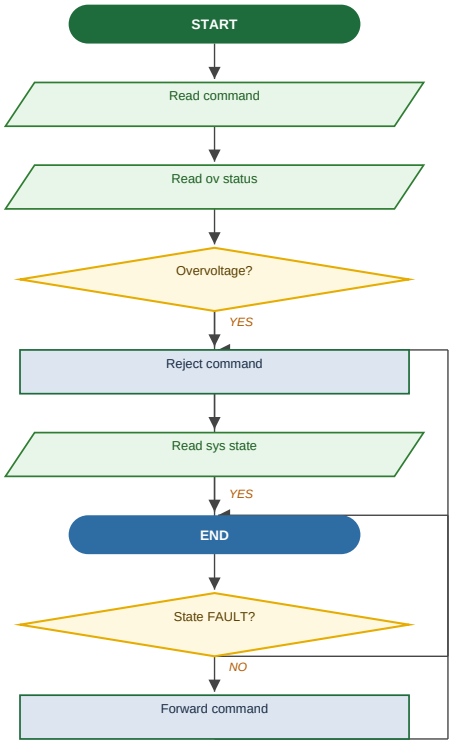
DD-COMP-051	**OvervoltageDriveDisabler** *(Class)*
Description	Disables actuator drive within one control loop cycle of overvoltage detection by acting on the overvoltage flag in the loop.
Inputs	overvoltage_flag, control_loop_tick
Outputs	drive_enable=false
Dependencies	OvervoltageDetector, ActuatorDriveService
Design Pattern	Observer
Traces to	SW-REQ-040

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



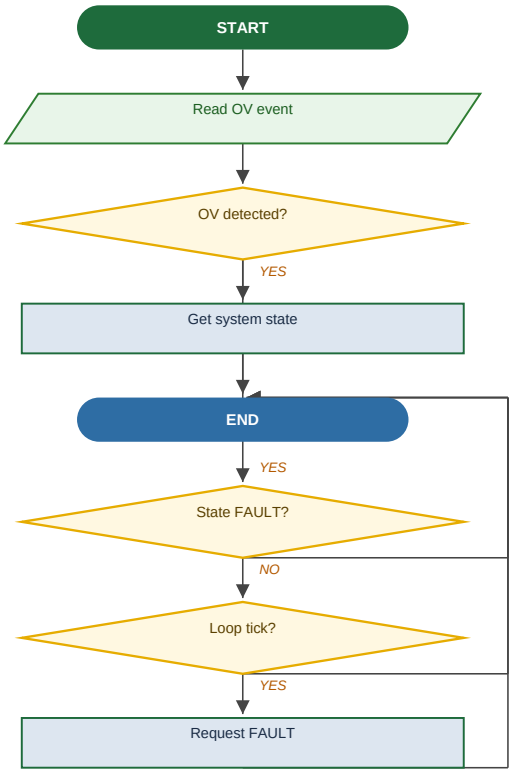
DD-COMP-052	**ActuatorCommandGate** *(Class)*
Description	Filters actuator movement commands and rejects them when the system is in overvoltage or FAULT state. The gate checks the OvervoltageMonitor status before allowing commands to reach the ActuatorController.
Inputs	Actuator command object; overvoltage status flag; system state
Outputs	Accepted command forwarded to ActuatorController or rejected status with reason
Dependencies	OvervoltageMonitor; StateMachine; ActuatorController
Design Pattern	Guard
Traces to	SW-REQ-041

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



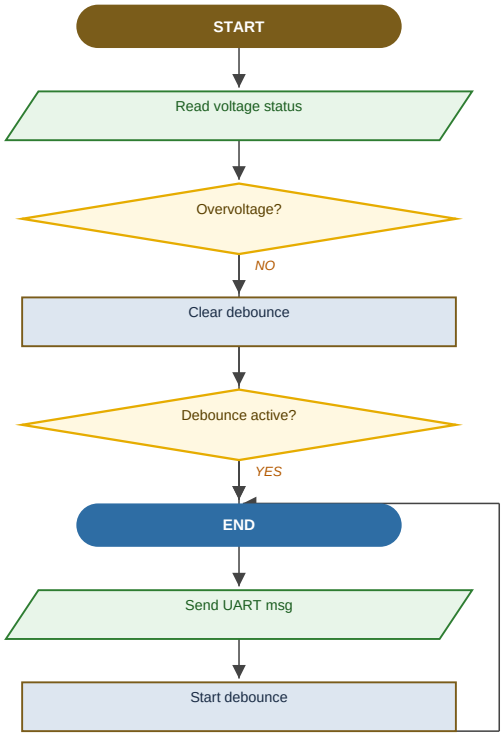
DD-COMP-053	**FaultTransitionHandler** *(Class)*
Description	Listens for overvoltage detection events and triggers an immediate transition to FAULT state within the next control loop cycle.
Inputs	Overvoltage event; current system state; control loop tick
Outputs	State transition request to FAULT
Dependencies	OvervoltageMonitor; StateMachine; ControlLoopScheduler
Design Pattern	Observer
Traces to	SW-REQ-042

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



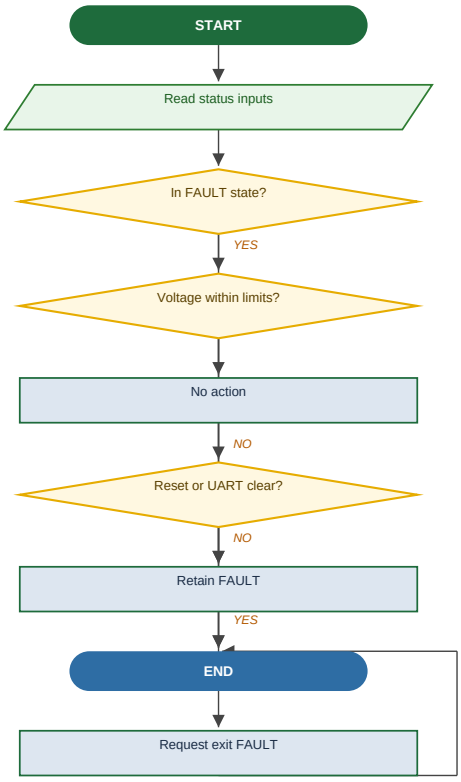
DD-COMP-054	**OvervoltageUartNotifier** *(Service)*
Description	Sends UART diagnostic message "OVERVOLTAGE_FAULT_ACTIVE" within 100 ms of detecting voltage >= Overvoltage_Threshold. Uses a debounce to avoid repeated transmissions until condition clears.
Inputs	Overvoltage status change; configured Overvoltage_Threshold
Outputs	UART message string
Dependencies	OvervoltageMonitor; UartDriver; ConfigurationService; TimerService
Design Pattern	Observer
Traces to	SW-REQ-043

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



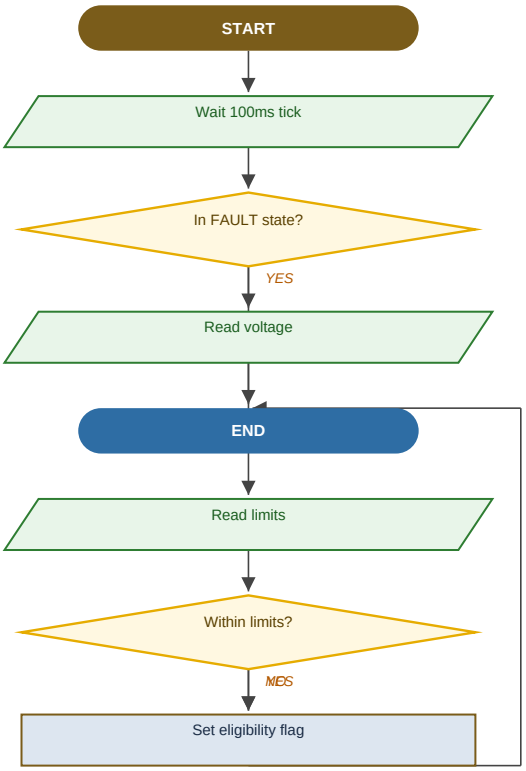
DD-COMP-055	**FaultStateRetentionLogic** *(Class)*
Description	Enforces FAULT state retention until voltage is within limits and a reset or authorized UART clear command is received. Validates both conditions before allowing exit.
Inputs	Current voltage status; reset signal; UART clear fault command; current system state
Outputs	State transition request to exit FAULT or retain FAULT
Dependencies	OvervoltageMonitor; StateMachine; UartCommandHandler; ResetManager
Design Pattern	State
Traces to	SW-REQ-044

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



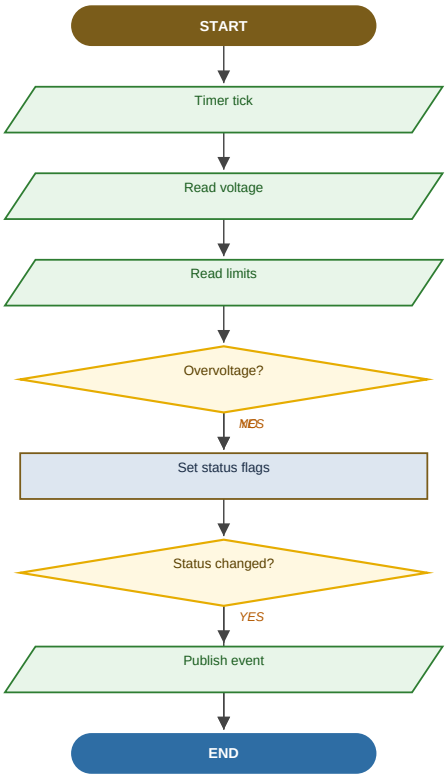
DD-COMP-056	**FaultExitVoltageChecker** *(Service)*
Description	Periodically evaluates supply voltage against acceptable limits at least every 100 ms while in FAULT state and updates eligibility to exit FAULT.
Inputs	Periodic timer tick; voltage sample; acceptable voltage limits
Outputs	Voltage-within-limits flag
Dependencies	VoltageSampler; ConfigurationService; TimerService
Design Pattern	Scheduler
Traces to	SW-REQ-045

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



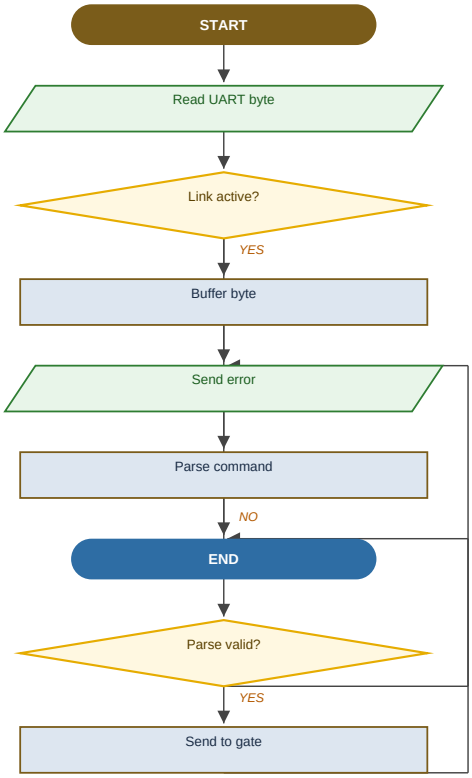
DD-COMP-057	**OvervoltageMonitor** *(Service)*
Description	Samples supply voltage and evaluates overvoltage status at least every 100 ms across all active states. Publishes status changes to subscribers.
Inputs	Voltage samples; Overvoltage_Threshold; acceptable limits; periodic timer tick
Outputs	Overvoltage status flag; voltage-within-limits flag; event notifications
Dependencies	VoltageSampler; ConfigurationService; TimerService; EventBus
Design Pattern	Observer
Traces to	SW-REQ-046

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



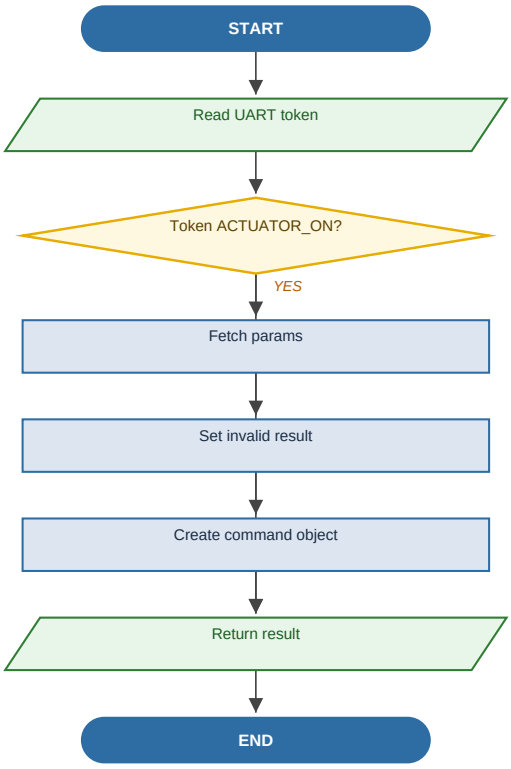
DD-COMP-058	**UartActuatorCommandInterface** *(Service)*
Description	Receives and parses actuator control commands over UART when link is active and forwards valid commands to the command gate.
Inputs	UART byte stream; link active status
Outputs	Parsed actuator command objects; error responses
Dependencies	UartDriver; CommandParser; ActuatorCommandGate; LinkMonitor
Design Pattern	Facade
Traces to	SW-REQ-047

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



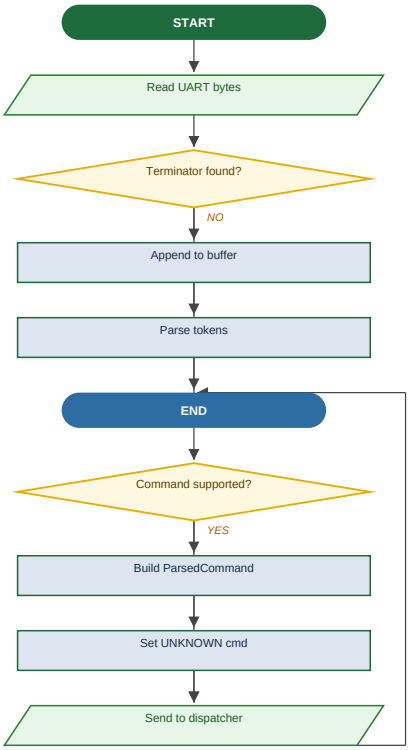
DD-COMP-059	**ActuatorOnCommandParser** *(Algorithm)*
Description	Recognizes the UART command token "ACTUATOR_ON" and maps it to an actuator enable command object with required parameters.
Inputs	Parsed UART command string
Outputs	ActuatorCommand object of type ACTUATOR_ON or invalid command result
Dependencies	CommandParser; CommandRegistry
Design Pattern	Factory
Traces to	SW-REQ-048

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



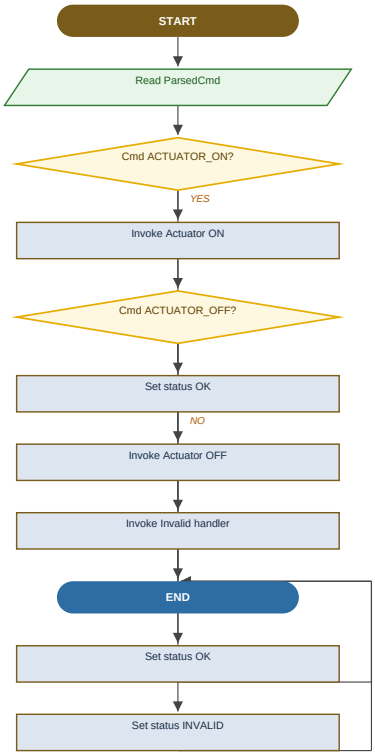
DD-COMP-060	**UART Command Parser** *(Class)*
Description	Parses raw UART input into command tokens and raw string payloads. Recognizes supported command strings including ACTUATOR_OFF and ACTUATOR_ON and forwards parsed results to the command dispatcher.
Inputs	raw_uart_bytes, message_terminator
Outputs	ParsedCommand {command_name, raw_string}
Dependencies	UartDriver, CommandDispatcher
Design Pattern	N/A
Traces to	SW-REQ-049

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



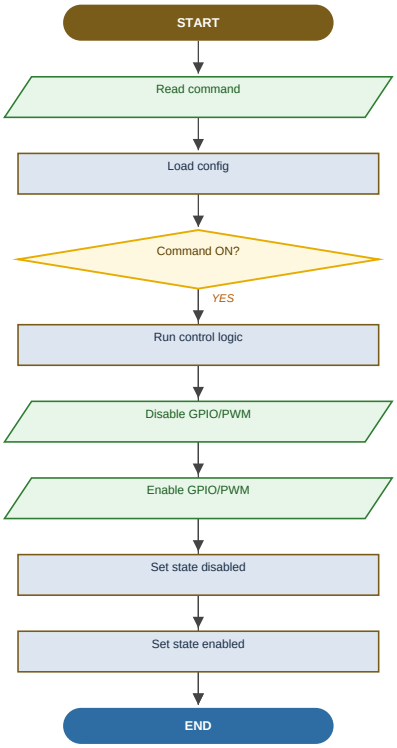
DD-COMP-061	**Command Dispatcher** *(Service)*
Description	Routes parsed commands to appropriate handlers. If command is ACTUATOR_ON, triggers actuator control logic within timing constraints; if ACTUATOR_OFF, triggers actuator disable; otherwise routes to invalid command handler without changing actuator state.
Inputs	ParsedCommand
Outputs	DispatchResult {status, handler_invoked}
Dependencies	ActuatorController, InvalidCommandHandler
Design Pattern	Command
Traces to	SW-REQ-050

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



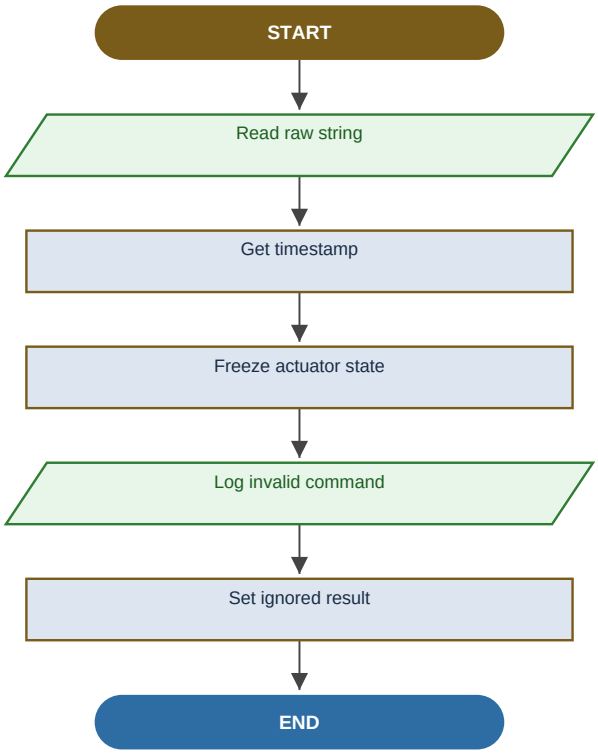
DD-COMP-062	**Actuator Controller** *(Service)*
Description	Implements actuator enable and disable operations. Executes configured control logic for ACTUATOR_ON and disables GPIO/PWM outputs for ACTUATOR_OFF within specified time limits.
Inputs	command_name, control_config
Outputs	ActuatorState {enabled, timestamp}
Dependencies	GpioDriver, PwmDriver, ControlLogic
Design Pattern	Facade
Traces to	SW-REQ-051

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



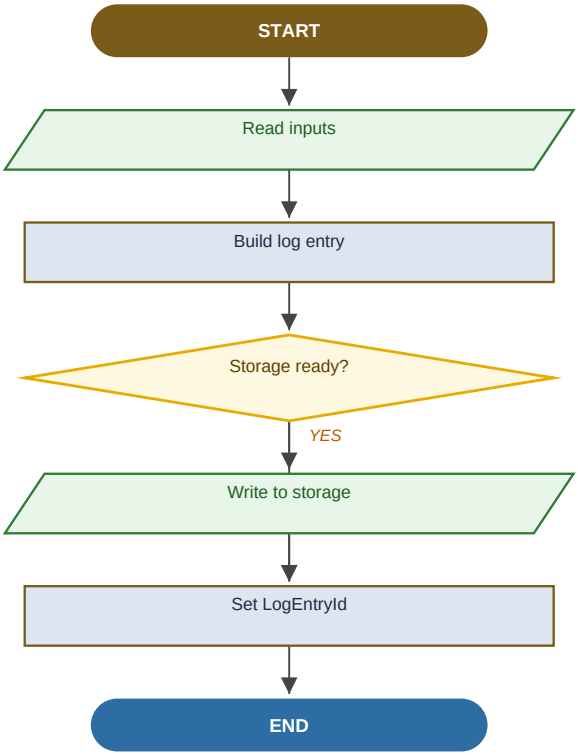
DD-COMP-063	**Invalid Command Handler** *(Service)*
Description	Handles invalid UART commands by ensuring no actuator state change occurs and by invoking the logging service with timestamp and raw command string.
Inputs	ParsedCommand.raw_string
Outputs	HandledResult {ignored:true}
Dependencies	ActuatorController, LoggingService, TimeService
Design Pattern	N/A
Traces to	SW-REQ-052

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



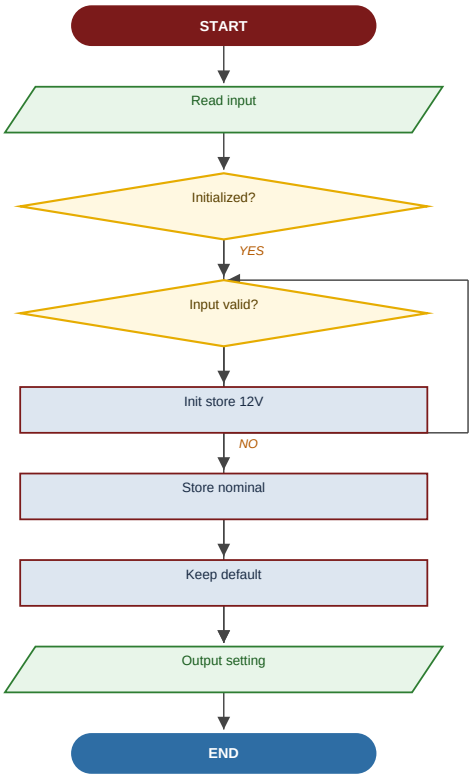
DD-COMP-064	**Invalid Command Logger** *(Service)*
Description	Logs invalid UART commands with timestamp and raw command string to persistent or ring-buffer storage.
Inputs	timestamp, raw_command_string
Outputs	LogEntryId
Dependencies	TimeService, LogStorage
Design Pattern	Repository
Traces to	SW-REQ-053

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



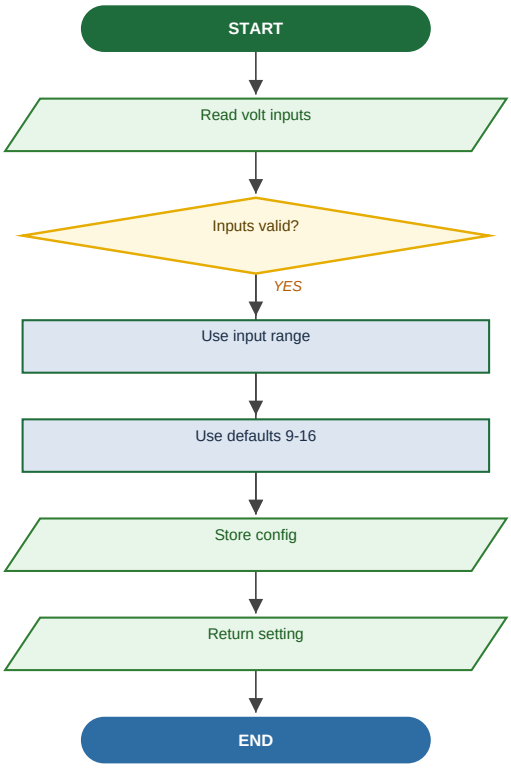
DD-COMP-065	**Voltage Configuration Store** *(Database)*
Description	Stores nominal and operating voltage configuration values used by diagnostics and range checks. Initializes nominal voltage to 12 V.
Inputs	nominal_voltage_value
Outputs	NominalVoltageSetting
Dependencies	ConfigInitializer
Design Pattern	Singleton
Traces to	SW-REQ-054

■ Flow Diagram *(data flow) — See Section 7 for full-page view*



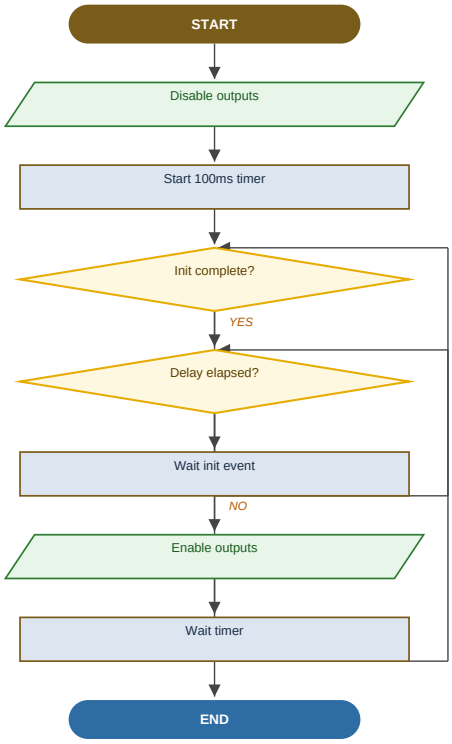
DD-COMP-066	**Voltage Range Configuration** *(Class)*
Description	Configures acceptable supply voltage limits for operating range checks. Sets min=9 V and max=16 V for validation use.
Inputs	min_voltage, max_voltage
Outputs	VoltageRangeSetting
Dependencies	VoltageConfigurationStore
Design Pattern	N/A
Traces to	SW-REQ-055

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



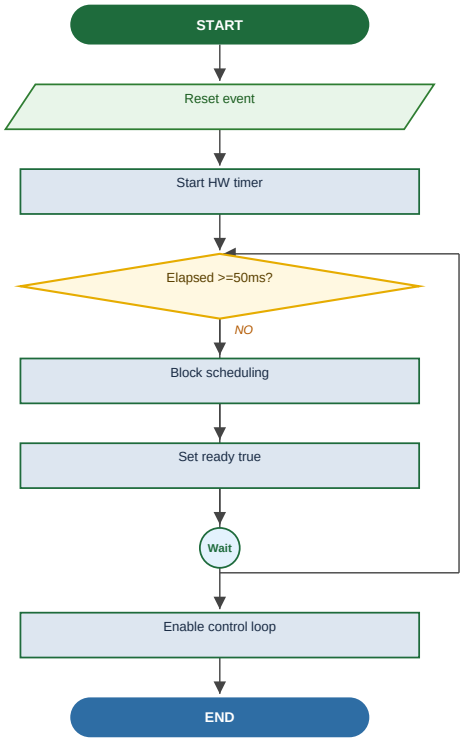
DD-COMP-067	**Startup Safety Manager** *(Service)*
Description	Ensures all actuator control outputs (PA5 GPIO and PWM outputs) are disabled or high-impedance until initialization completes and a 100 ms post-reset stabilization delay has elapsed. Enables outputs only after delay and initialization success.
Inputs	reset_event, init_complete_event
Outputs	SafetyState {outputs_enabled}
Dependencies	GpioDriver, PwmDriver, TimeService, SystemInitializer
Design Pattern	State
Traces to	SW-REQ-056

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



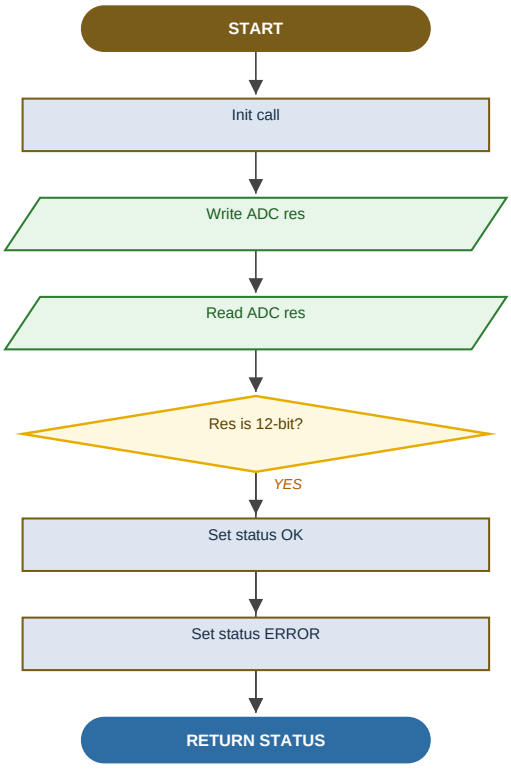
DD-COMP-068	**StartupDelayManager** *(Class)*
Description	Implements a post-reset delay gate. On system reset, it starts a hardware timer and blocks control loop scheduling and actuator enable until the elapsed time is at least 50 ms, then signals readiness.
Inputs	Reset event, system tick/timer count
Outputs	ready_to_enable boolean, enable_control_loop event
Dependencies	SystemTimerService, ControlLoopScheduler, ActuatorControlService
Design Pattern	State
Traces to	SW-REQ-057

■ Flow Diagram *(state flow) — See Section 7 for full-page view*



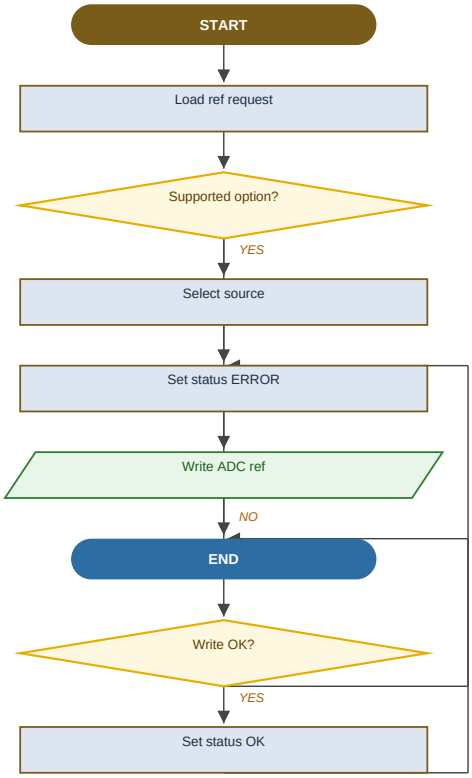
DD-COMP-069	**ADCResolutionConfigurator** *(Service)*
Description	Configures the ADC peripheral registers to 12-bit resolution during initialization, verifying the setting and reporting status.
Inputs	Init call, desired resolution constant (12-bit)
Outputs	Status code (OK/ERROR)
Dependencies	MCU_HAL_ADC
Design Pattern	N/A
Traces to	SW-REQ-058

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



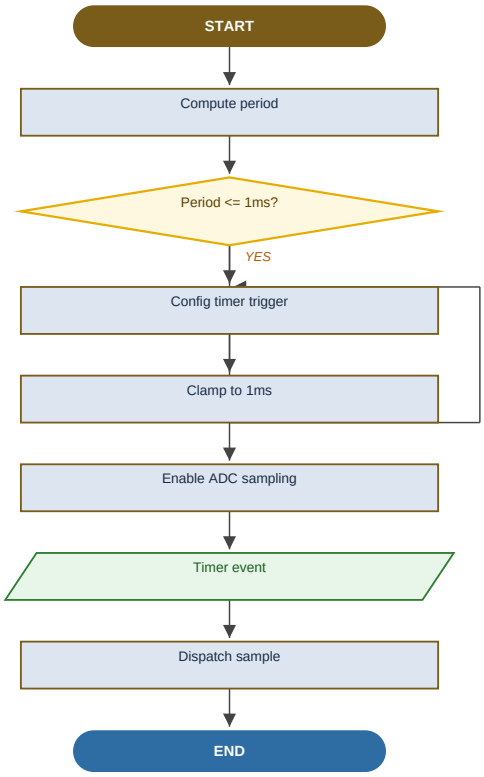
DD-COMP-070	**ADCReferenceConfigurator** *(Service)*
Description	Sets ADC reference voltage to 3.3 V by configuring the reference source selection and validates it against supported reference options.
Inputs	Init call, reference selection (3.3V internal/external)
Outputs	Status code (OK/ERROR)
Dependencies	MCU_HAL_ADC, BoardConfig
Design Pattern	N/A
Traces to	SW-REQ-059

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



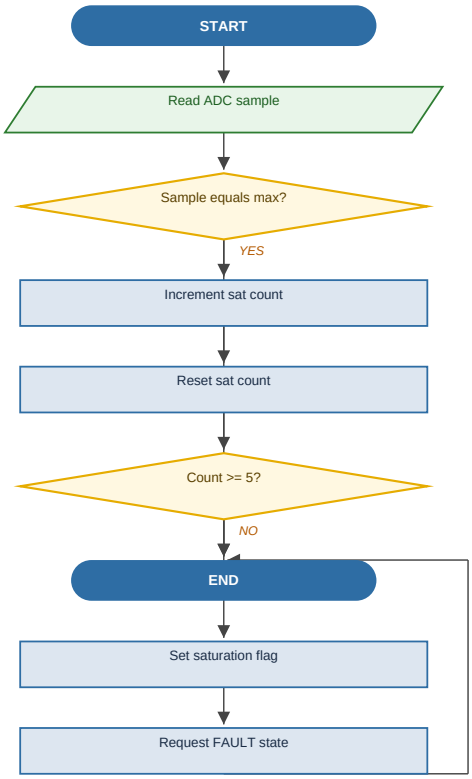
DD-COMP-071	**ADCSamplingScheduler** *(Service)*
Description	Schedules ADC sampling at a minimum of 1 kHz using a hardware timer trigger or periodic task. Ensures sample period <= 1 ms and dispatches samples to processing pipeline.
Inputs	System clock, desired sample rate
Outputs	ADC sample ready events, sample timestamps
Dependencies	SystemTimerService, MCU_HAL_ADC, ADCDriver
Design Pattern	Observer
Traces to	SW-REQ-060

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



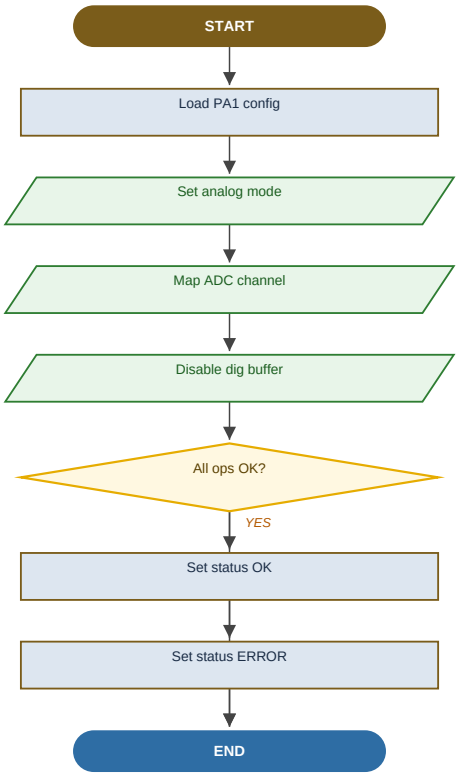
DD-COMP-072	**ADCSaturationDetector** *(Algorithm)*
Description	Tracks consecutive ADC samples equal to maximum count. If 5 consecutive maximum readings are observed, flags saturation and requests transition to FAULT state.
Inputs	ADC sample value, ADC max count
Outputs	saturation_detected boolean, fault_request event
Dependencies	StateManager, ADCDriver
Design Pattern	State
Traces to	SW-REQ-061

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



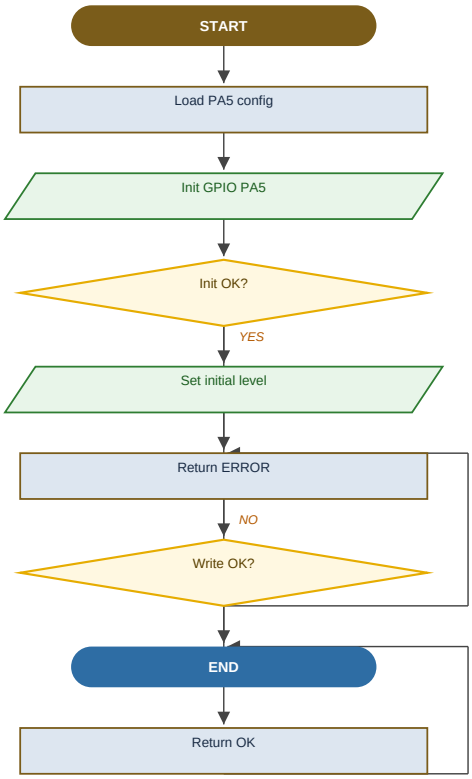
DD-COMP-073	**ADCPinInitializer** *(Service)*
Description	Initializes GPIO pin PA1 in analog mode, configures it for ADC input channel mapping, and disables digital input buffers to reduce noise.
Inputs	Init call, pin identifier PA1
Outputs	Status code (OK/ERROR)
Dependencies	MCU_HAL_GPIO, MCU_HAL_ADC, BoardConfig
Design Pattern	N/A
Traces to	SW-REQ-062

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



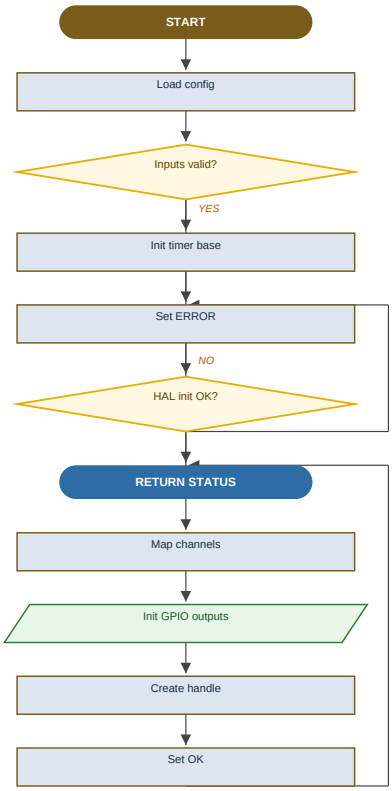
DD-COMP-074	**ActuatorGPIOInitializer** *(Service)*
Description	Configures pin PA5 as a push-pull GPIO output for actuator control and sets an initial safe output level.
Inputs	Init call, pin identifier PA5, initial level
Outputs	Status code (OK/ERROR)
Dependencies	MCU_HAL_GPIO, BoardConfig
Design Pattern	N/A
Traces to	SW-REQ-063

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



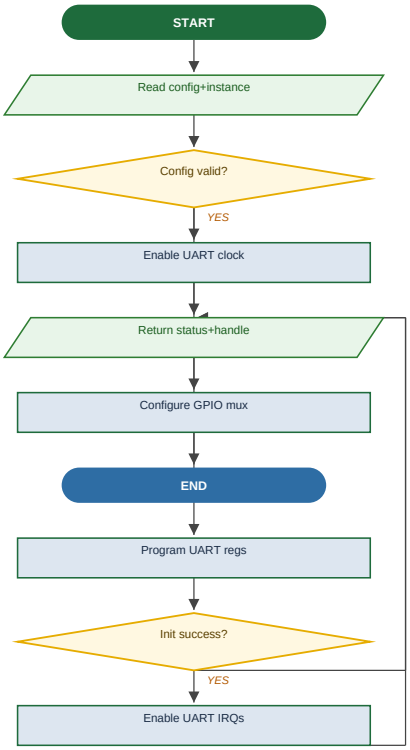
DD-COMP-075	**PWMTimerInitializer** *(Service)*
Description	Initializes an advanced PWM timer with configured frequency, duty cycle resolution, and output channel mapping to generate actuator control signals.
Inputs	Init call, PWM frequency, duty cycle resolution, channel mapping
Outputs	Status code (OK/ERROR), PWM timer handle
Dependencies	MCU_HAL_TIM, BoardConfig, ActuatorGPIOInitializer
Design Pattern	Factory
Traces to	SW-REQ-064

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



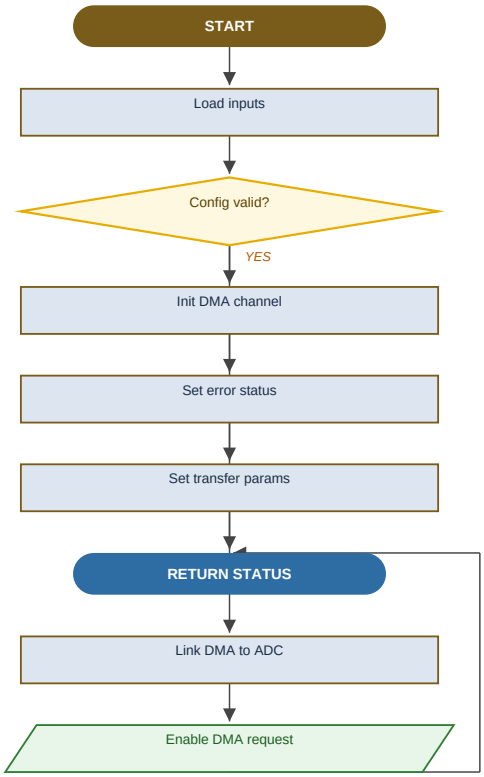
DD-COMP-076	**UART Peripheral Initializer** *(Class)*
Description	Configures UART peripheral registers (baud rate, data bits, parity, stop bits, GPIO muxing) and enables the UART clock and interrupts as required for external communication.
Inputs	UART configuration structure (baud_rate, data_bits, parity, stop_bits, flow_control), UART instance identifier
Outputs	Initialization status (success/failure), configured UART handle
Dependencies	ClockManager, GPIOConfigurator, InterruptController
Design Pattern	Factory
Traces to	SW-REQ-065

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



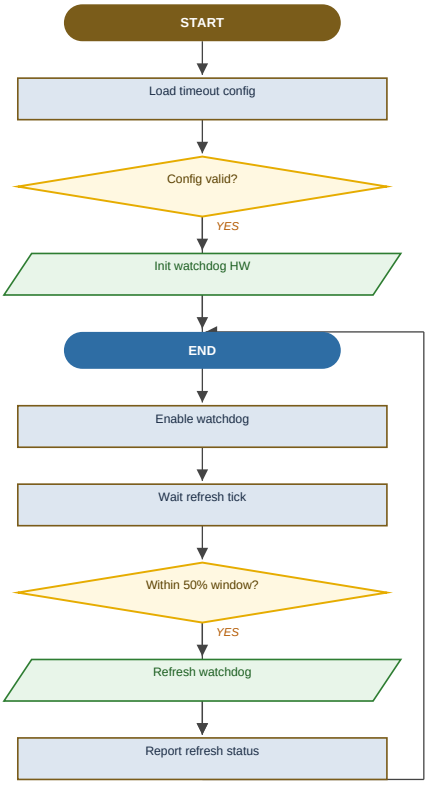
DD-COMP-077	**ADC DMA Configuration Service** *(Service)*
Description	Sets up DMA controller channels for ADC data transfers, configures transfer size, circular/normal mode, memory buffer address, and enables DMA requests from ADC to reduce CPU load.
Inputs	ADC handle, DMA channel config (buffer_address, length, mode, priority), sampling trigger source
Outputs	DMA configuration status, DMA handle linked to ADC
Dependencies	HAL.ADCPeripheral, HAL.DMAController, BufferManager
Design Pattern	Facade
Traces to	SW-REQ-066

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



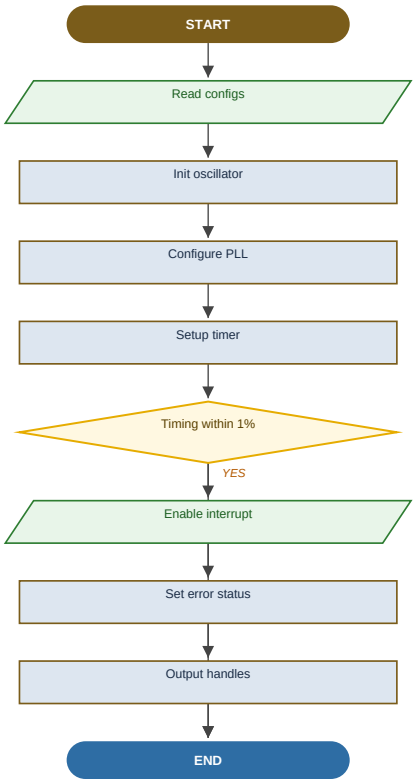
DD-COMP-078	**Independent Watchdog Service** *(Service)*
Description	Enables independent watchdog with configured timeout and provides periodic refresh within 50% of timeout during normal operation; exposes refresh API used by application scheduler.
Inputs	Watchdog timeout configuration, periodic refresh tick
Outputs	Watchdog enable status, refresh success/failure
Dependencies	HAL.WatchdogPeripheral, SystemTimer
Design Pattern	Singleton
Traces to	SW-REQ-067

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



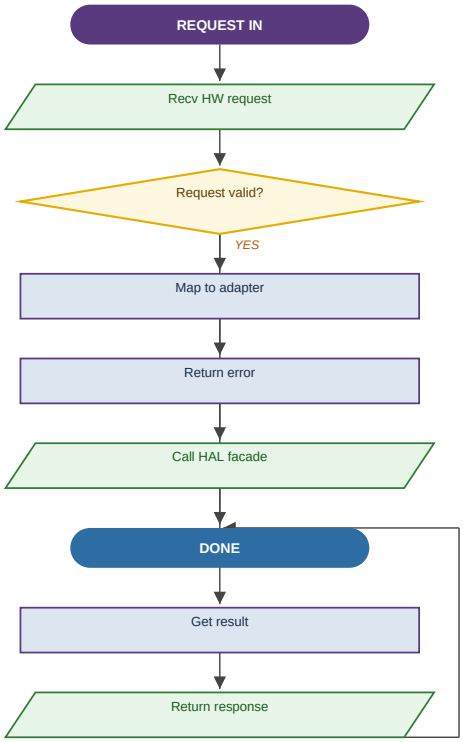
DD-COMP-079	**Clock and Timer Configuration** *(Service)*
Description	Configures system clock source, PLL settings, and timer prescalers to produce a 1 ms control loop period within ±1% tolerance; provides timer interrupt for control loop scheduling.
Inputs	Clock source config, PLL multipliers/dividers, timer target period (1 ms)
Outputs	Configured system clock and timer handles, timing accuracy status
Dependencies	OscillatorDriver, TimerPeripheral, InterruptController
Design Pattern	Factory
Traces to	SW-REQ-068

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



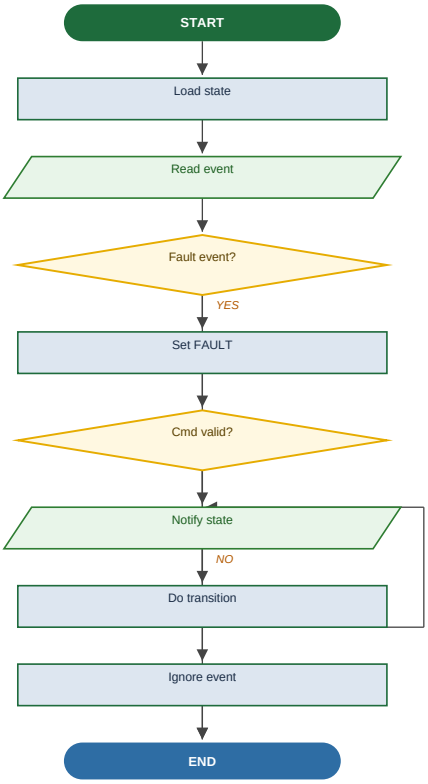
DD-COMP-080	**Layered Architecture Enforcement Interface** *(Interface)*
Description	Defines abstract hardware access interfaces used by the Application layer; all hardware interactions are routed through Middleware/Driver/HAL implementations to prevent direct hardware access.
Inputs	Hardware access requests (e.g., read_adc, write_uart, set_gpio)
Outputs	Hardware operation results via interface methods
Dependencies	Middleware.HardwareAbstractionFacade
Design Pattern	Adapter
Traces to	SW-REQ-069

■ Flow Diagram *(request response) — See Section 7 for full-page view*



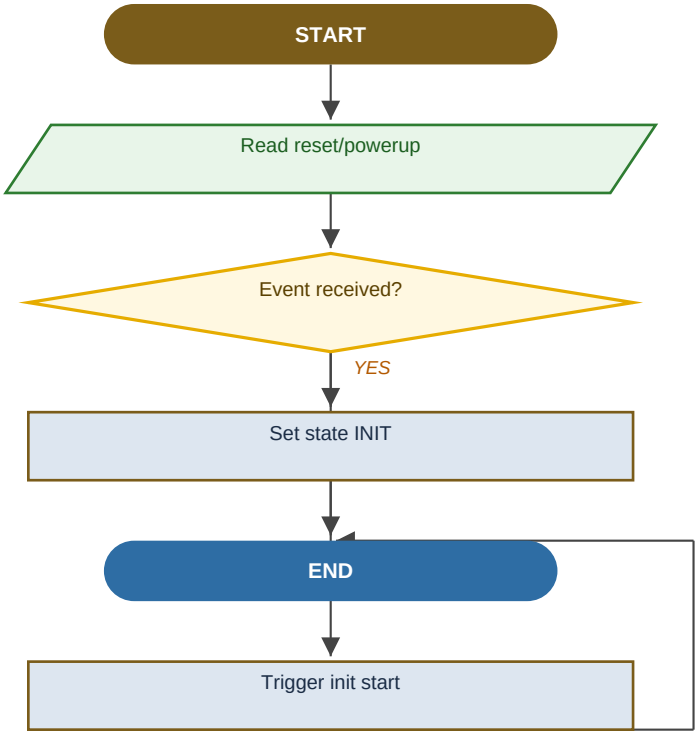
DD-COMP-081	**System State Machine** *(Class)*
Description	Implements finite state machine with states INIT, IDLE, ACTIVE, SHIPPING_LOCK, and FAULT; processes events/commands and triggers state transitions and entry/exit actions.
Inputs	Events (reset, init_complete, activate_cmd, shipping_lock_cmd, unlock_cmd, fault_detected)
Outputs	Current state, state transition notifications
Dependencies	CommandParser, FaultManager, InitializationManager
Design Pattern	State
Traces to	SW-REQ-070

■ Flow Diagram *(state flow) — See Section 7 for full-page view*



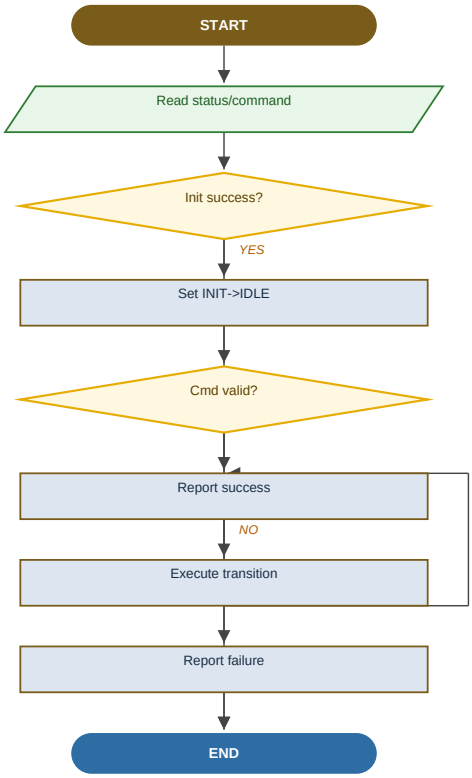
DD-COMP-082	**Reset Initialization Handler** *(Service)*
Description	On reset or power-up, forces state machine to INIT state and initiates initialization sequence.
Inputs	Reset event signal, power-up event
Outputs	State machine set to INIT, initialization start trigger
Dependencies	SystemStateMachine, InitializationManager
Design Pattern	Observer
Traces to	SW-REQ-071

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



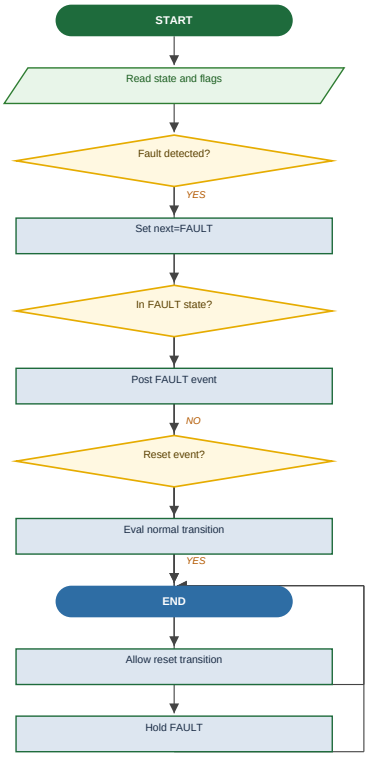
DD-COMP-083	**Nominal Transition Controller** *(Service)*
Description	Validates and executes nominal transitions: INIT->IDLE after initialization success, IDLE->ACTIVE on activate command, ACTIVE->SHIPPING_LOCK on shipping lock command, SHIPPING_LOCK->IDLE on unlock command.
Inputs	Initialization status, command inputs (activate, shipping_lock, unlock)
Outputs	Requested state transition, transition success/failure
Dependencies	SystemStateMachine, CommandParser, InitializationManager
Design Pattern	Command
Traces to	SW-REQ-072

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



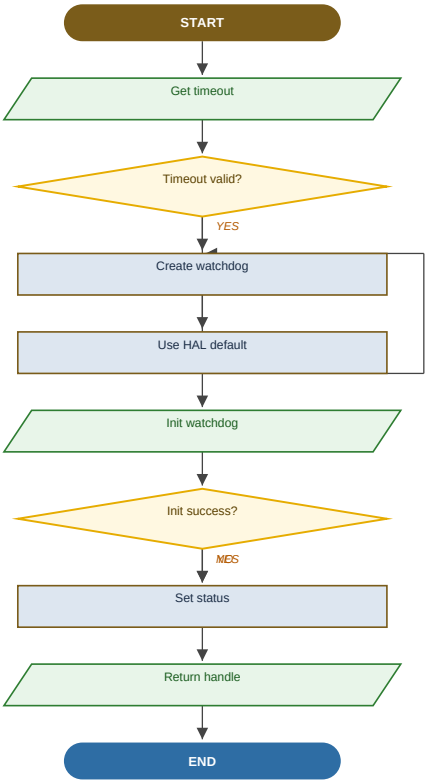
DD-COMP-084	**FaultStateManager** *(Class)*
Description	Implements system state machine logic to transition to FAULT from any current state when a fault condition is flagged, and to remain in FAULT until an explicit reset event occurs. The manager checks fault flags each cycle and gates all non-reset transitions while in FAULT.
Inputs	Current system state, fault_detected flag, reset_event flag
Outputs	Next system state, state transition events
Dependencies	FaultMonitor, ResetController, StateEventBus
Design Pattern	State
Traces to	SW-REQ-073

■ Flow Diagram *(state flow) — See Section 7 for full-page view*



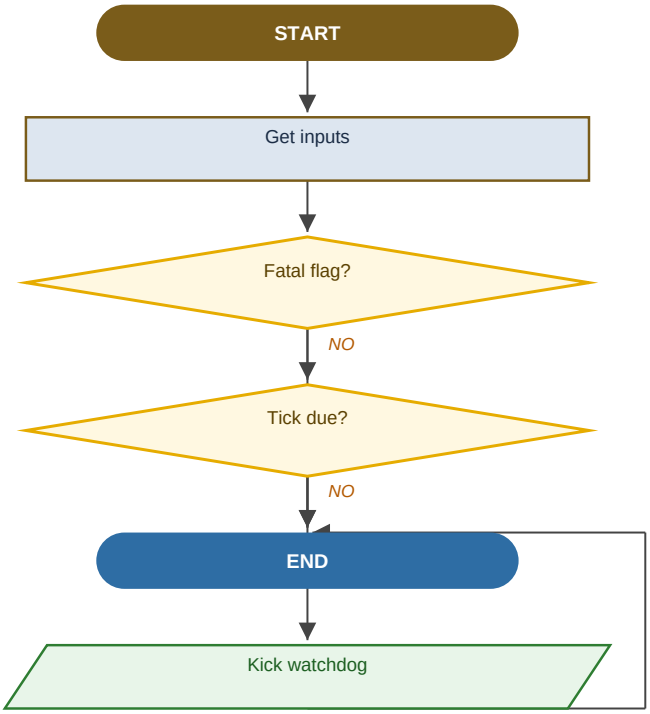
DD-COMP-085	**HardwareWatchdogInitializer** *(Service)*
Description	Initializes the independent hardware watchdog during system startup and configures it with the platform-supported timeout value retrieved from platform configuration or HAL.
Inputs	Platform timeout value, startup lifecycle event
Outputs	Configured watchdog handle, initialization status
Dependencies	HALWatchdogDriver, PlatformConfigService
Design Pattern	Factory
Traces to	SW-REQ-074

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



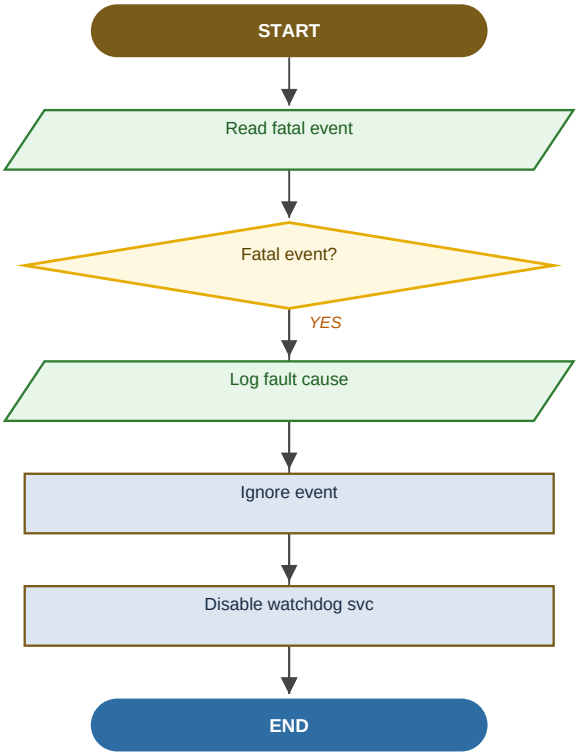
DD-COMP-086	**WatchdogServicer** *(Service)*
Description	Services (kicks) the independent watchdog on a periodic schedule during normal operation, ensuring at least one service within each watchdog timeout period. The service is disabled when a fatal malfunction is declared.
Inputs	System run state, scheduler ticks, fatal_malfunction flag
Outputs	Watchdog service (kick) signal
Dependencies	HALWatchdogDriver, SchedulerService, HealthStatusService
Design Pattern	Observer
Traces to	SW-REQ-075

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



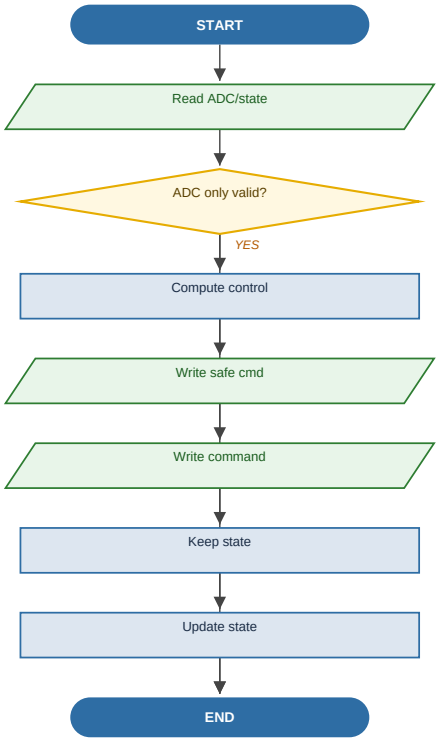
DD-COMP-087	**FatalMalfunctionResetHandler** *(Service)*
Description	On detection of fatal software malfunction, inhibits WatchdogServicer and allows watchdog timeout to occur, triggering a system reset. Logs the malfunction cause before inhibiting servicing.
Inputs	Fatal malfunction event, malfunction diagnostics
Outputs	Disable-watchdog-service command, fault log entry
Dependencies	WatchdogServicer, DiagnosticsLogger, FaultMonitor
Design Pattern	Command
Traces to	SW-REQ-076

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



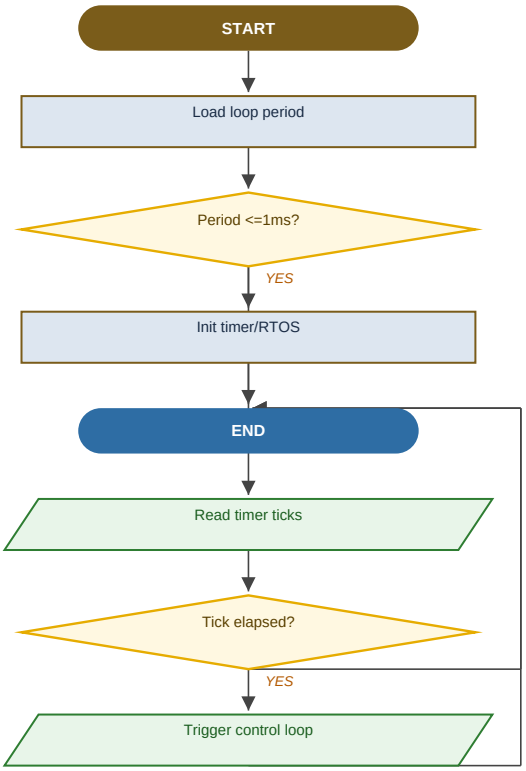
DD-COMP-088	**ADCOnlyControlAlgorithm** *(Algorithm)*
Description	Computes closed-loop control outputs using only raw ADC feedback values and internal control state. The algorithm explicitly excludes any non-ADC sensor inputs by design, enforcing input validation at the interface boundary.
Inputs	Raw ADC samples, internal controller state, configuration gains
Outputs	Actuator command values, updated controller state
Dependencies	ADCReader, ControlStateStore, ControlConfig
Design Pattern	Strategy
Traces to	SW-REQ-077

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



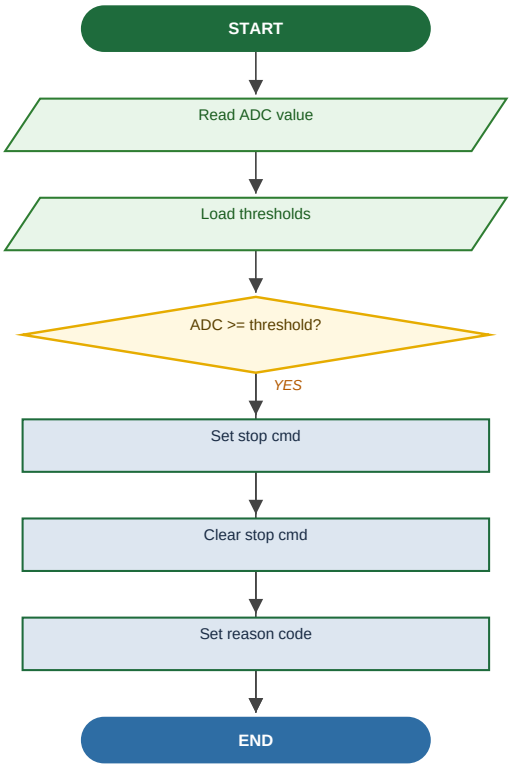
DD-COMP-089	**ControlLoopScheduler** *(Service)*
Description	Schedules and triggers execution of the control loop at a fixed period of 1.0 ms or less. Uses a high-resolution timer or RTOS task period to ensure ≥1 kHz execution rate.
Inputs	System clock/timer ticks, configured loop period
Outputs	Control loop execution triggers
Dependencies	RTOScheduler, HighResTimer
Design Pattern	Singleton
Traces to	SW-REQ-078

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



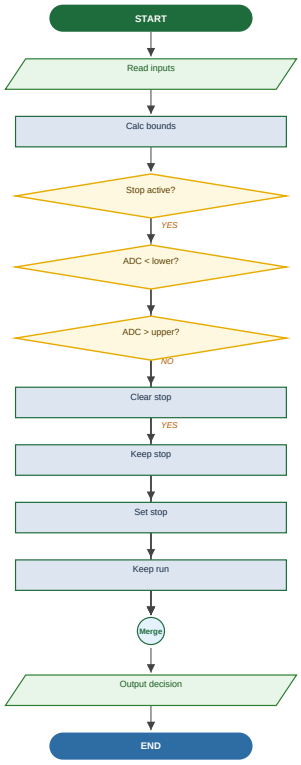
DD-COMP-090	**ThresholdStopEvaluator** *(Class)*
Description	Evaluates raw ADC values against configured stop thresholds each control cycle and commands an actuator stop within the same cycle when thresholds are met or exceeded.
Inputs	Raw ADC value, stop thresholds
Outputs	Stop command boolean, reason code
Dependencies	ActuatorController, ControlLoopScheduler, ThresholdConfig
Design Pattern	N/A
Traces to	SW-REQ-079

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



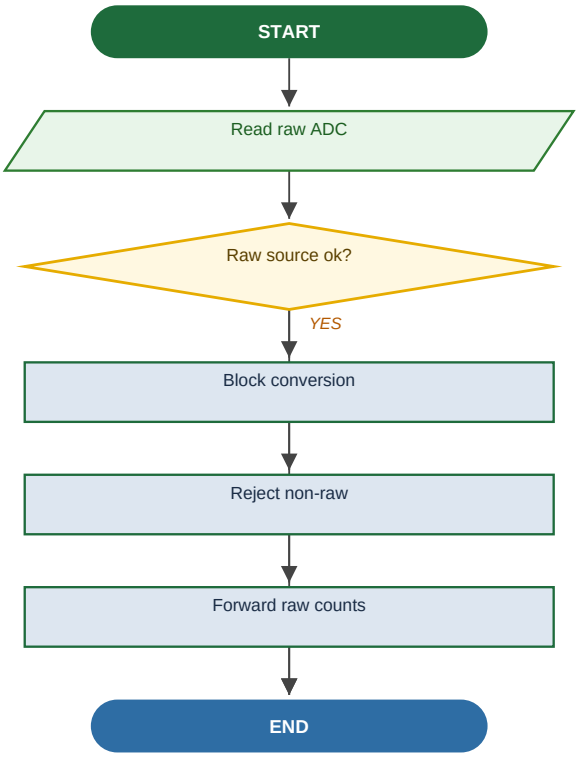
DD-COMP-091	**HysteresisBandManager** *(Class)*
Description	Applies a configurable hysteresis band around stop thresholds to prevent rapid toggling of actuator commands. Maintains last stop state and uses upper/lower bounds for re-enable decisions.
Inputs	Raw ADC value, hysteresis band configuration, last stop state
Outputs	Filtered stop command decision
Dependencies	ThresholdStopEvaluator, ThresholdConfig
Design Pattern	State
Traces to	SW-REQ-080

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



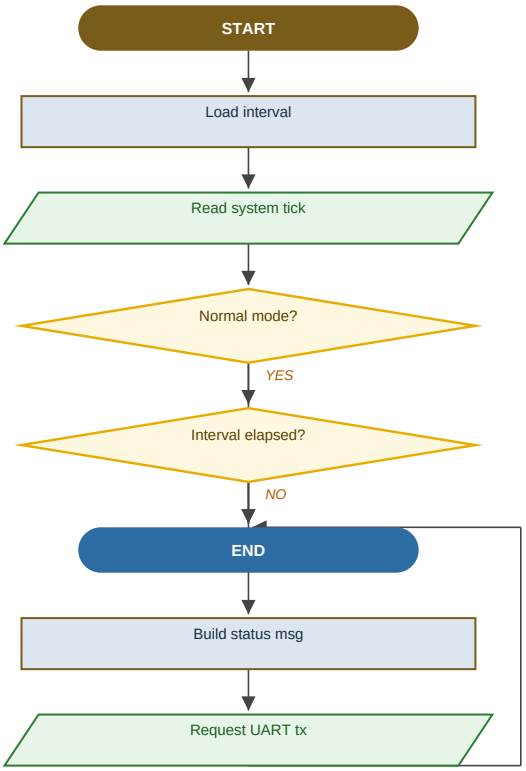
DD-COMP-092	**RawAdcDataPolicy** *(Class)*
Description	Enforces use of raw ADC counts in control and decision logic by exposing only raw ADC values to control algorithms and preventing any conversion to physical units within the control path.
Inputs	Raw ADC counts from AdcDriver
Outputs	Raw ADC counts forwarded to control/decision logic
Dependencies	AdcDriver, ControlStateMachine
Design Pattern	Facade
Traces to	SW-REQ-081

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



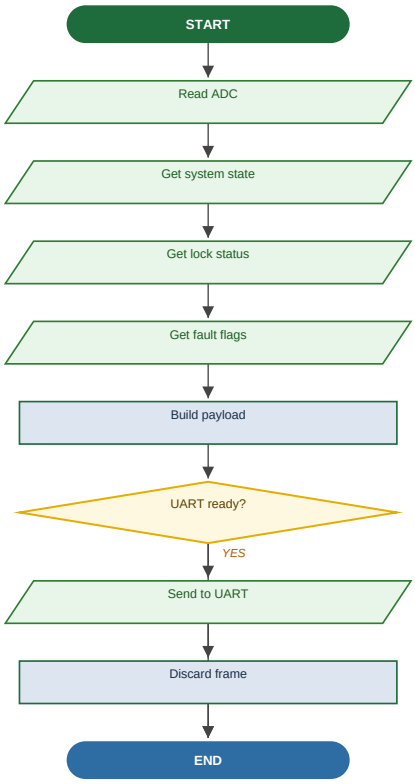
DD-COMP-093	**UartStatusScheduler** *(Service)*
Description	Schedules and triggers periodic UART status message transmission at a configurable interval during normal operation.
Inputs	Configured interval (ms), system tick timer, normal operation flag
Outputs	Transmit request to UartStatusMessageBuilder
Dependencies	SystemTimer, UartStatusMessageBuilder, UartDriver, ConfigurationStore, SystemStateService
Design Pattern	Observer
Traces to	SW-REQ-082

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



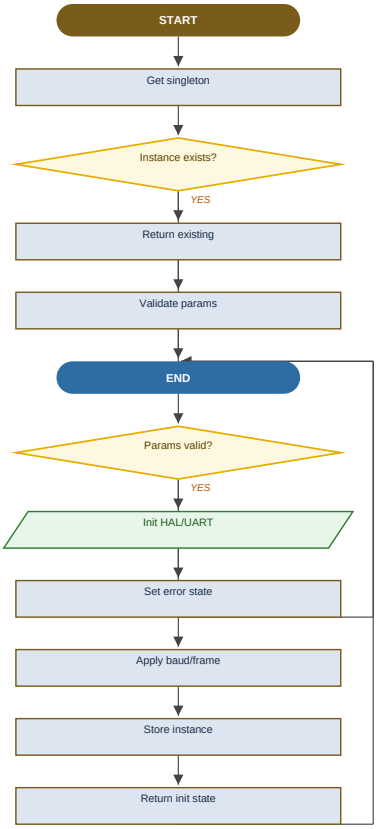
DD-COMP-094	**UartStatusMessageBuilder** *(Class)*
Description	Builds the UART status payload containing raw ADC value, system state, shipping lock status, and fault flags, and hands it to the UART driver for transmission.
Inputs	Raw ADC counts, system state enum, shipping lock status, fault flags bitmask
Outputs	Serialized UART frame buffer
Dependencies	AdcDriver, SystemStateService, ShippingLockService, FaultManager, UartDriver
Design Pattern	Builder
Traces to	SW-REQ-082

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



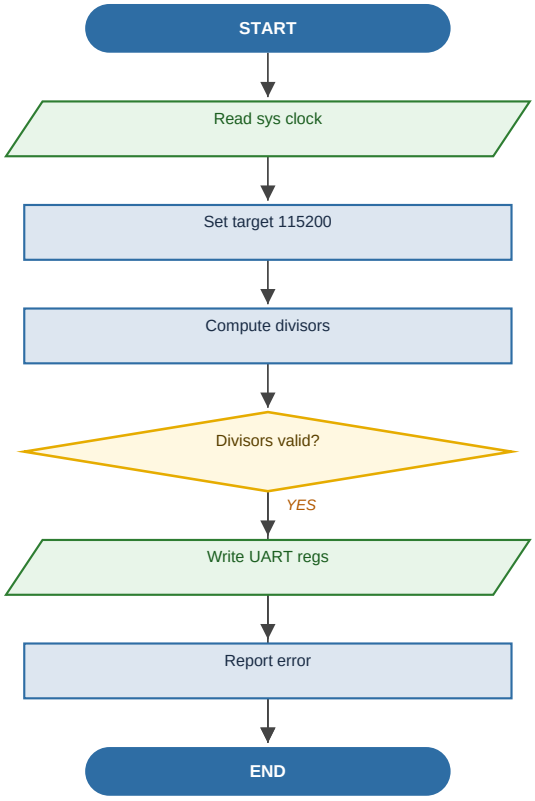
DD-COMP-095	**UartConfigService** *(Service)*
Description	Configures UART peripheral with required baud rate and frame format during initialization.
Inputs	UART peripheral handle, desired configuration parameters
Outputs	Initialized UART peripheral state
Dependencies	UartDriver, HardwareAbstractionLayer
Design Pattern	Singleton
Traces to	SW-REQ-083

Flow Diagram *(function flow) — See Section 7 for full-page view*



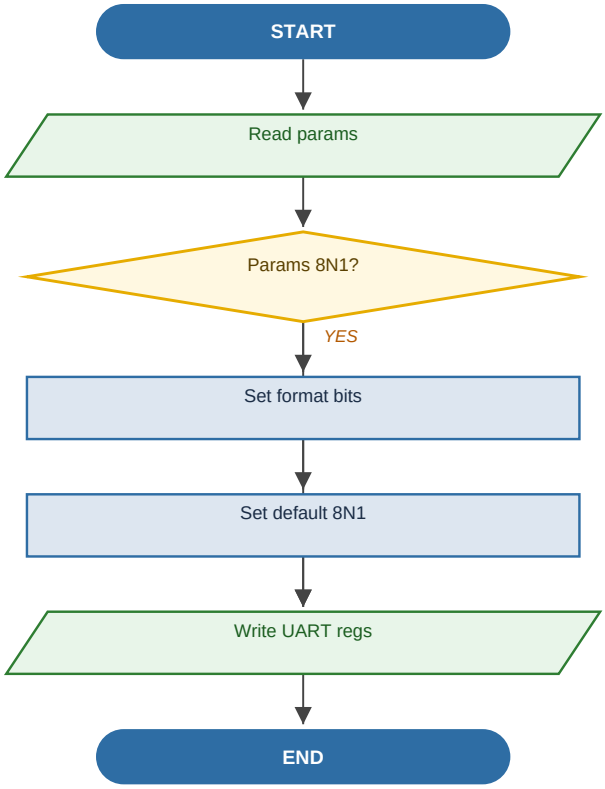
DD-COMP-096	**UartBaudRateConfig** *(Algorithm)*
Description	Sets UART baud rate to 115200 by computing and applying peripheral divisor values.
Inputs	System clock frequency, target baud rate (115200)
Outputs	UART register settings for baud rate
Dependencies	UartDriver, HardwareAbstractionLayer
Design Pattern	N/A
Traces to	SW-REQ-083

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



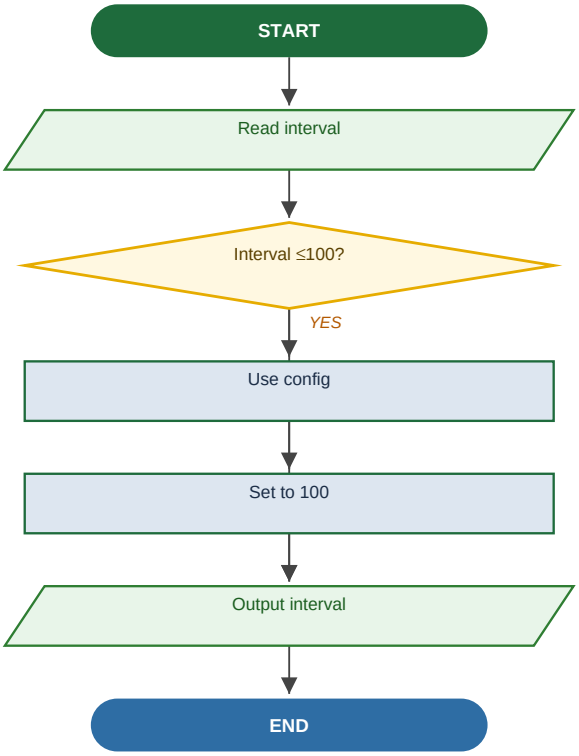
DD-COMP-097	**UartFrameFormatConfig** *(Algorithm)*
Description	Configures UART frame format to 8 data bits, no parity, and 1 stop bit (8N1).
Inputs	Frame format parameters (8N1)
Outputs	UART register settings for frame format
Dependencies	UartDriver, HardwareAbstractionLayer
Design Pattern	N/A
Traces to	SW-REQ-084

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



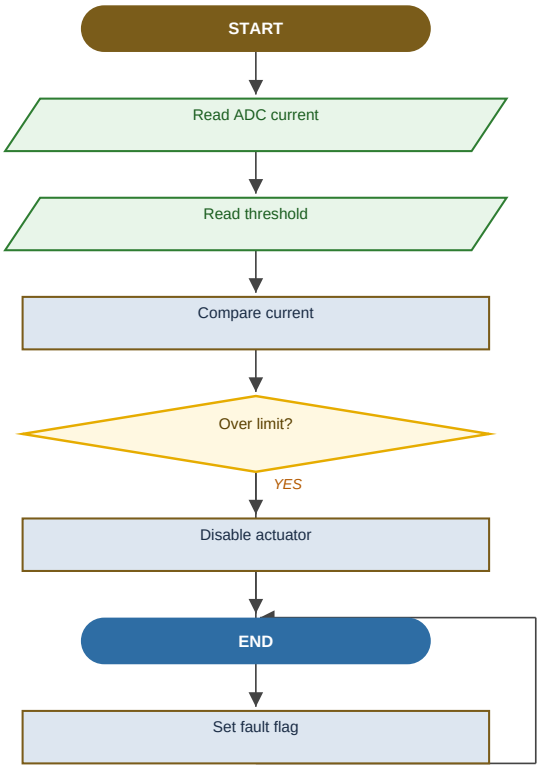
DD-COMP-098	**UartTransmitIntervalValidator** *(Class)*
Description	Validates configured transmit interval to be less than or equal to 100 ms and enforces fallback to 100 ms if out of range.
Inputs	Configured interval (ms)
Outputs	Validated interval (ms)
Dependencies	ConfigurationStore
Design Pattern	N/A
Traces to	SW-REQ-085

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



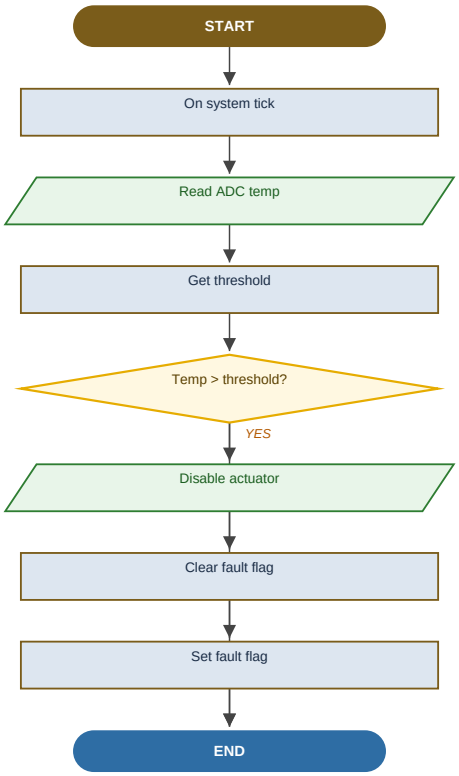
DD-COMP-099	**OvercurrentMonitor** *(Service)*
Description	Continuously monitors motor current and triggers actuator shutdown and fault when current exceeds configurable threshold.
Inputs	Motor current ADC counts, overcurrent threshold, system tick
Outputs	Actuator disable command, overcurrent fault flag
Dependencies	AdcDriver, ConfigurationStore, ActuatorControlService, FaultManager
Design Pattern	Observer
Traces to	SW-REQ-086

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



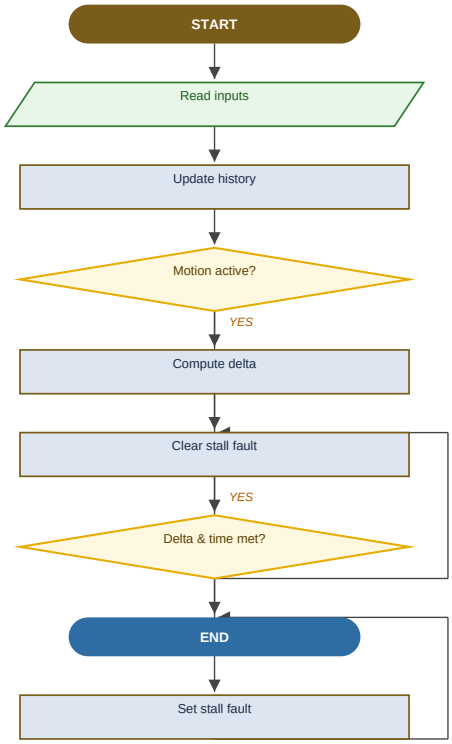
DD-COMP-100	**ThermalMonitor** *(Service)*
Description	Monitors MCU internal temperature sensor and triggers actuator shutdown and fault when temperature exceeds configurable threshold.
Inputs	Temperature sensor ADC counts, thermal threshold, system tick
Outputs	Actuator disable command, thermal fault flag
Dependencies	AdcDriver, ConfigurationStore, ActuatorControlService, FaultManager
Design Pattern	Observer
Traces to	SW-REQ-087

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



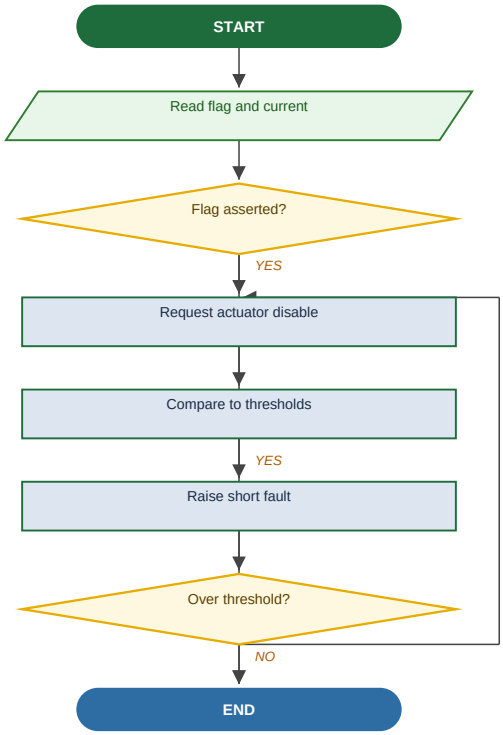
DD-COMP-101	**StallDetectionService** *(Service)*
Description	Detects stall by tracking commanded motion and ensuring ADC feedback changes by at least a configurable delta within a configurable duration; raises stall fault when not met.
Inputs	Commanded motion state, raw ADC counts history, delta threshold, duration threshold, system tick
Outputs	Stall fault flag
Dependencies	ControlStateMachine, AdcDriver, ConfigurationStore, FaultManager, SystemTimer
Design Pattern	State
Traces to	SW-REQ-088

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



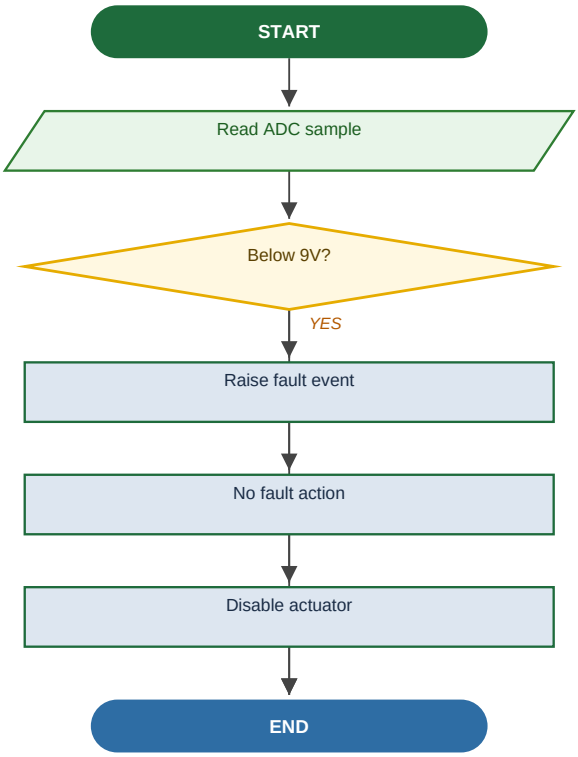
DD-COMP-102	**ShortCircuitDetector** *(Class)*
Description	Evaluates short-circuit indicators using either dedicated hardware flag inputs or current measurement thresholds. On detection, it requests actuator shutdown and raises a short-circuit fault in the fault manager.
Inputs	current_measurement, short_circuit_flag_gpio, detection_thresholds
Outputs	fault_event=SHORT_CIRCUIT, actuator_disable_request
Dependencies	AdcAcquisitionService, GpioInputService, FaultManager, ActuatorControlService
Design Pattern	Observer
Traces to	SW-REQ-089

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



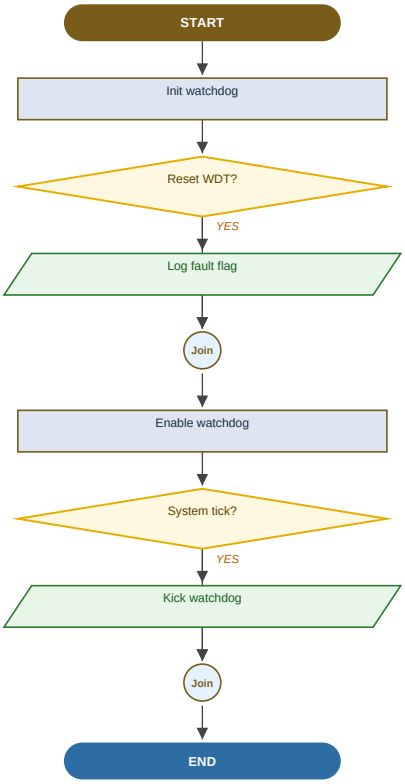
DD-COMP-103	**UndervoltageMonitor** *(Class)*
Description	Monitors supply voltage from ADC samples and triggers undervoltage fault when value falls below 9 V. Ensures actuator output is disabled upon fault.
Inputs	supply_voltage_sample
Outputs	fault_event=UNDERVOLTAGE, actuator_disable_request
Dependencies	AdcAcquisitionService, FaultManager, ActuatorControlService
Design Pattern	Observer
Traces to	SW-REQ-090

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



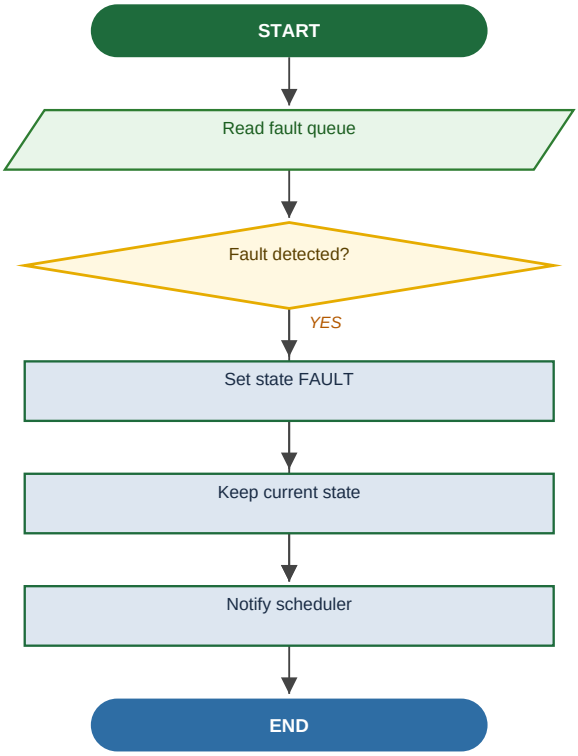
DD-COMP-104	**WatchdogService** *(Service)*
Description	Initializes and enables the hardware watchdog, periodically services it during normal execution, and on reset records a watchdog fault flag in non-volatile fault log on next boot.
Inputs	system_tick, boot_reset_cause
Outputs	watchdog_kick, fault_event=WATCHDOG_TIMEOUT
Dependencies	HardwareWatchdogDriver, FaultLogStorage, SystemResetService
Design Pattern	Singleton
Traces to	SW-REQ-091

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



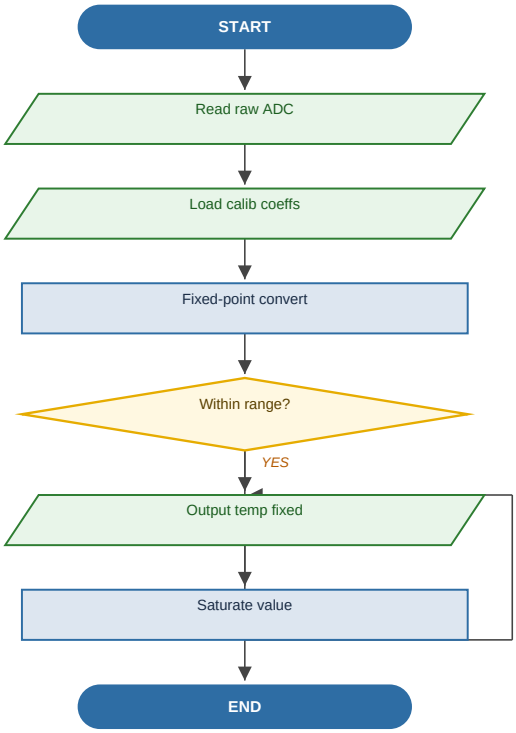
DD-COMP-105	**FaultStateTransitionHandler** *(Class)*
Description	Receives fault events within the current control cycle and forces the state machine to transition to FAULT state before cycle completion. Ensures deterministic same-cycle transition.
Inputs	fault_event_queue, current_state
Outputs	next_state=FAULT when any fault detected
Dependencies	FaultManager, ControlLoopScheduler
Design Pattern	State
Traces to	SW-REQ-092

■ Flow Diagram *(state flow) — See Section 7 for full-page view*



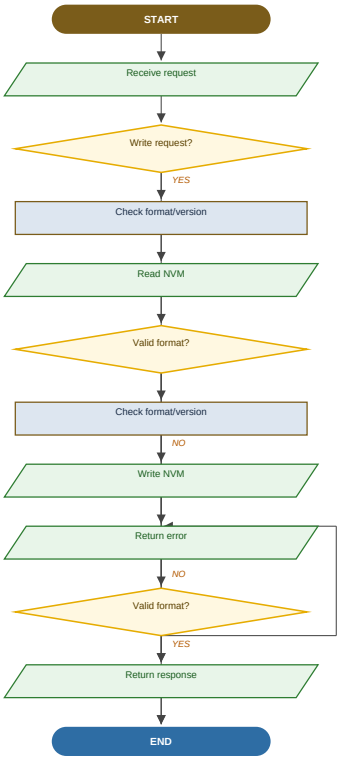
DD-COMP-106	**TemperatureProcessing** *(Algorithm)*
Description	Processes raw temperature sensor ADC values into engineering units using fixed-point scaling that supports -40°C to +85°C without overflow or resolution loss. Uses calibrated coefficients and saturates out-of-range values only after conversion.
Inputs	raw_temperature_adc, calibration_coefficients
Outputs	temperature_celsius_fixed_point
Dependencies	AdcAcquisitionService, CalibrationDataStore
Design Pattern	N/A
Traces to	SW-REQ-093

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



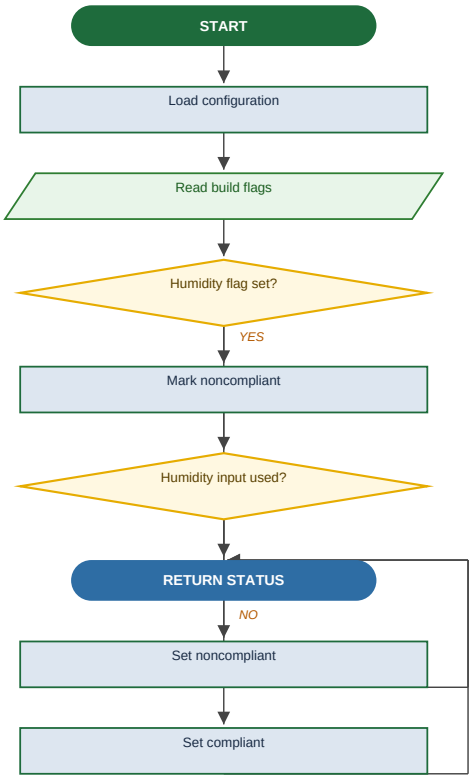
DD-COMP-107	**NvmConfigStorage** *(Service)*
Description	Manages configuration and calibration data in non-volatile memory without software-enforced invalidation due to storage temperature range. Provides integrity checks only on data format/version, not temperature.
Inputs	config_write_requests, config_read_requests
Outputs	config_data, write_status
Dependencies	NvmDriver, DataIntegrityChecker
Design Pattern	Repository
Traces to	SW-REQ-094

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



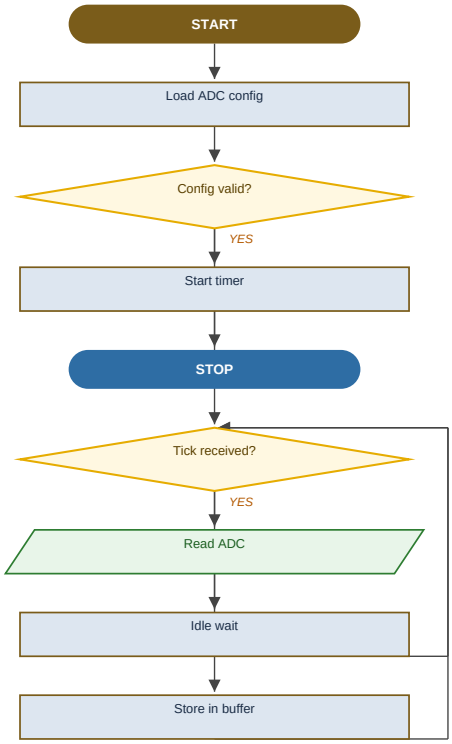
DD-COMP-108	**HumidityIndependencePolicy** *(Class)*
Description	Defines operational policy indicating no humidity sensor dependency or condensation-detection logic. Ensures build-time and runtime checks prevent use of humidity inputs.
Inputs	system_configuration
Outputs	policy_compliance_status
Dependencies	ConfigurationManager, BuildFeatureFlags
Design Pattern	N/A
Traces to	SW-REQ-095

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



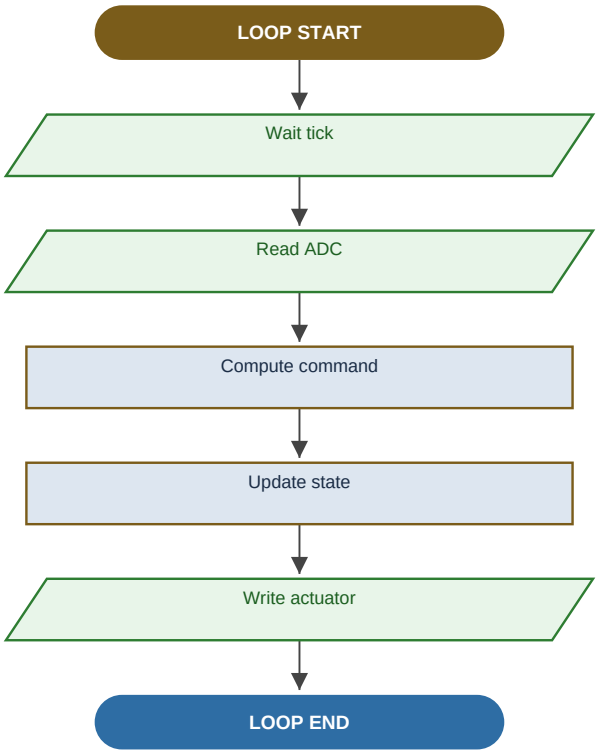
DD-COMP-109	**AdcAcquisitionService** *(Service)*
Description	Continuously samples ADC channels at a fixed periodic rate using a timer/interrupt-driven loop. Stores the latest sample per channel in a lock-free buffer accessible by the control loop each cycle.
Inputs	timer_tick, adc_channels_config
Outputs	latest_adc_samples
Dependencies	AdcDriver, TimerService, SampleBuffer
Design Pattern	Producer-Consumer
Traces to	SW-REQ-096

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



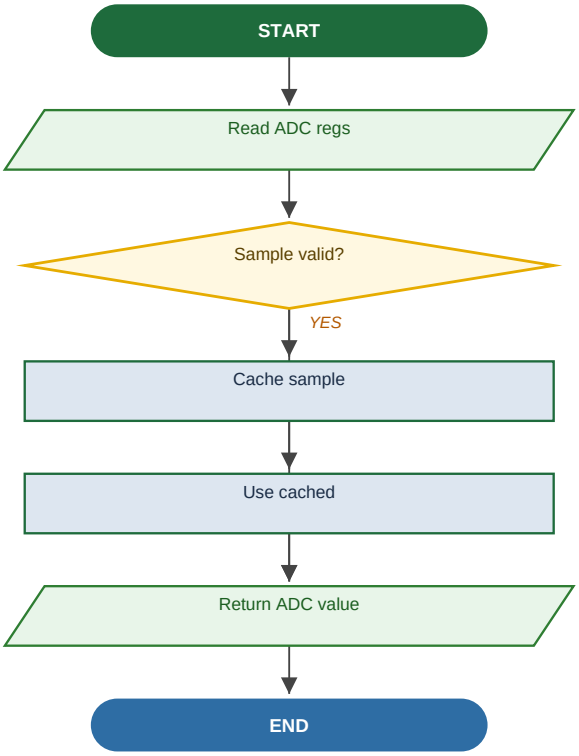
DD-COMP-110	**ControlLoopScheduler** *(Service)*
Description	Executes the periodic control loop. On each iteration it reads the latest ADC feedback, invokes the actuator command computation, and issues updated commands to the actuator driver.
Inputs	Timer tick, latest ADC feedback sample
Outputs	Actuator command updates
Dependencies	ADCReader, CommandCalculator, ActuatorDriver
Design Pattern	Observer
Traces to	SW-REQ-097

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



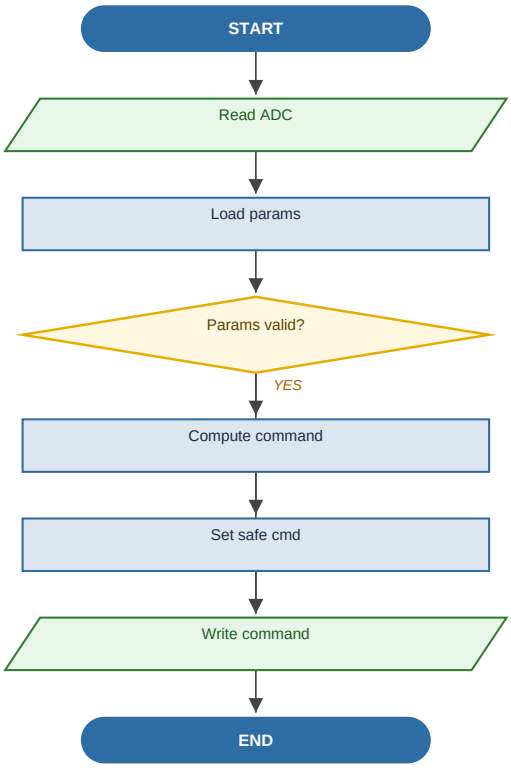
DD-COMP-111	**ADCReader** *(Class)*
Description	Provides the most recent ADC feedback sample from the ADC hardware abstraction and caches it for control loop use.
Inputs	ADC hardware data registers
Outputs	Latest ADC value
Dependencies	ADCHardwareHAL
Design Pattern	Singleton
Traces to	SW-REQ-097

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



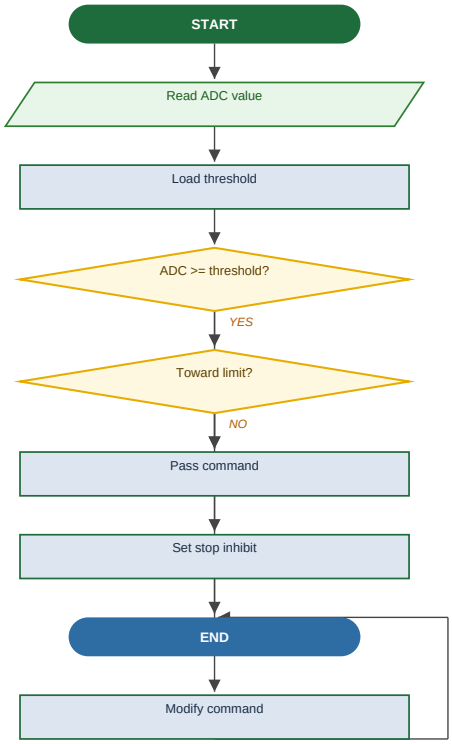
DD-COMP-112	**CommandCalculator** *(Algorithm)*
Description	Computes actuator command values based on the most recent ADC feedback and control law parameters. Produces updated command signals on each loop iteration.
Inputs	ADC value, control parameters, current actuator state
Outputs	Computed actuator command
Dependencies	ControlParamsStore
Design Pattern	Strategy
Traces to	SW-REQ-097

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



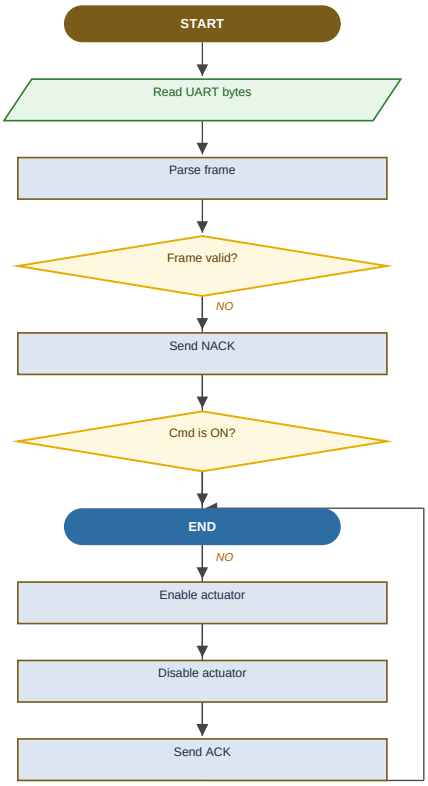
DD-COMP-113	**ShippingLimitGuard** *(Class)*
Description	Checks ADC feedback against the configured Shipping_ADC_Value threshold and prevents actuator motion toward the limit by issuing a stop command and inhibiting further commands in the limit direction.
Inputs	ADC value, Shipping_ADC_Value threshold, requested actuator direction
Outputs	Stop/inhibit flags, modified actuator command
Dependencies	ConfigStore, ActuatorDriver
Design Pattern	Decorator
Traces to	SW-REQ-098

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



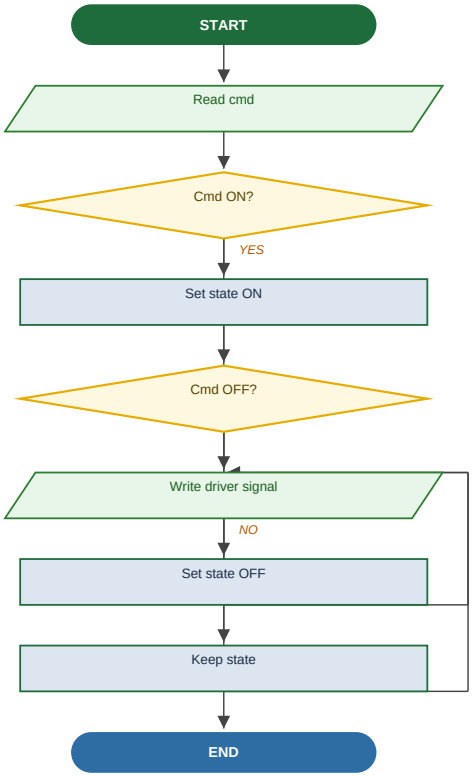
DD-COMP-114	**UARTCommandHandler** *(Service)*
Description	Parses incoming UART frames and dispatches actuator ON/OFF commands. Updates actuator enable state based on valid commands.
Inputs	UART RX bytes, frame metadata
Outputs	Actuator enable state updates, command acknowledgments
Dependencies	UARTDriver, ActuatorEnableController
Design Pattern	Command
Traces to	SW-REQ-099

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



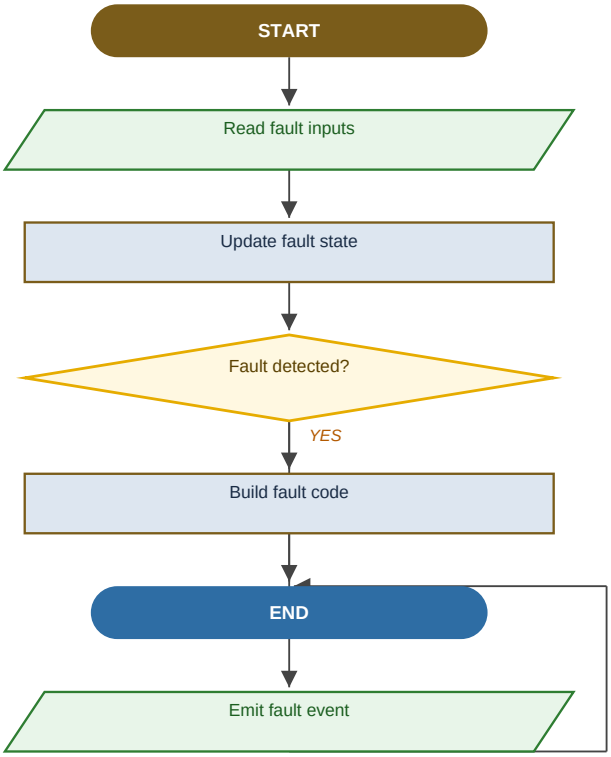
DD-COMP-115	**ActuatorEnableController** *(Class)*
Description	Maintains actuator enable/disable state and applies state to the actuator driver. Ensures ON/OFF commands are reflected in hardware control signals.
Inputs	Enable/disable commands
Outputs	Actuator enable control signal
Dependencies	ActuatorDriver
Design Pattern	State
Traces to	SW-REQ-099

■ Flow Diagram *(state flow) — See Section 7 for full-page view*



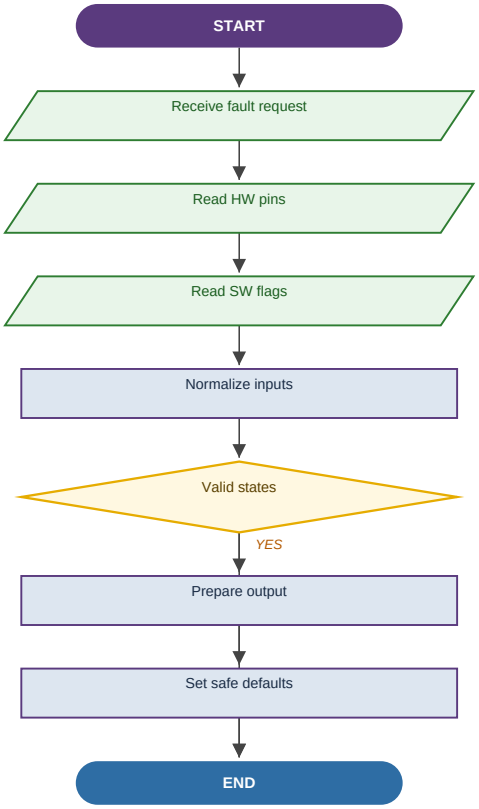
DD-COMP-116	**FaultMonitor** *(Service)*
Description	Polls or receives fault input changes and detects fault conditions within 100 ms. Emits fault events upon detection.
Inputs	Fault input signals, timer tick
Outputs	Fault detected events with fault code
Dependencies	FaultInputProvider, FaultEventBus
Design Pattern	Observer
Traces to	SW-REQ-100

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



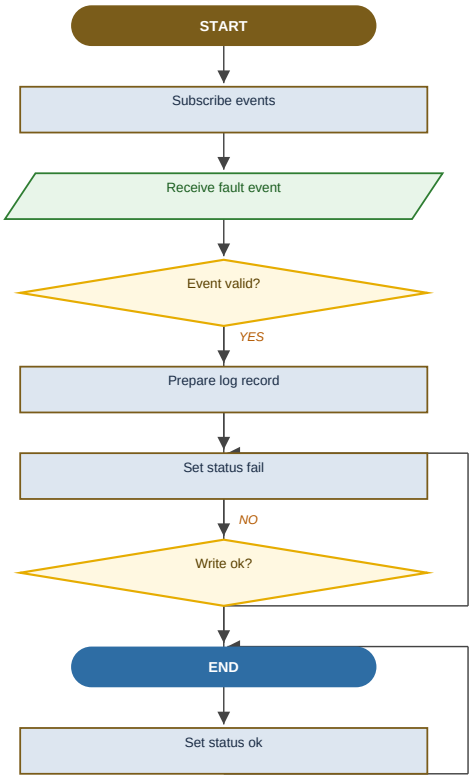
DD-COMP-117	**FaultInputProvider** *(Interface)*
Description	Abstracts access to hardware or software fault inputs for the fault monitor to evaluate.
Inputs	Hardware fault pins, software fault flags
Outputs	Normalized fault input states
Dependencies	HardwareIOHAL
Design Pattern	Adapter
Traces to	SW-REQ-100

■ Flow Diagram *(request response) — See Section 7 for full-page view*



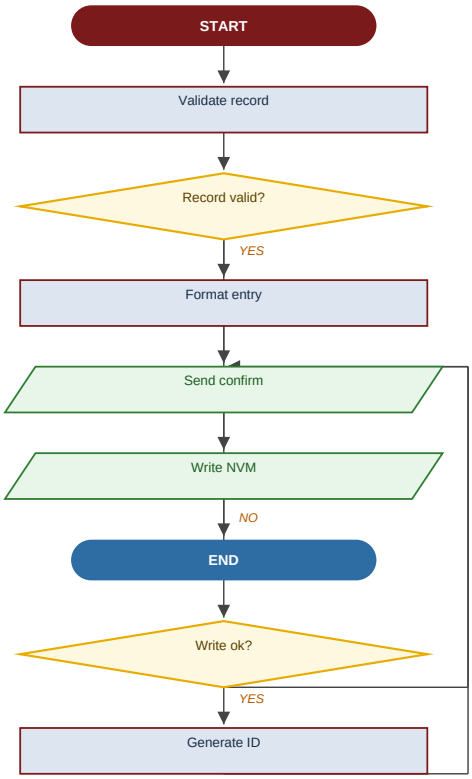
DD-COMP-118	**FaultLogger** *(Service)*
Description	Logs detected faults to non-volatile storage with timestamp and fault code within 1 second of detection. Subscribes to fault events.
Inputs	Fault event (timestamp, fault code)
Outputs	Persisted log record, logging status
Dependencies	FaultEventBus, NonVolatileStorage, RealTimeClock
Design Pattern	Observer
Traces to	SW-REQ-101

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



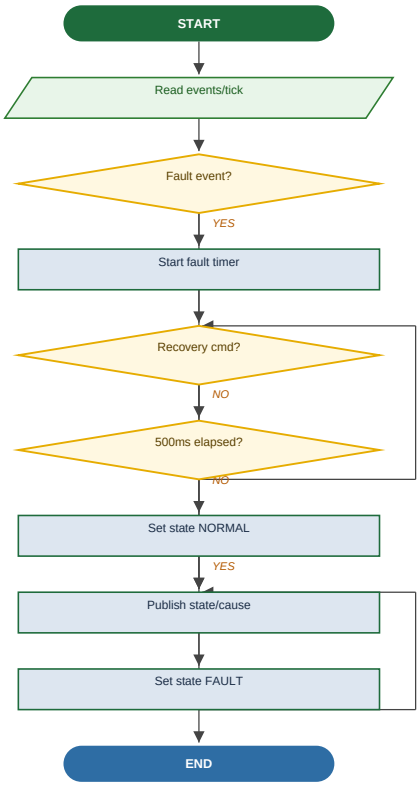
DD-COMP-119	**FaultLogStore** *(Database)*
Description	Non-volatile storage schema for fault log records including timestamp and fault code. Provides append-only write access.
Inputs	Log record
Outputs	Write confirmation, stored record ID
Dependencies	NonVolatileStorageDriver
Design Pattern	Repository
Traces to	SW-REQ-101

■ Flow Diagram *(data flow) — See Section 7 for full-page view*



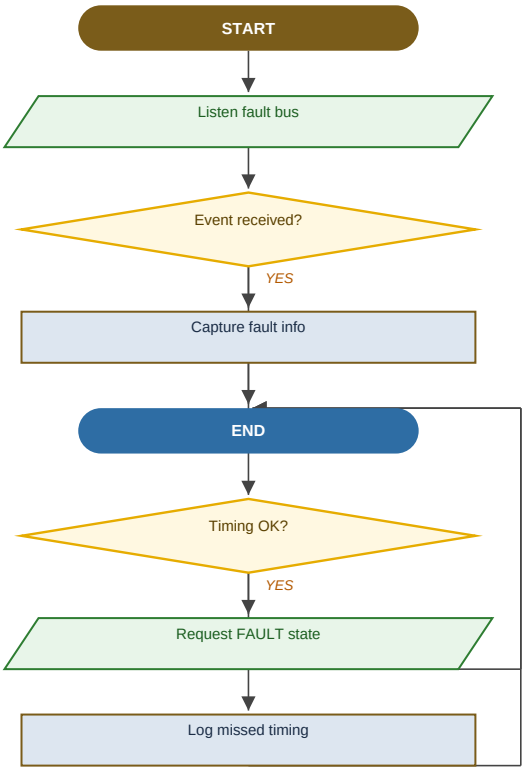
DD-COMP-120	**SystemStateMachine** *(Class)*
Description	Controls system state transitions. Transitions from NORMAL to FAULT within 500 ms after any fault event. Maintains current state and transition causes.
Inputs	Fault events, recovery commands, timer tick
Outputs	Current state, transition cause updates
Dependencies	FaultEventBus, RecoveryCommandHandler, StateStatusProvider
Design Pattern	State
Traces to	SW-REQ-102

■ Flow Diagram *(state flow) — See Section 7 for full-page view*



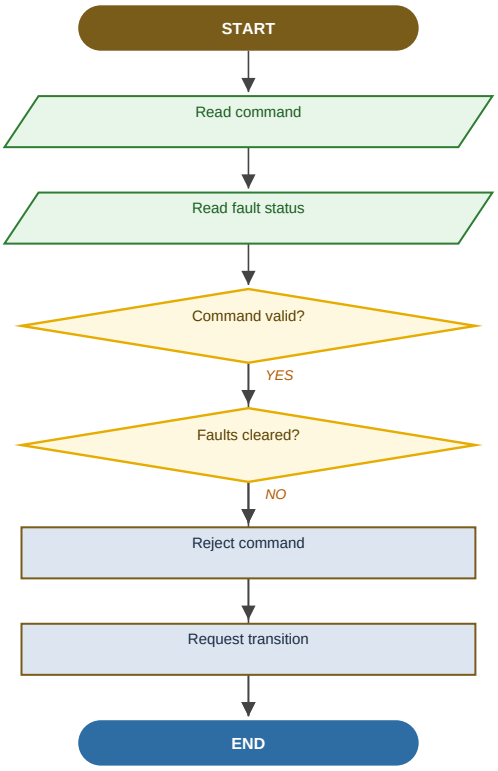
DD-COMP-121	**FaultToStateTransitionService** *(Service)*
Description	Listens for fault events and triggers the SystemStateMachine to transition to FAULT within required timing constraints, providing fault cause details.
Inputs	Fault events
Outputs	State transition requests
Dependencies	FaultEventBus, SystemStateMachine
Design Pattern	Observer
Traces to	SW-REQ-102

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



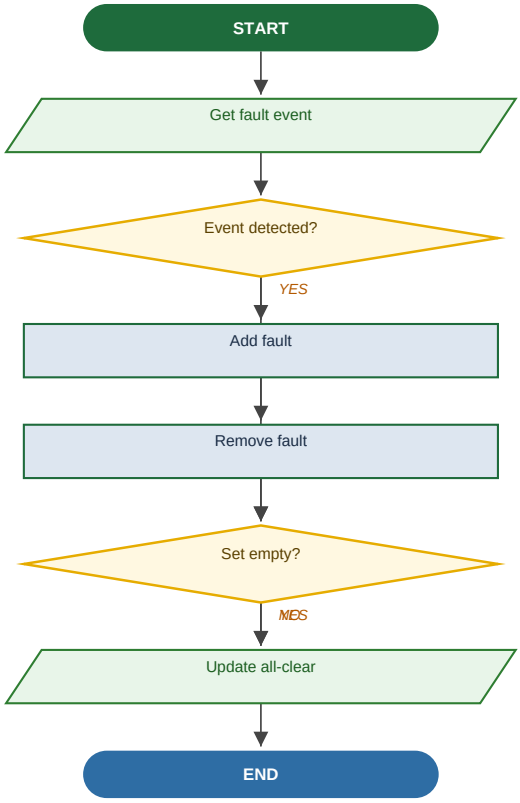
DD-COMP-122	**RecoveryCommandHandler** *(Service)*
Description	Accepts recovery commands and validates that all faults are cleared before requesting transition from FAULT to NORMAL.
Inputs	Recovery command, current fault status
Outputs	Recovery transition request
Dependencies	FaultStatusTracker, SystemStateMachine
Design Pattern	Command
Traces to	SW-REQ-103

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



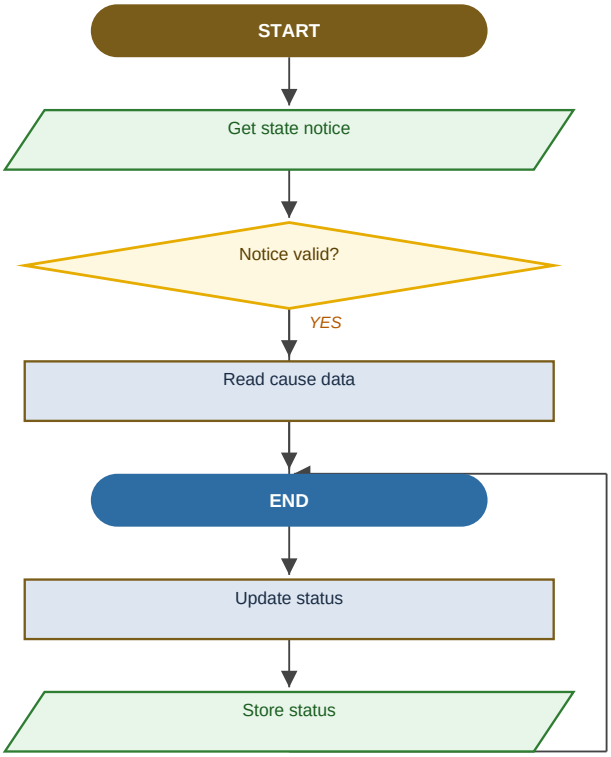
DD-COMP-123	**FaultStatusTracker** *(Class)*
Description	Maintains the set of active faults and provides a query to determine whether all faults are cleared.
Inputs	Fault detected events, fault cleared events
Outputs	Active fault set, all-clear status
Dependencies	FaultEventBus
Design Pattern	Observer
Traces to	SW-REQ-103

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



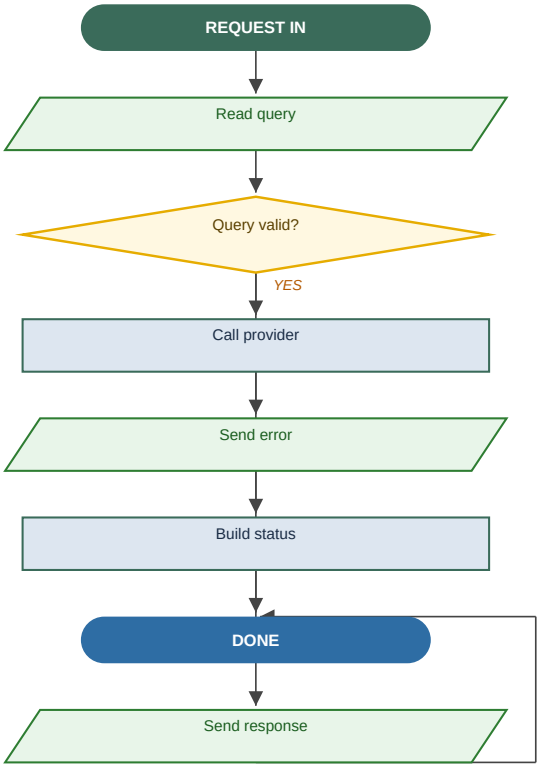
DD-COMP-124	**StateStatusProvider** *(Service)*
Description	Exposes software-accessible status indicator for current system state and last transition cause. Updated on each state transition.
Inputs	State change notifications, transition cause data
Outputs	State status structure for software access
Dependencies	SystemStateMachine
Design Pattern	Observer
Traces to	SW-REQ-104

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



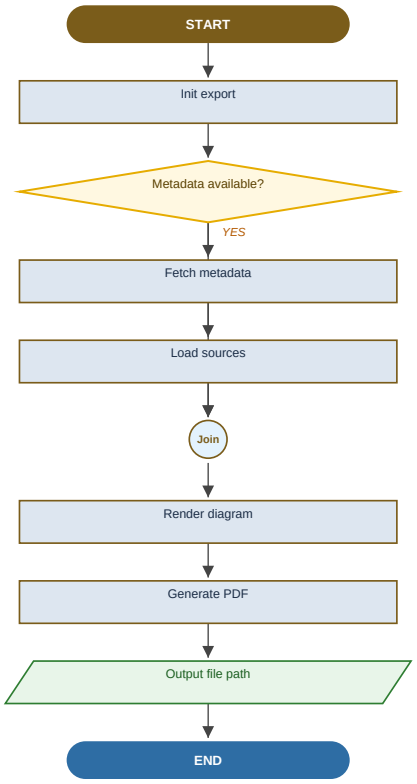
DD-COMP-125	**StateStatusAPI** *(API Endpoint)*
Description	Provides read access to current system state and last transition cause for other software components.
Inputs	Status query request
Outputs	State status structure
Dependencies	StateStatusProvider
Design Pattern	Facade
Traces to	SW-REQ-104

■ Flow Diagram *(request response) — See Section 7 for full-page view*



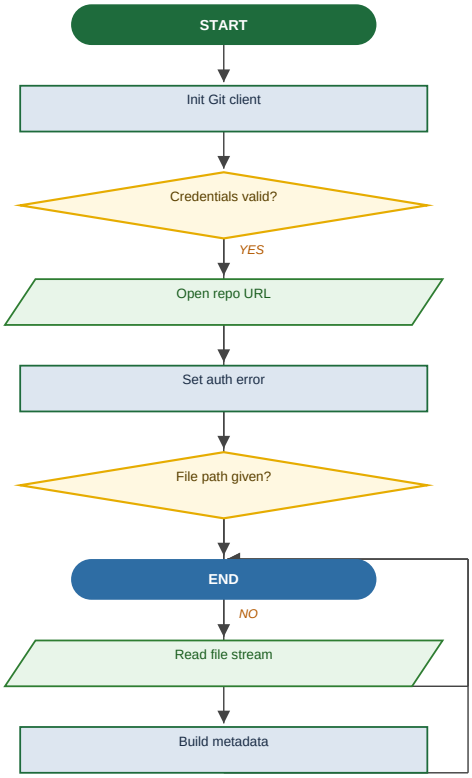
DD-COMP-126	**ArchitectureDiagramExporter** *(Service)*
Description	Generates an exportable architecture diagram PDF by pulling architecture metadata or diagram sources from the documentation repository and rendering to PDF using a diagram renderer.
Inputs	project_id, doc_repo_path, diagram_source_refs, export_options
Outputs	architecture_diagram.pdf file path
Dependencies	DocumentationRepositoryAdapter, DiagramRenderer, PdfGenerator
Design Pattern	Facade
Traces to	SW-REQ-105

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



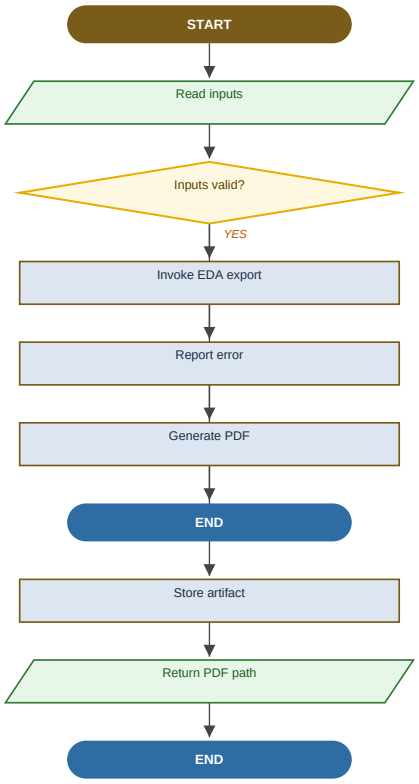
DD-COMP-127	**DocumentationRepositoryAdapter** *(Class)*
Description	Provides access to project documentation repository contents, including diagram sources and PDFs.
Inputs	repo_url, credentials, file_path
Outputs	file stream or file metadata
Dependencies	GitClientLibrary
Design Pattern	Repository
Traces to	SW-REQ-105

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



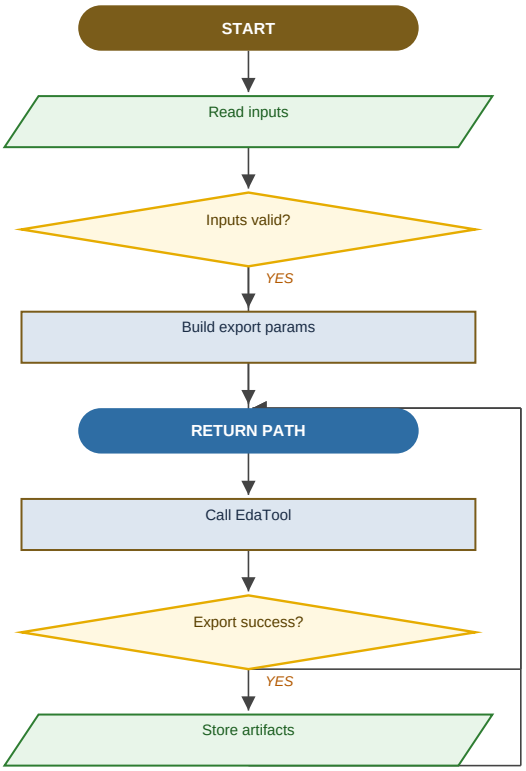
DD-COMP-128	**SchematicPdfExporter** *(Service)*
Description	Exports the electrical schematic to PDF using EDA tool automation or schematic source files and writes the result to the release artifacts directory.
Inputs	project_id, schematic_source_path, export_options
Outputs	schematic.pdf file path
Dependencies	EdaToolAdapter, PdfGenerator, ArtifactStore
Design Pattern	Facade
Traces to	SW-REQ-106

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



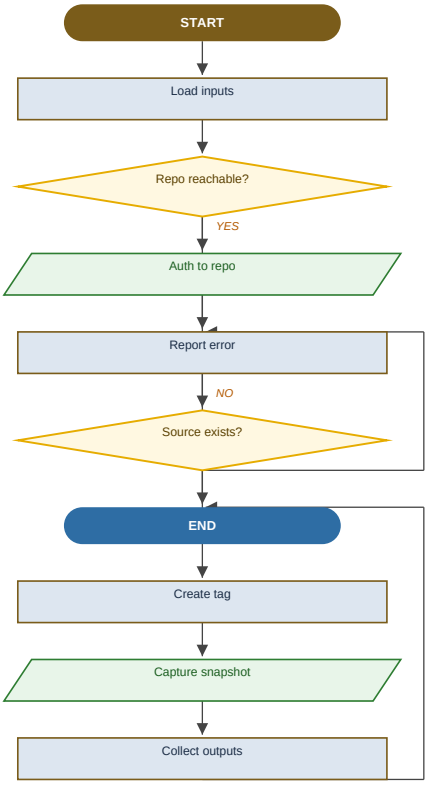
DD-COMP-129	**PcbGerberExporter** *(Service)*
Description	Automates PCB layout export to Gerber format and stores the generated files in the release artifacts directory.
Inputs	project_id, pcb_layout_source_path, layer_config, export_options
Outputs	gerber_files directory path
Dependencies	EdaToolAdapter, ArtifactStore
Design Pattern	Facade
Traces to	SW-REQ-107

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



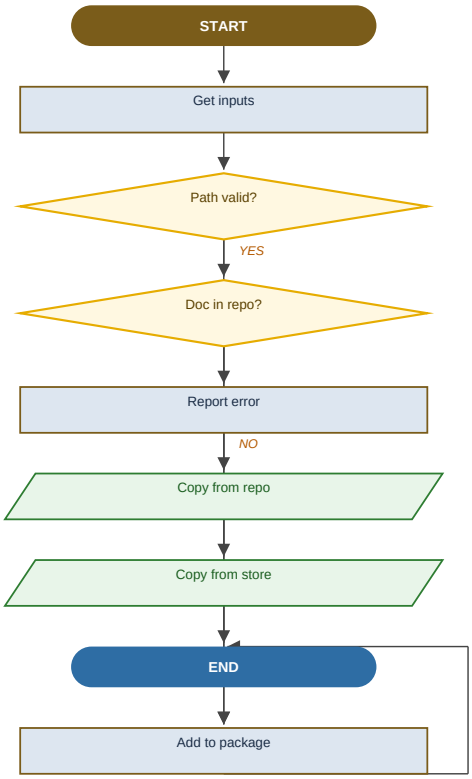
DD-COMP-130	**FirmwareReleaseTagger** *(Service)*
Description	Ensures firmware source code exists in a version-controlled repository, creates an annotated tag for the release, and captures a snapshot reference for delivery.
Inputs	repo_url, release_version, tag_message, credentials
Outputs	tag_name, commit_hash, repository_snapshot_ref
Dependencies	GitClientLibrary
Design Pattern	Command
Traces to	SW-REQ-108

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



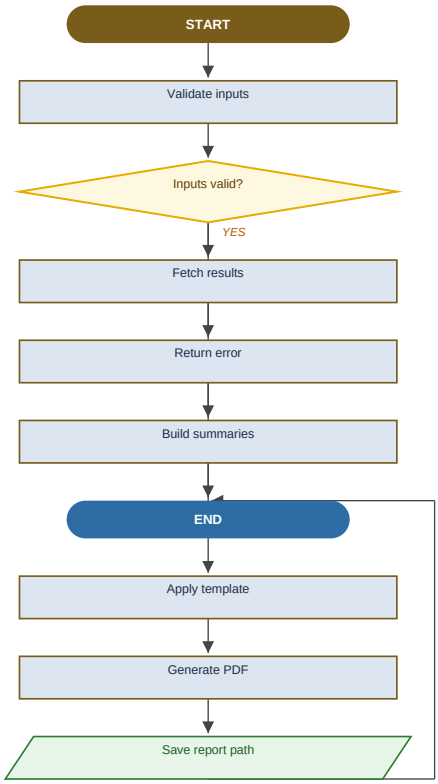
DD-COMP-131	**CalibrationDocIncluder** *(Service)*
Description	Validates and includes the calibration procedure PDF in the release package, copying from documentation repository or artifact store.
Inputs	project_id, calibration_doc_path
Outputs	release_package_entry for calibration_procedure.pdf
Dependencies	DocumentationRepositoryAdapter, ArtifactStore, ReleasePackageBuilder
Design Pattern	Facade
Traces to	SW-REQ-109

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



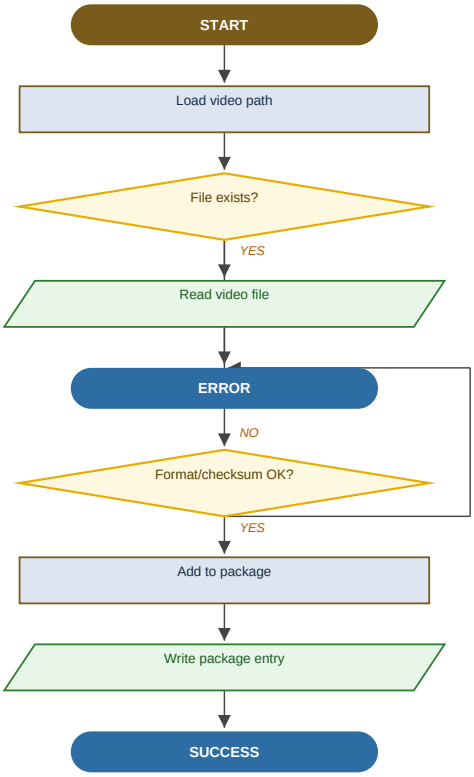
DD-COMP-132	**TestReportGenerator** *(Service)*
Description	Aggregates test results for a build, formats test case summaries with pass/fail status, and renders a PDF test report.
Inputs	build_id, test_results_data, template_id
Outputs	test_report.pdf file path
Dependencies	TestResultsRepository, PdfGenerator, TemplateEngine
Design Pattern	Builder
Traces to	SW-REQ-110

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



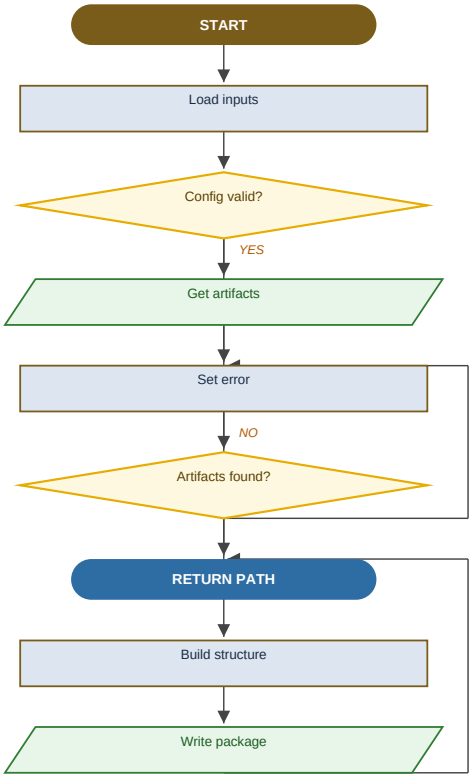
DD-COMP-133	**DemonstrationVideoIncluder** *(Service)*
Description	Validates and includes the demonstration MP4 video file in the release package, ensuring format and checksum.
Inputs	project_id, video_file_path
Outputs	release_package_entry for demo_video.mp4
Dependencies	ArtifactStore, ReleasePackageBuilder, MediaValidator
Design Pattern	Facade
Traces to	SW-REQ-111

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



DD-COMP-134	**ReleasePackageBuilder** *(Service)*
Description	Assembles exported artifacts and documentation into a structured release package directory or archive.
Inputs	release_id, artifact_paths, package_structure_config
Outputs	release_package_path
Dependencies	ArtifactStore, FileSystemAdapter
Design Pattern	Builder
Traces to	SW-REQ-106

■ Flow Diagram *(function flow) — See Section 7 for full-page view*



5. Traceability Matrix

The table below maps each Design Component (DD-COMP) back to its parent Software Requirement (SW-REQ) for full design-to-requirement traceability.

DD-COMP ID	Component	Type	Module	SW-REQ ID
DD-COMP-001	ControlLoopTask	Service	ControlModule	SW-REQ-001
DD-COMP-002	PIDControlAlgorithm	Algorithm	ControlModule	SW-REQ-001
DD-COMP-003	ADCSampler	Service	SensorModule	SW-REQ-002
DD-COMP-004	UARTCommandHandler	Service	CommModule	SW-REQ-003
DD-COMP-005	CommandParser	Class	CommModule	SW-REQ-003
DD-COMP-006	UARTDiagnosticsService	Service	CommModule	SW-REQ-004
DD-COMP-007	UARTDriverConfig	Class	CommModule	SW-REQ-005
DD-COMP-008	ConfigService	Service	ConfigModule	SW-REQ-006
DD-COMP-009	ConfigStore	Database	ConfigModule	SW-REQ-006
DD-COMP-010	ADCValidationService	Service	SafetyModule	SW-REQ-007
DD-COMP-011	UARTFaultHandler	Service	SafetyModule	SW-REQ-007

DD-COMP-012	FaultManager	Service	SafetyModule	SW-REQ-007
DD-COMP-013	SafeStateController	Service	SafetyModule	SW-REQ-008
DD-COMP-014	CalibrationStore	Database	SafetyModule	SW-REQ-008
DD-COMP-015	MISRA Compliance Process	Module	ComplianceModule	SW-REQ-009
DD-COMP-016	Supply Voltage Assumption	Class	ControlModule	SW-REQ-010
DD-COMP-017	ADC Channel Configuration	Service	HALModule	SW-REQ-011
DD-COMP-018	ADC Read Service	Service	HALModule	SW-REQ-012
DD-COMP-019	Position Scaling Algorithm	Algorithm	SignalProcessingModule	SW-REQ-013
DD-COMP-020	Control Output Calculator	Class	ControlModule	SW-REQ-014
DD-COMP-021	Control Loop Scheduler	Service	SchedulerModule	SW-REQ-015
DD-COMP-022	Feedback Fault Monitor	Class	SafetyModule	SW-REQ-016
DD-COMP-023	Fault Reporter	Service	SafetyModule	SW-REQ-016
DD-COMP-024	Actuator Drive Output	Service	HALModule	SW-REQ-014
DD-COMP-025	ThresholdConfigRepository	Class	ConfigModule	SW-REQ-017

DD-COMP-026	ThresholdConfigService	Service	ConfigModule	SW-REQ-017
DD-COMP-027	ShippingPositionStopLogic	Algorithm	ControlModule	SW-REQ-018
DD-COMP-028	UartCommandParser	Service	UartModule	SW-REQ-019
DD-COMP-029	UartCommandParser_OFF	Service	UartModule	SW-REQ-020
DD-COMP-030	ActuatorStateMachine	Class	ControlModule	SW-REQ-019
DD-COMP-031	ActuatorStateMachine_OFF	Class	ControlModule	SW-REQ-020
DD-COMP-032	UartAckService	Service	UartModule	SW-REQ-021
DD-COMP-033	AdcDriverConfig	Class	AdcModule	SW-REQ-023
DD-COMP-034	AdcSamplingScheduler	Service	AdcModule	SW-REQ-024
DD-COMP-035	AdcContinuousAcquisition	Service	AdcModule	SW-REQ-022
DD-COMP-036	ADC Sample Buffer	Class	SensingModule	SW-REQ-025
DD-COMP-037	Flap Control Computation	Service	ControlModule	SW-REQ-026
DD-COMP-038	Actuation Range Guard	Algorithm	SafetyModule	SW-REQ-027
DD-COMP-039	Real-Time ADC Freshness Check	Algorithm	SensingModule	SW-REQ-028

DD-COMP-040	Open-Loop Inhibit Gate	Service	SafetyModule	SW-REQ-029
DD-COMP-041	ADC Validity Monitor	Service	SensingModule	SW-REQ-030
DD-COMP-042	Drive Permission Check	Algorithm	SafetyModule	SW-REQ-031
DD-COMP-043	Target ADC Validator	Service	ConfigModule	SW-REQ-032
DD-COMP-044	ActuatorSafetyMonitor	Class	ControlLoopModule	SW-REQ-033
DD-COMP-045	ActuatorTargetStopController	Class	ControlLoopModule	SW-REQ-034
DD-COMP-046	SupplyVoltageADCChannelConfig	Service	HardwareAbstractionLayer	SW-REQ-035
DD-COMP-047	SupplyVoltageScaler	Algorithm	VoltageMonitorModule	SW-REQ-036
DD-COMP-048	SupplyVoltageSampler	Service	VoltageMonitorModule	SW-REQ-037
DD-COMP-049	OvervoltageThresholdStore	Service	ConfigModule	SW-REQ-038
DD-COMP-050	OvervoltageDetector	Class	VoltageMonitorModule	SW-REQ-039
DD-COMP-051	OvervoltageDriveDisabler	Class	ControlLoopModule	SW-REQ-040
DD-COMP-052	ActuatorCommandGate	Class	ControlModule	SW-REQ-041
DD-COMP-053	FaultTransitionHandler	Class	FaultManagementModule	SW-REQ-042

DD-COMP-054	OvervoltageUartNotifier	Service	DiagnosticsModule	SW-REQ-043
DD-COMP-055	FaultStateRetentionLogic	Class	FaultManagementModule	SW-REQ-044
DD-COMP-056	FaultExitVoltageChecker	Service	FaultManagementModule	SW-REQ-045
DD-COMP-057	OvervoltageMonitor	Service	PowerSafetyModule	SW-REQ-046
DD-COMP-058	UartActuatorCommandInterface	Service	UartModule	SW-REQ-047
DD-COMP-059	ActuatorOnCommandParser	Algorithm	UartModule	SW-REQ-048
DD-COMP-060	UART Command Parser	Class	UartCommModule	SW-REQ-049
DD-COMP-061	Command Dispatcher	Service	CommandModule	SW-REQ-050
DD-COMP-062	Actuator Controller	Service	ActuatorModule	SW-REQ-051
DD-COMP-063	Invalid Command Handler	Service	CommandModule	SW-REQ-052
DD-COMP-064	Invalid Command Logger	Service	LoggingModule	SW-REQ-053
DD-COMP-065	Voltage Configuration Store	Database	ConfigModule	SW-REQ-054
DD-COMP-066	Voltage Range Configuration	Class	ConfigModule	SW-REQ-055
DD-COMP-067	Startup Safety Manager	Service	StartupModule	SW-REQ-056

DD-COMP-068	StartupDelayManager	Class	StartupModule	SW-REQ-057
DD-COMP-069	ADCResolutionConfigurator	Service	ADCModule	SW-REQ-058
DD-COMP-070	ADCReferenceConfigurator	Service	ADCModule	SW-REQ-059
DD-COMP-071	ADCSamplingScheduler	Service	ADCModule	SW-REQ-060
DD-COMP-072	ADCSaturationDetector	Algorithm	ADCModule	SW-REQ-061
DD-COMP-073	ADCPinInitializer	Service	BoardSupportModule	SW-REQ-062
DD-COMP-074	ActuatorGPIOInitializer	Service	BoardSupportModule	SW-REQ-063
DD-COMP-075	PWMTimerInitializer	Service	ActuatorControlModule	SW-REQ-064
DD-COMP-076	UART Peripheral Initializer	Class	HAL	SW-REQ-065
DD-COMP-077	ADC DMA Configuration Service	Service	Driver	SW-REQ-066
DD-COMP-078	Independent Watchdog Service	Service	Middleware	SW-REQ-067
DD-COMP-079	Clock and Timer Configuration	Service	HAL	SW-REQ-068
DD-COMP-080	Layered Architecture Enforcement Interface	Interface	Application	SW-REQ-069
DD-COMP-081	System State Machine	Class	Application	SW-REQ-070

DD-COMP-082	Reset Initialization Handler	Service	Application	SW-REQ-071
DD-COMP-083	Nominal Transition Controller	Service	Application	SW-REQ-072
DD-COMP-084	FaultStateManager	Class	StateManagementModule	SW-REQ-073
DD-COMP-085	HardwareWatchdogInitializer	Service	WatchdogModule	SW-REQ-074
DD-COMP-086	WatchdogServicer	Service	WatchdogModule	SW-REQ-075
DD-COMP-087	FatalMalfunctionResetHandler	Service	FaultHandlingModule	SW-REQ-076
DD-COMP-088	ADCOnlyControlAlgorithm	Algorithm	ControlModule	SW-REQ-077
DD-COMP-089	ControlLoopScheduler	Service	ControlModule	SW-REQ-078
DD-COMP-090	ThresholdStopEvaluator	Class	SafetyControlModule	SW-REQ-079
DD-COMP-091	HysteresisBandManager	Class	SafetyControlModule	SW-REQ-080
DD-COMP-092	RawAdcDataPolicy	Class	ControlModule	SW-REQ-081
DD-COMP-093	UartStatusScheduler	Service	CommsModule	SW-REQ-082
DD-COMP-094	UartStatusMessageBuilder	Class	CommsModule	SW-REQ-082
DD-COMP-095	UartConfigService	Service	CommsModule	SW-REQ-083

DD-COMP-096	UartBaudRateConfig	Algorithm	CommsModule	SW-REQ-083
DD-COMP-097	UartFrameFormatConfig	Algorithm	CommsModule	SW-REQ-084
DD-COMP-098	UartTransmitIntervalValidator	Class	CommsModule	SW-REQ-085
DD-COMP-099	OvercurrentMonitor	Service	SafetyModule	SW-REQ-086
DD-COMP-100	ThermalMonitor	Service	SafetyModule	SW-REQ-087
DD-COMP-101	StallDetectionService	Service	SafetyModule	SW-REQ-088
DD-COMP-102	ShortCircuitDetector	Class	FaultMonitoringModule	SW-REQ-089
DD-COMP-103	UndervoltageMonitor	Class	PowerMonitoringModule	SW-REQ-090
DD-COMP-104	WatchdogService	Service	ReliabilityModule	SW-REQ-091
DD-COMP-105	FaultStateTransitionHandler	Class	StateMachineModule	SW-REQ-092
DD-COMP-106	TemperatureProcessing	Algorithm	SensorProcessingModule	SW-REQ-093
DD-COMP-107	NvmConfigStorage	Service	DataPersistenceModule	SW-REQ-094
DD-COMP-108	HumidityIndependencePolicy	Class	SystemConstraintsModule	SW-REQ-095
DD-COMP-109	AdcAcquisitionService	Service	AcquisitionModule	SW-REQ-096

DD-COMP-110	ControlLoopScheduler	Service	ControlModule	SW-REQ-097
DD-COMP-111	ADCReader	Class	SensorModule	SW-REQ-097
DD-COMP-112	CommandCalculator	Algorithm	ControlModule	SW-REQ-097
DD-COMP-113	ShippingLimitGuard	Class	SafetyModule	SW-REQ-098
DD-COMP-114	UARTCommandHandler	Service	CommunicationModule	SW-REQ-099
DD-COMP-115	ActuatorEnableController	Class	ControlModule	SW-REQ-099
DD-COMP-116	FaultMonitor	Service	FaultModule	SW-REQ-100
DD-COMP-117	FaultInputProvider	Interface	FaultModule	SW-REQ-100
DD-COMP-118	FaultLogger	Service	FaultModule	SW-REQ-101
DD-COMP-119	FaultLogStore	Database	DataLayer	SW-REQ-101
DD-COMP-120	SystemStateMachine	Class	StateModule	SW-REQ-102
DD-COMP-121	FaultToStateTransitionService	Service	StateModule	SW-REQ-102
DD-COMP-122	RecoveryCommandHandler	Service	StateModule	SW-REQ-103
DD-COMP-123	FaultStatusTracker	Class	FaultModule	SW-REQ-103

DD-COMP-124	StateStatusProvider	Service	StateModule	SW-REQ-104
DD-COMP-125	StateStatusAPI	API Endpoint	InterfaceModule	SW-REQ-104
DD-COMP-126	ArchitectureDiagramExporter	Service	ArtifactGenerationModule	SW-REQ-105
DD-COMP-127	DocumentationRepositoryAdapter	Class	DocRepositoryModule	SW-REQ-105
DD-COMP-128	SchematicPdfExporter	Service	ArtifactGenerationModule	SW-REQ-106
DD-COMP-129	PcbGerberExporter	Service	ArtifactGenerationModule	SW-REQ-107
DD-COMP-130	FirmwareReleaseTagger	Service	VCSModule	SW-REQ-108
DD-COMP-131	CalibrationDocIncluder	Service	ReleasePackagingModule	SW-REQ-109
DD-COMP-132	TestReportGenerator	Service	TestReportingModule	SW-REQ-110
DD-COMP-133	DemonstrationVideoIncluder	Service	ReleasePackagingModule	SW-REQ-111
DD-COMP-134	ReleasePackageBuilder	Service	ReleasePackagingModule	SW-REQ-106

6. Design Components Summary

DD-COMP ID	Title	Module	Type
DD-COMP-001	ControlLoopTask	ControlModule	Service
DD-COMP-002	PIDControlAlgorithm	ControlModule	Algorithm
DD-COMP-003	ADCSampler	SensorModule	Service
DD-COMP-004	UARTCommandHandler	CommModule	Service
DD-COMP-005	CommandParser	CommModule	Class
DD-COMP-006	UARTDiagnosticsService	CommModule	Service
DD-COMP-007	UARTDriverConfig	CommModule	Class
DD-COMP-008	ConfigService	ConfigModule	Service
DD-COMP-009	ConfigStore	ConfigModule	Database
DD-COMP-010	ADCValidationService	SafetyModule	Service
DD-COMP-011	UARTFaultHandler	SafetyModule	Service
DD-COMP-012	FaultManager	SafetyModule	Service
DD-COMP-013	SafeStateController	SafetyModule	Service
DD-COMP-014	CalibrationStore	SafetyModule	Database
DD-COMP-015	MISRA Compliance Process	ComplianceModule	Module
DD-COMP-016	Supply Voltage Assumption	ControlModule	Class
DD-COMP-017	ADC Channel Configuration	HALModule	Service
DD-COMP-018	ADC Read Service	HALModule	Service
DD-COMP-019	Position Scaling Algorithm	SignalProcessingModule	Algorithm
DD-COMP-020	Control Output Calculator	ControlModule	Class

DD-COMP-021	Control Loop Scheduler	SchedulerModule	Service
DD-COMP-022	Feedback Fault Monitor	SafetyModule	Class
DD-COMP-023	Fault Reporter	SafetyModule	Service
DD-COMP-024	Actuator Drive Output	HALModule	Service
DD-COMP-025	ThresholdConfigRepository	ConfigModule	Class
DD-COMP-026	ThresholdConfigService	ConfigModule	Service
DD-COMP-027	ShippingPositionStopLogic	ControlModule	Algorithm
DD-COMP-028	UartCommandParser	UartModule	Service
DD-COMP-029	UartCommandParser_OFF	UartModule	Service
DD-COMP-030	ActuatorStateMachine	ControlModule	Class
DD-COMP-031	ActuatorStateMachine_OFF	ControlModule	Class
DD-COMP-032	UartAckService	UartModule	Service
DD-COMP-033	AdcDriverConfig	AdcModule	Class
DD-COMP-034	AdcSamplingScheduler	AdcModule	Service
DD-COMP-035	AdcContinuousAcquisition	AdcModule	Service
DD-COMP-036	ADC Sample Buffer	SensingModule	Class
DD-COMP-037	Flap Control Computation	ControlModule	Service
DD-COMP-038	Actuation Range Guard	SafetyModule	Algorithm
DD-COMP-039	Real-Time ADC Freshness Check	SensingModule	Algorithm
DD-COMP-040	Open-Loop Inhibit Gate	SafetyModule	Service
DD-COMP-041	ADC Validity Monitor	SensingModule	Service
DD-COMP-042	Drive Permission Check	SafetyModule	Algorithm
DD-COMP-043	Target ADC Validator	ConfigModule	Service

DD-COMP-044	ActuatorSafetyMonitor	ControlLoopModule	Class
DD-COMP-045	ActuatorTargetStopController	ControlLoopModule	Class
DD-COMP-046	SupplyVoltageADCChannelConfig	HardwareAbstractionLayer	Service
DD-COMP-047	SupplyVoltageScaler	VoltageMonitorModule	Algorithm
DD-COMP-048	SupplyVoltageSampler	VoltageMonitorModule	Service
DD-COMP-049	OvervoltageThresholdStore	ConfigModule	Service
DD-COMP-050	OvervoltageDetector	VoltageMonitorModule	Class
DD-COMP-051	OvervoltageDriveDisabler	ControlLoopModule	Class
DD-COMP-052	ActuatorCommandGate	ControlModule	Class
DD-COMP-053	FaultTransitionHandler	FaultManagementModule	Class
DD-COMP-054	OvervoltageUartNotifier	DiagnosticsModule	Service
DD-COMP-055	FaultStateRetentionLogic	FaultManagementModule	Class
DD-COMP-056	FaultExitVoltageChecker	FaultManagementModule	Service
DD-COMP-057	OvervoltageMonitor	PowerSafetyModule	Service
DD-COMP-058	UartActuatorCommandInterface	UartModule	Service
DD-COMP-059	ActuatorOnCommandParser	UartModule	Algorithm
DD-COMP-060	UART Command Parser	UartCommModule	Class
DD-COMP-061	Command Dispatcher	CommandModule	Service
DD-COMP-062	Actuator Controller	ActuatorModule	Service
DD-COMP-063	Invalid Command Handler	CommandModule	Service
DD-COMP-064	Invalid Command Logger	LoggingModule	Service
DD-COMP-065	Voltage Configuration Store	ConfigModule	Database
DD-COMP-066	Voltage Range Configuration	ConfigModule	Class

DD-COMP-067	Startup Safety Manager	StartupModule	Service
DD-COMP-068	StartupDelayManager	StartupModule	Class
DD-COMP-069	ADCResolutionConfigurator	ADCModule	Service
DD-COMP-070	ADCReferenceConfigurator	ADCModule	Service
DD-COMP-071	ADCSamplingScheduler	ADCModule	Service
DD-COMP-072	ADCSaturationDetector	ADCModule	Algorithm
DD-COMP-073	ADCPinInitializer	BoardSupportModule	Service
DD-COMP-074	ActuatorGPIOInitializer	BoardSupportModule	Service
DD-COMP-075	PWMTimerInitializer	ActuatorControlModule	Service
DD-COMP-076	UART Peripheral Initializer	HAL	Class
DD-COMP-077	ADC DMA Configuration Service	Driver	Service
DD-COMP-078	Independent Watchdog Service	Middleware	Service
DD-COMP-079	Clock and Timer Configuration	HAL	Service
DD-COMP-080	Layered Architecture Enforcement Interface	Application	Interface
DD-COMP-081	System State Machine	Application	Class
DD-COMP-082	Reset Initialization Handler	Application	Service
DD-COMP-083	Nominal Transition Controller	Application	Service
DD-COMP-084	FaultStateManager	StateManagementModule	Class
DD-COMP-085	HardwareWatchdogInitializer	WatchdogModule	Service
DD-COMP-086	WatchdogServicer	WatchdogModule	Service
DD-COMP-087	FatalMalfunctionResetHandler	FaultHandlingModule	Service
DD-COMP-088	ADCOnlyControlAlgorithm	ControlModule	Algorithm
DD-COMP-089	ControlLoopScheduler	ControlModule	Service

DD-COMP-090	ThresholdStopEvaluator	SafetyControlModule	Class
DD-COMP-091	HysteresisBandManager	SafetyControlModule	Class
DD-COMP-092	RawAdcDataPolicy	ControlModule	Class
DD-COMP-093	UartStatusScheduler	CommsModule	Service
DD-COMP-094	UartStatusMessageBuilder	CommsModule	Class
DD-COMP-095	UartConfigService	CommsModule	Service
DD-COMP-096	UartBaudRateConfig	CommsModule	Algorithm
DD-COMP-097	UartFrameFormatConfig	CommsModule	Algorithm
DD-COMP-098	UartTransmitIntervalValidator	CommsModule	Class
DD-COMP-099	OvercurrentMonitor	SafetyModule	Service
DD-COMP-100	ThermalMonitor	SafetyModule	Service
DD-COMP-101	StallDetectionService	SafetyModule	Service
DD-COMP-102	ShortCircuitDetector	FaultMonitoringModule	Class
DD-COMP-103	UndervoltageMonitor	PowerMonitoringModule	Class
DD-COMP-104	WatchdogService	ReliabilityModule	Service
DD-COMP-105	FaultStateTransitionHandler	StateMachineModule	Class
DD-COMP-106	TemperatureProcessing	SensorProcessingModule	Algorithm
DD-COMP-107	NvmConfigStorage	DataPersistenceModule	Service
DD-COMP-108	HumidityIndependencePolicy	SystemConstraintsModule	Class
DD-COMP-109	AdcAcquisitionService	AcquisitionModule	Service
DD-COMP-110	ControlLoopScheduler	ControlModule	Service
DD-COMP-111	ADCReader	SensorModule	Class
DD-COMP-112	CommandCalculator	ControlModule	Algorithm

DD-COMP-113	ShippingLimitGuard	SafetyModule	Class
DD-COMP-114	UARTCommandHandler	CommunicationModule	Service
DD-COMP-115	ActuatorEnableController	ControlModule	Class
DD-COMP-116	FaultMonitor	FaultModule	Service
DD-COMP-117	FaultInputProvider	FaultModule	Interface
DD-COMP-118	FaultLogger	FaultModule	Service
DD-COMP-119	FaultLogStore	DataLayer	Database
DD-COMP-120	SystemStateMachine	StateModule	Class
DD-COMP-121	FaultToStateTransitionService	StateModule	Service
DD-COMP-122	RecoveryCommandHandler	StateModule	Service
DD-COMP-123	FaultStatusTracker	FaultModule	Class
DD-COMP-124	StateStatusProvider	StateModule	Service
DD-COMP-125	StateStatusAPI	InterfaceModule	API Endpoint
DD-COMP-126	ArchitectureDiagramExporter	ArtifactGenerationModule	Service
DD-COMP-127	DocumentationRepositoryAdapter	DocRepositoryModule	Class
DD-COMP-128	SchematicPdfExporter	ArtifactGenerationModule	Service
DD-COMP-129	PcbGerberExporter	ArtifactGenerationModule	Service
DD-COMP-130	FirmwareReleaseTagger	VCSModule	Service
DD-COMP-131	CalibrationDocIncluder	ReleasePackagingModule	Service
DD-COMP-132	TestReportGenerator	TestReportingModule	Service
DD-COMP-133	DemonstrationVideoIncluder	ReleasePackagingModule	Service
DD-COMP-134	ReleasePackageBuilder	ReleasePackagingModule	Service

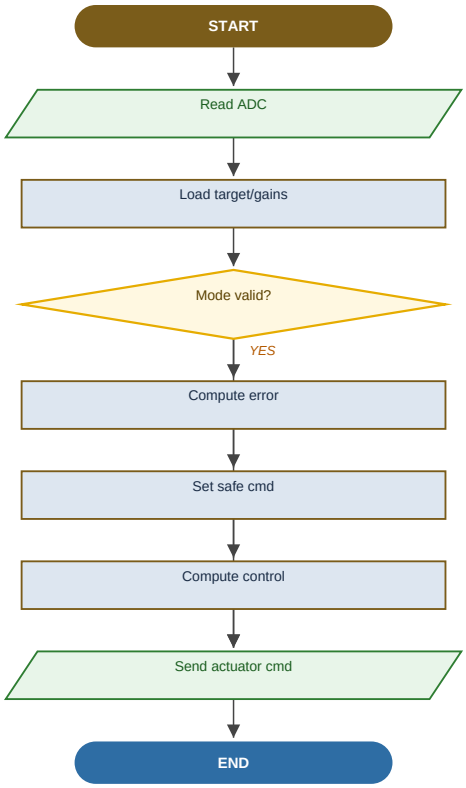
7. Component Flow Diagrams

Full-page flow diagrams for every design component. Each diagram uses standard flowchart symbols: rounded rectangle = Start/End, rectangle = Process, diamond = Decision, parallelogram = I/O, circle = Connector.

DD-COMP-001	**ControlLoopTask** (Service) | Module: ControlModule | SW-REQ-001

Diagram Type: **Function Flow**

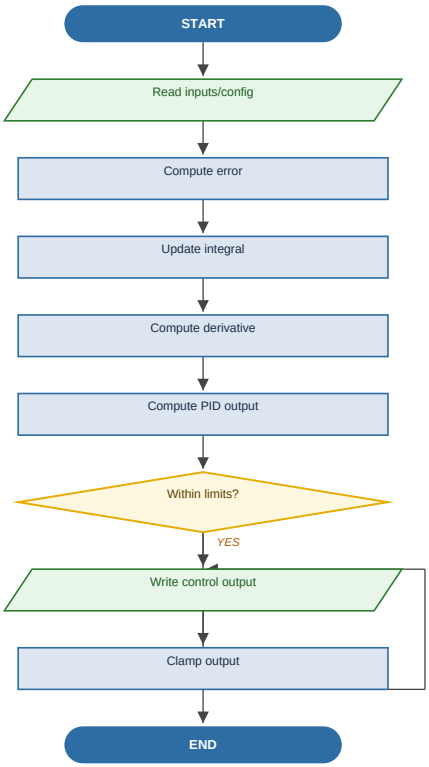
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-002	**PIDControlAlgorithm** (Algorithm) | Module: ControlModule | SW-REQ-001

Diagram Type: **Function Flow**

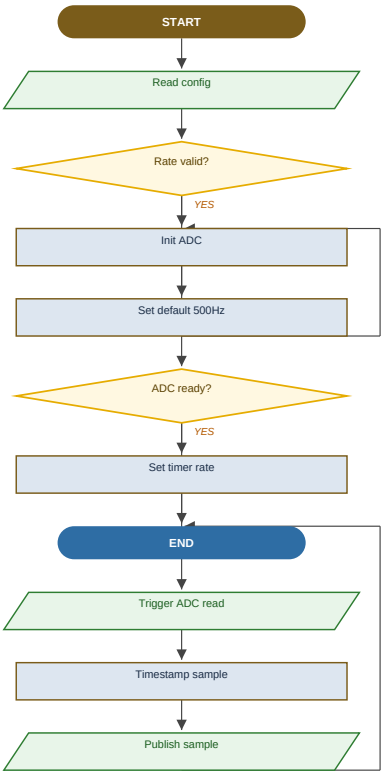
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-003	**ADCSampler** (Service) | Module: SensorModule | SW-REQ-002

Diagram Type: **Function Flow**

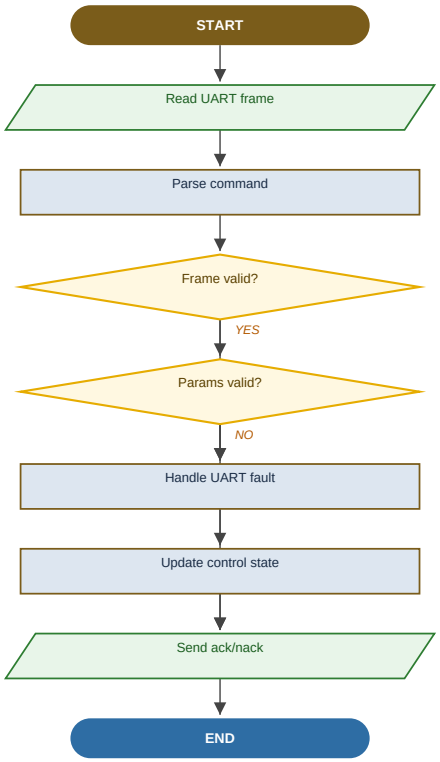
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-004	**UARTCommandHandler** (Service) | Module: CommModule | SW-REQ-003

Diagram Type: **Function Flow**

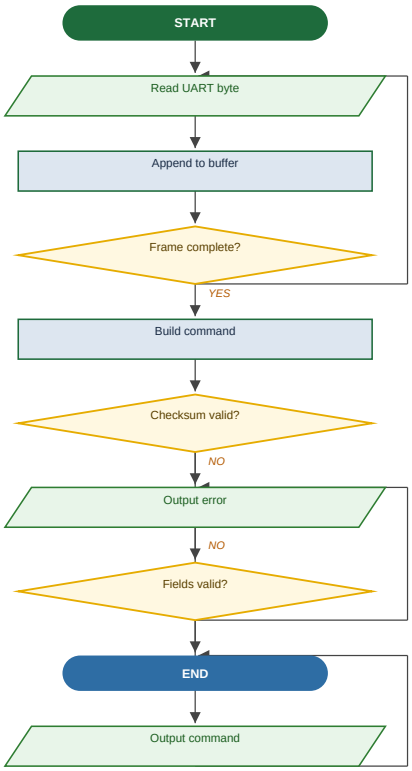
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-005	**CommandParser** (Class) | Module: CommModule | SW-REQ-003

Diagram Type: **Function Flow**

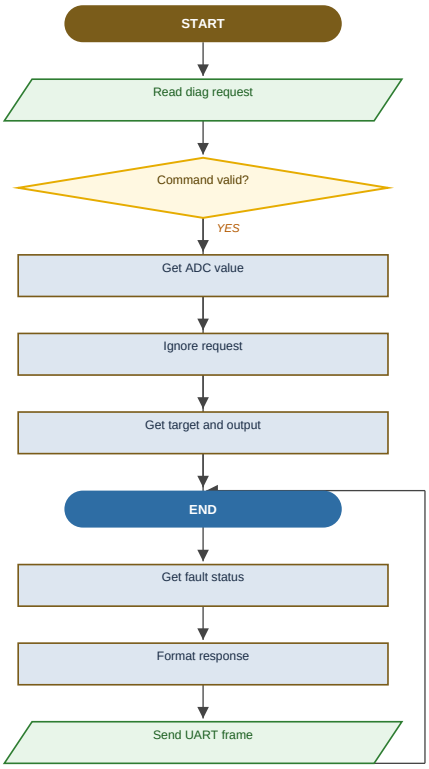
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-006	**UARTDiagnosticsService** (Service) | Module: CommModule | SW-REQ-004

Diagram Type: **Function Flow**

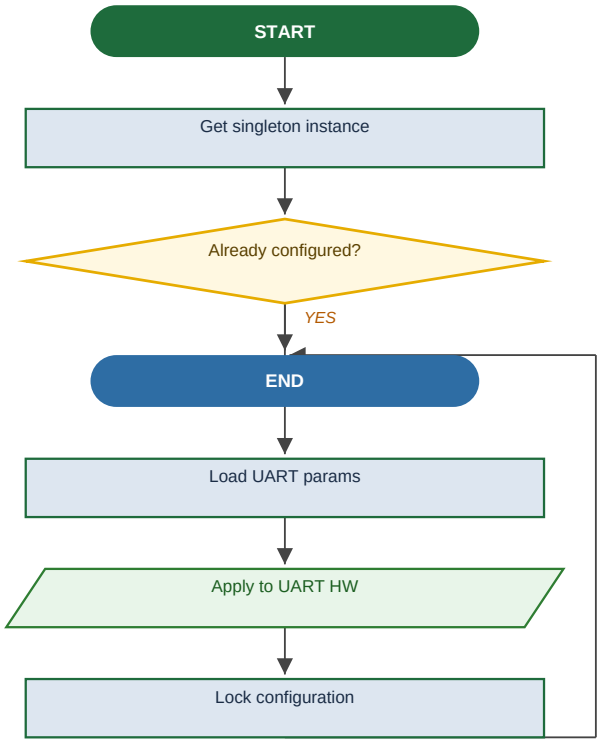
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-007	**UARTDriverConfig** (Class) | Module: CommModule | SW-REQ-005

Diagram Type: **Function Flow**

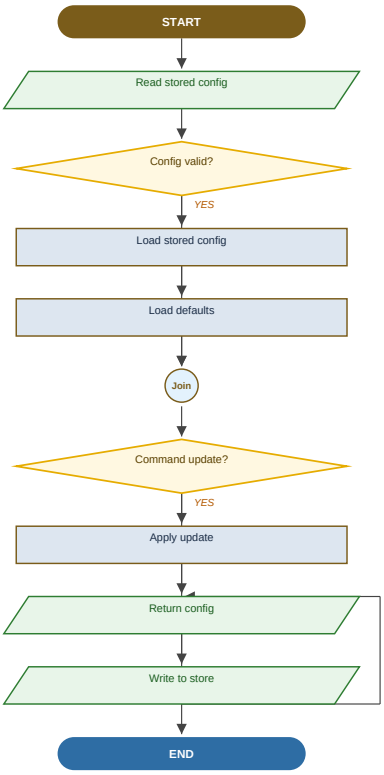
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-008	**ConfigService** (Service) | Module: ConfigModule | SW-REQ-006

Diagram Type: **Function Flow**

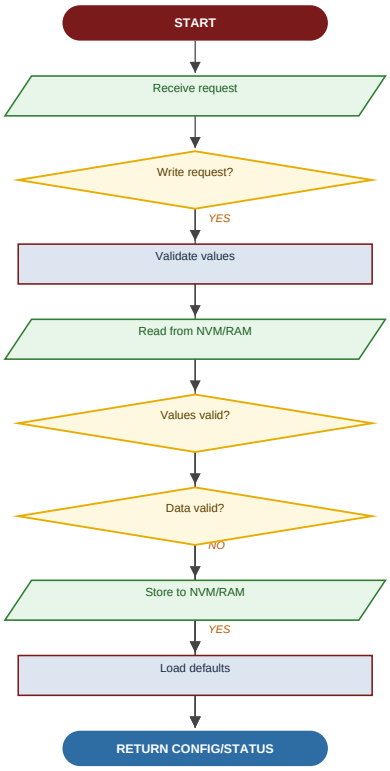
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-009	**ConfigStore** (Database) | Module: ConfigModule | SW-REQ-006

Diagram Type: **Data Flow**

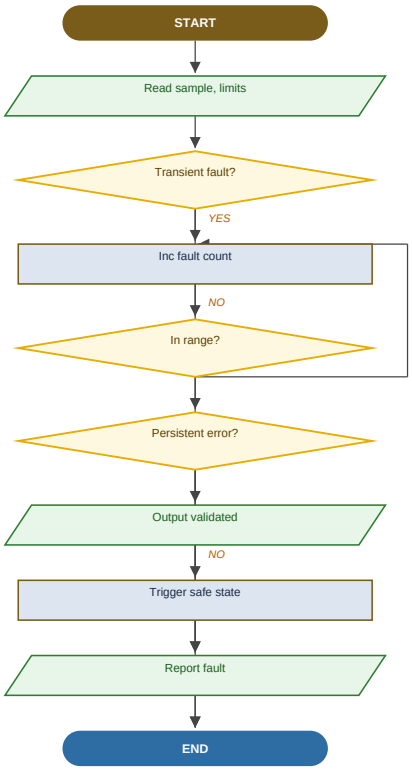
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-010	**ADCValidationService** (Service) | Module: SafetyModule | SW-REQ-007

Diagram Type: **Function Flow**

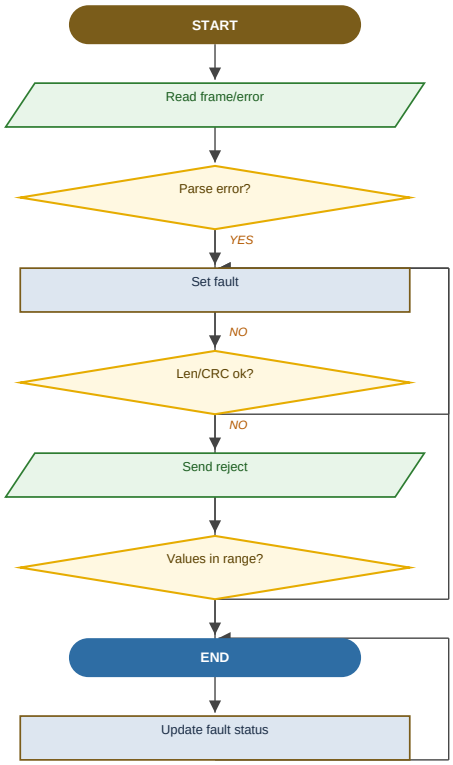
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-011	**UARTFaultHandler** (Service) | Module: SafetyModule | SW-REQ-007

Diagram Type: **Function Flow**

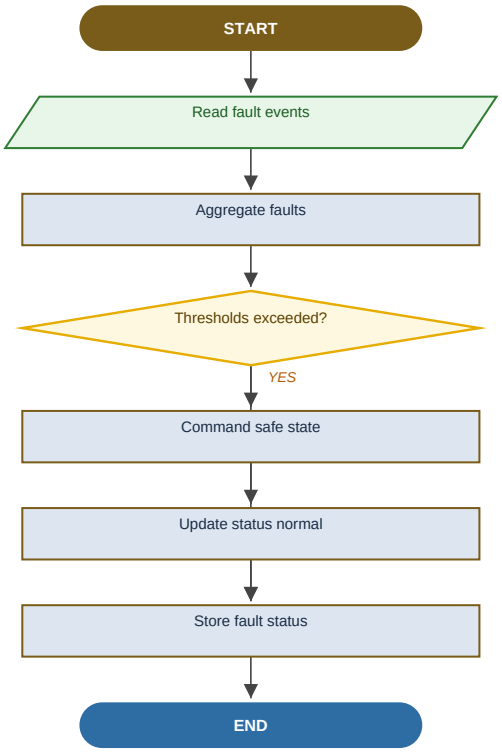
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-012	**FaultManager** (Service) | Module: SafetyModule | SW-REQ-007

Diagram Type: **Function Flow**

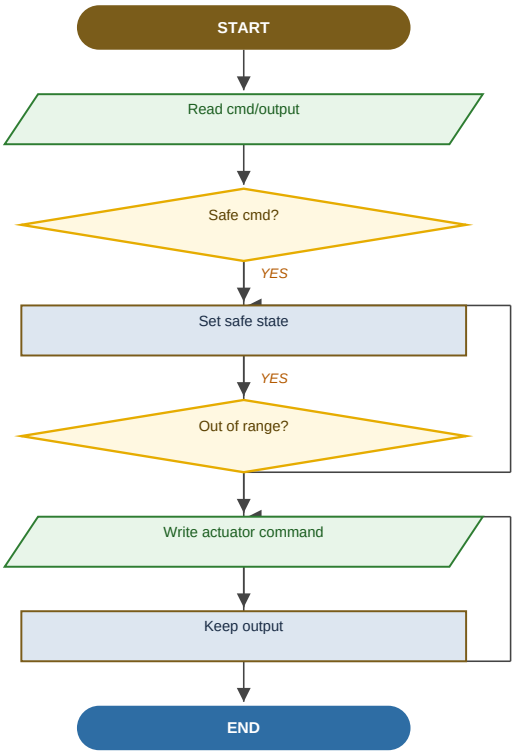
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-013	**SafeStateController** (Service) | Module: SafetyModule | SW-REQ-008

Diagram Type: **Function Flow**

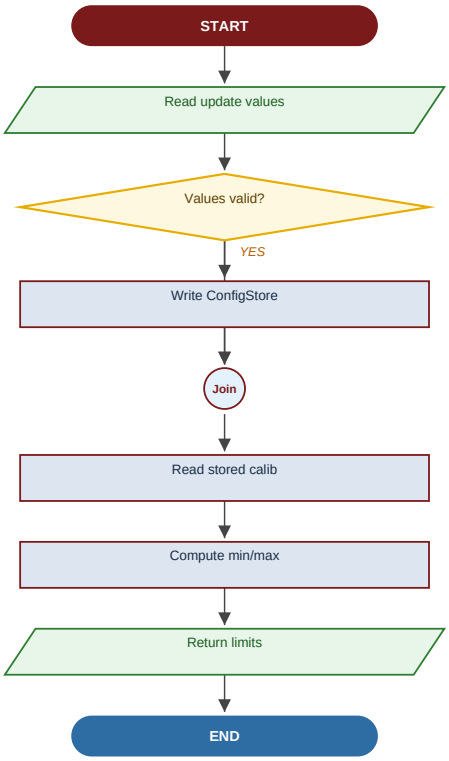
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-014	**CalibrationStore** (Database) | Module: SafetyModule | SW-REQ-008

Diagram Type: **Data Flow**

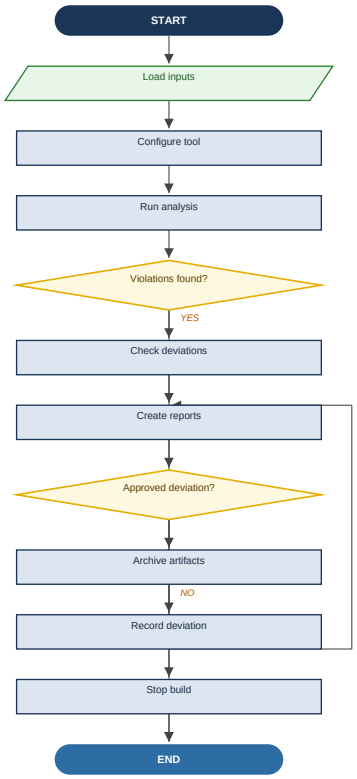
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-015	**MISRA Compliance Process** (Module) | Module: ComplianceModule | SW-REQ-009

Diagram Type: **Function Flow**

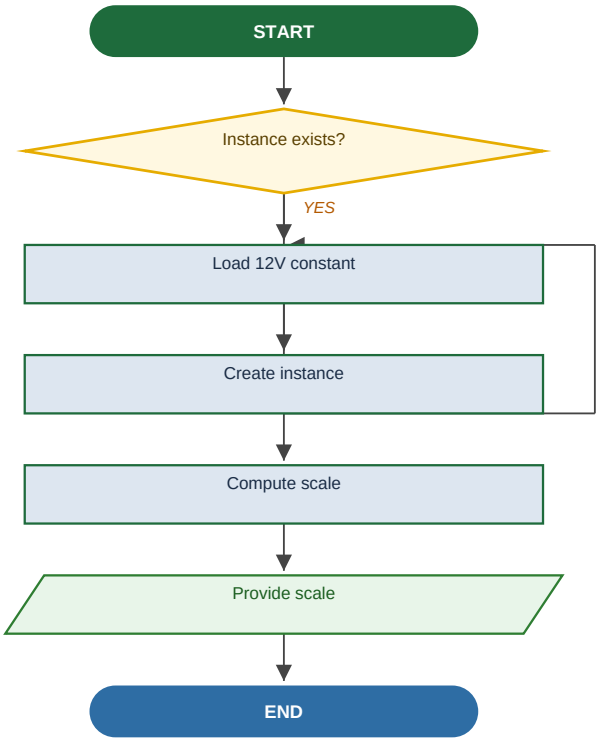
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-016	**Supply Voltage Assumption** (Class) | Module: ControlModule | SW-REQ-010

Diagram Type: **Function Flow**

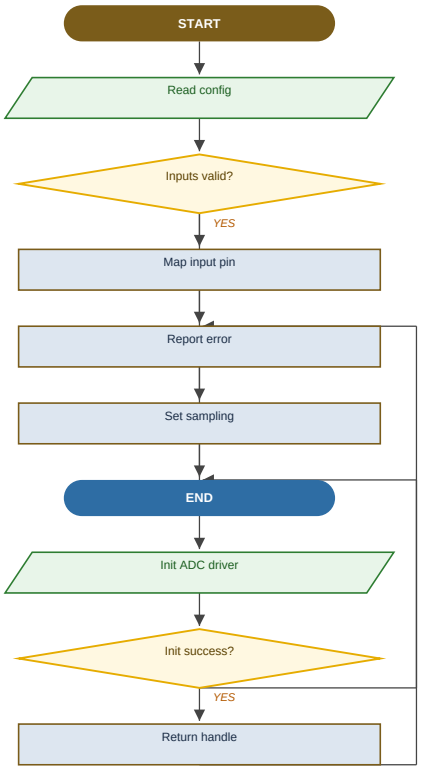
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-017	**ADC Channel Configuration** (Service) | Module: HALModule | SW-REQ-011

Diagram Type: **Function Flow**

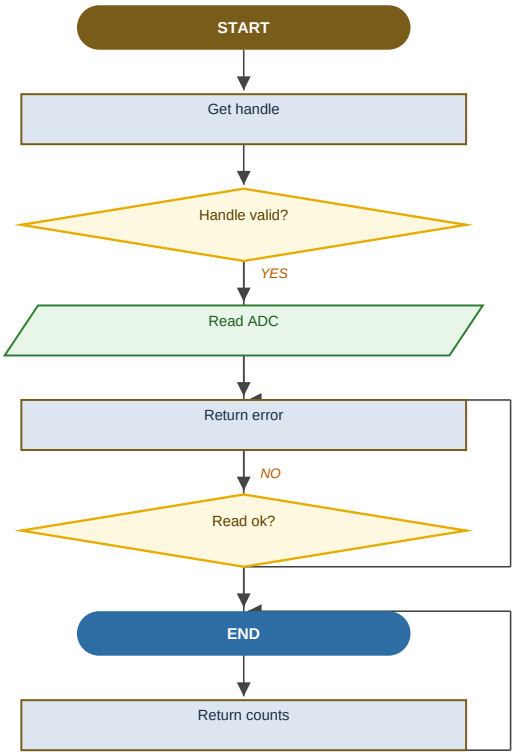
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-018	**ADC Read Service** (Service) | Module: HALModule | SW-REQ-012

Diagram Type: **Function Flow**

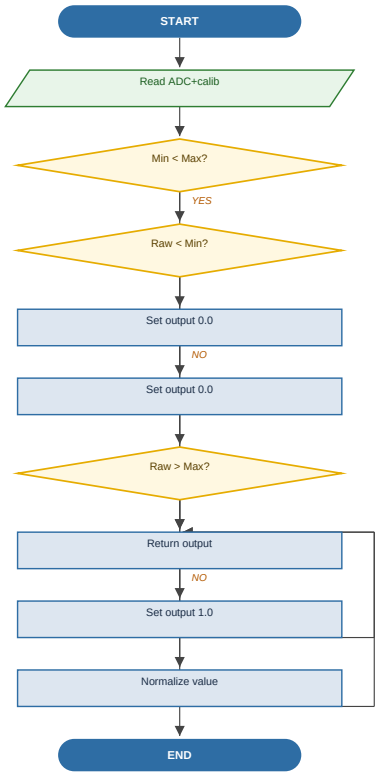
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-019	**Position Scaling Algorithm** (Algorithm) | Module: SignalProcessingModule | SW-REQ-013

Diagram Type: **Function Flow**

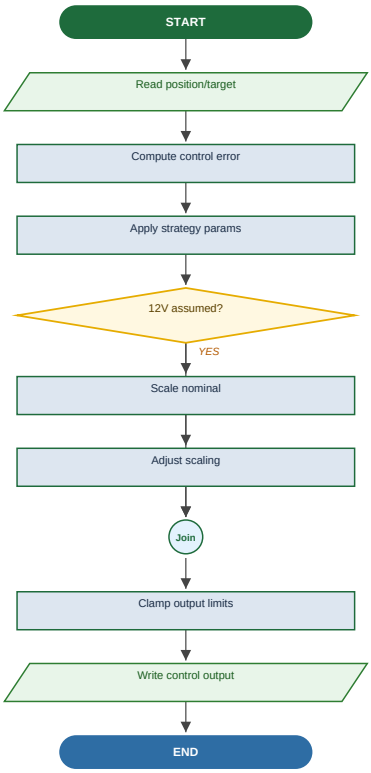
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-020	**Control Output Calculator** (Class) | Module: ControlModule | SW-REQ-014

Diagram Type: **Function Flow**

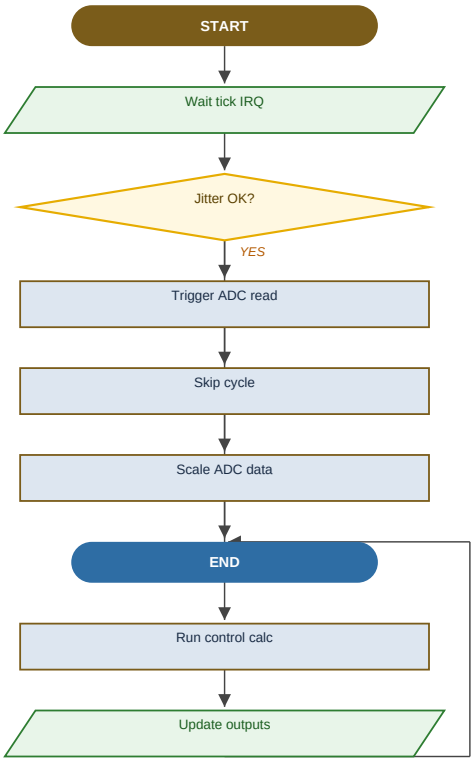
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-021	**Control Loop Scheduler** (Service) | Module: SchedulerModule | SW-REQ-015

Diagram Type: **Function Flow**

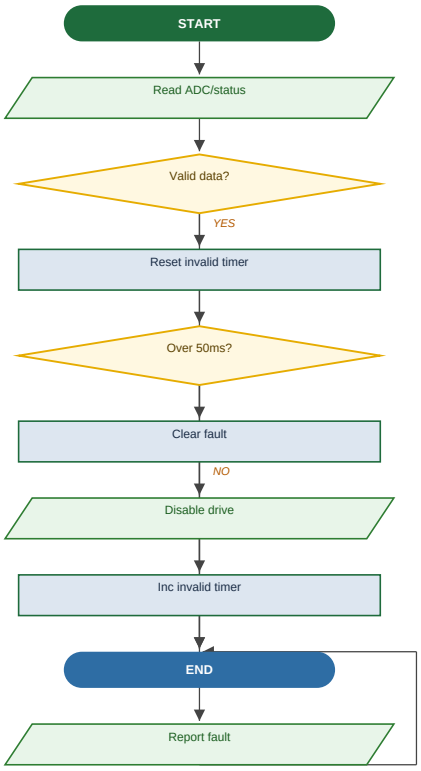
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-022	**Feedback Fault Monitor** (Class) | Module: SafetyModule | SW-REQ-016

Diagram Type: **Function Flow**

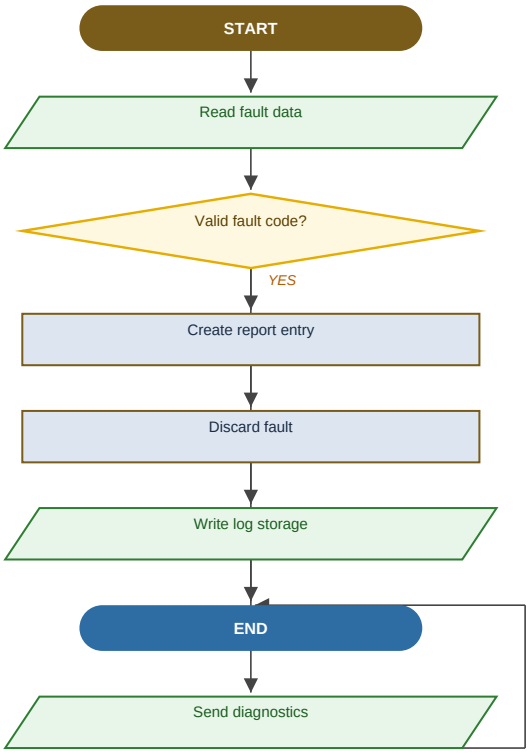
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-023	**Fault Reporter** (Service) | Module: SafetyModule | SW-REQ-016

Diagram Type: **Function Flow**

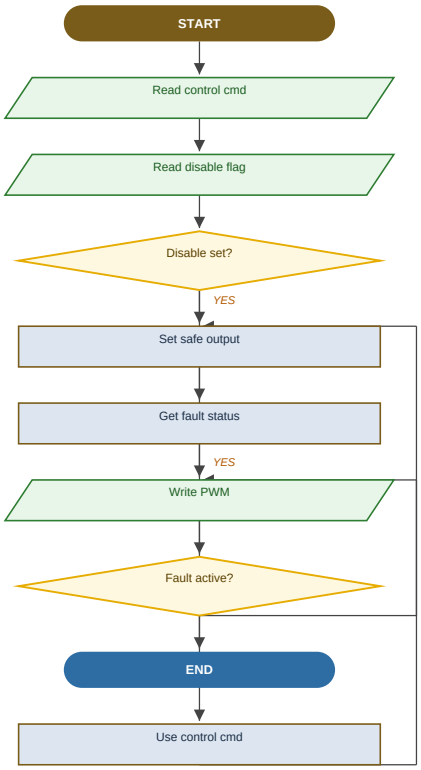
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-024	**Actuator Drive Output** (Service) | Module: HALModule | SW-REQ-014

Diagram Type: **Function Flow**

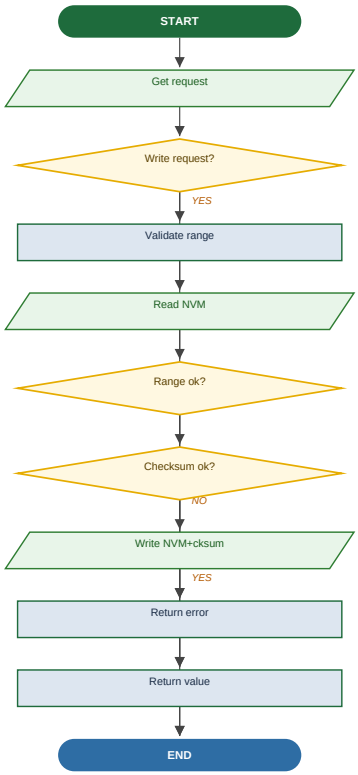
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-025	**ThresholdConfigRepository** (Class) | Module: ConfigModule | SW-REQ-017

Diagram Type: **Function Flow**

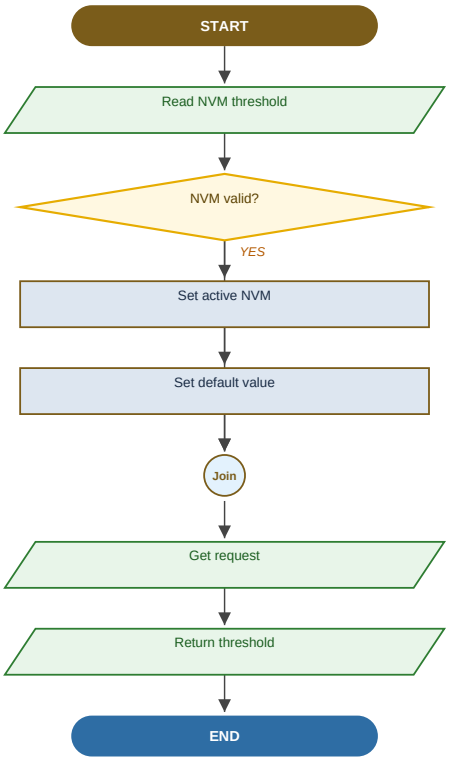
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-026	**ThresholdConfigService** (Service) | Module: ConfigModule | SW-REQ-017

Diagram Type: **Function Flow**

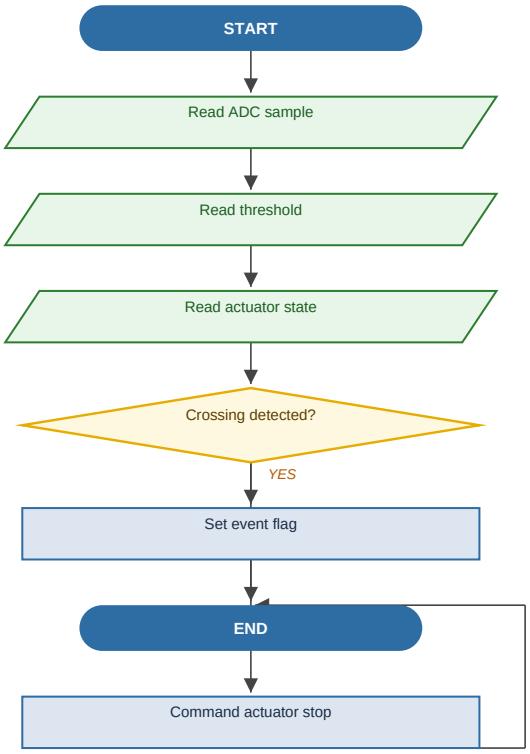
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-027	**ShippingPositionStopLogic** (Algorithm) | Module: ControlModule | SW-REQ-018

Diagram Type: **Function Flow**

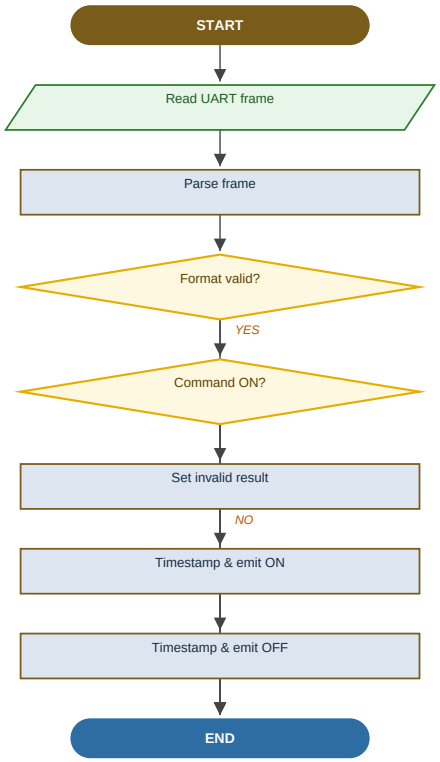
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-028	**UartCommandParser** (Service) | Module: UartModule | SW-REQ-019

Diagram Type: **Function Flow**

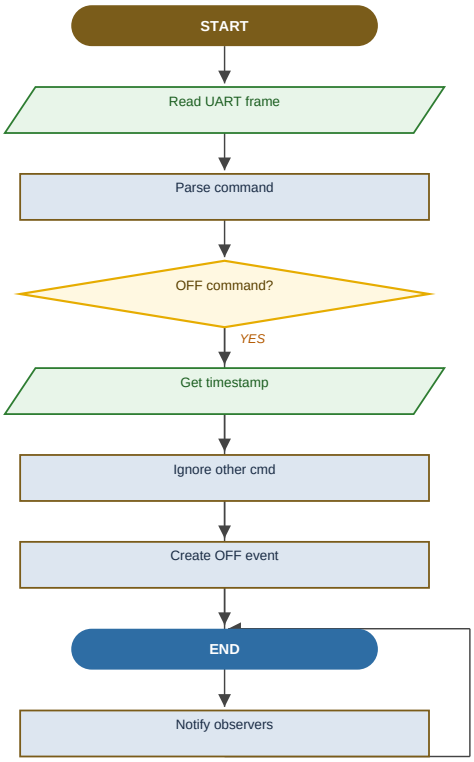
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-029	**UartCommandParser_OFF** (Service) | Module: UartModule | SW-REQ-020

Diagram Type: **Function Flow**

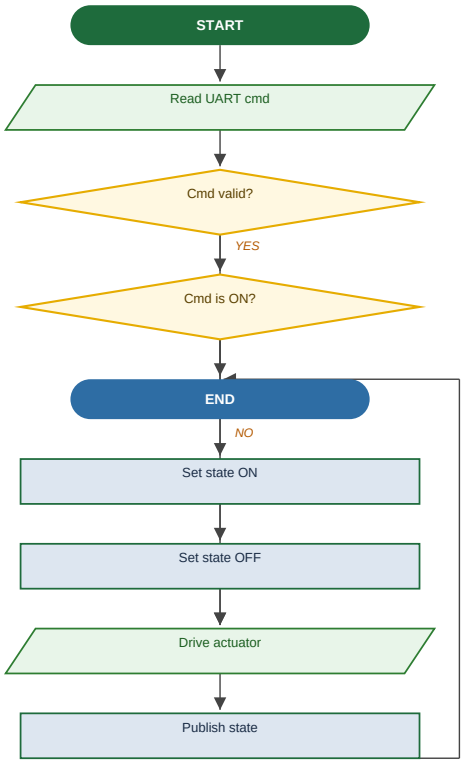
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-030	**ActuatorStateMachine** (Class) | Module: ControlModule | SW-REQ-019

Diagram Type: **State Flow**

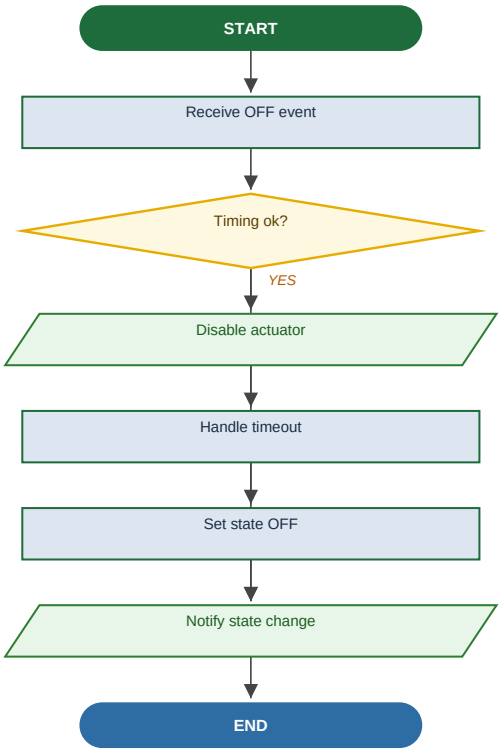
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-031	**ActuatorStateMachine_OFF** (Class) | Module: ControlModule | SW-REQ-020

Diagram Type: **State Flow**

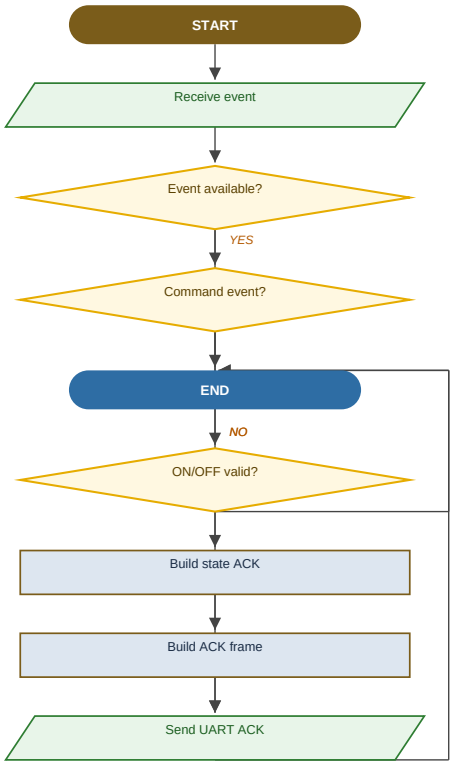
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-032	**UartAckService** (Service) | Module: UartModule | SW-REQ-021

Diagram Type: **Function Flow**

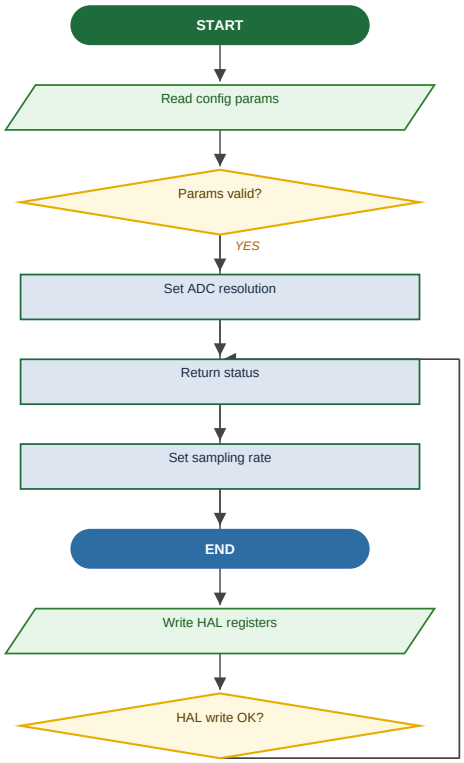
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-033	**AdcDriverConfig** (Class) | Module: AdcModule | SW-REQ-023

Diagram Type: **Function Flow**

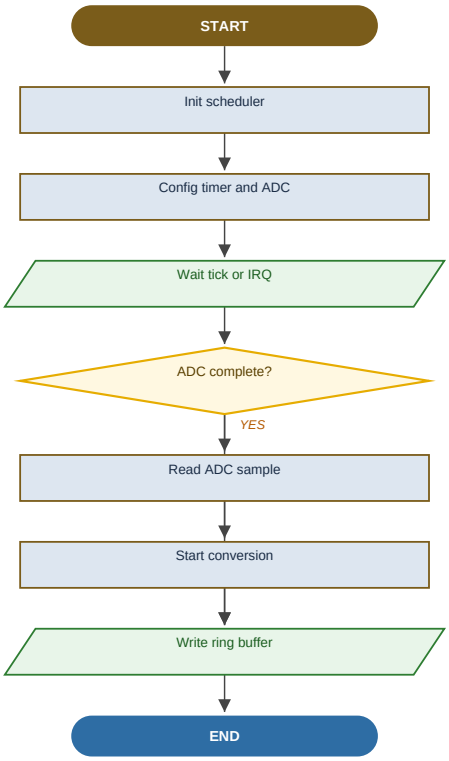
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-034	**AdcSamplingScheduler** (Service) | Module: AdcModule | SW-REQ-024

Diagram Type: **Function Flow**

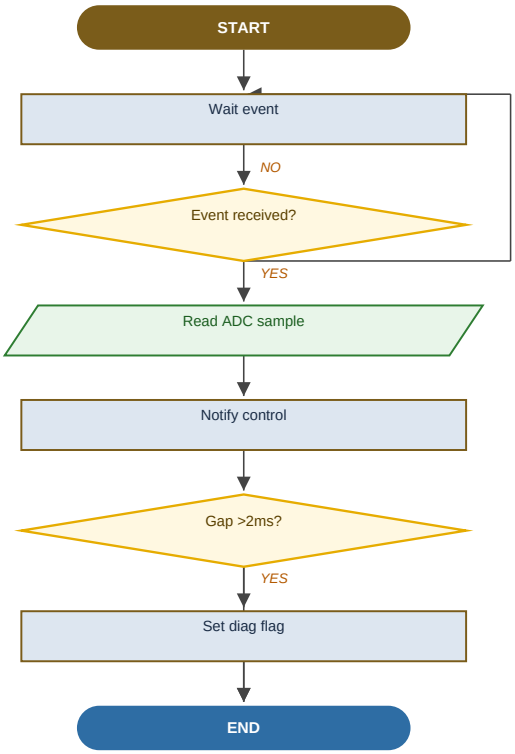
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-035	**AdcContinuousAcquisition** (Service) | Module: AdcModule | SW-REQ-022

Diagram Type: **Function Flow**

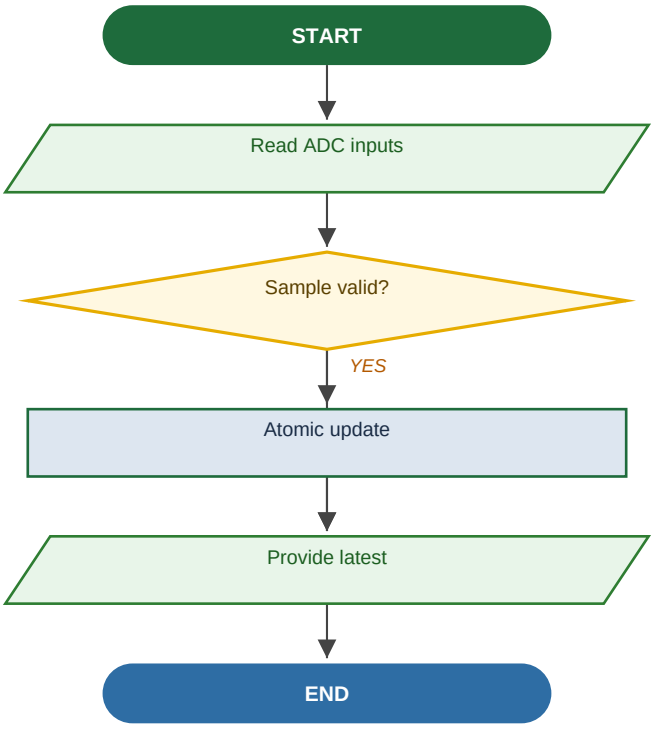
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-036	**ADC Sample Buffer** (Class) | Module: SensingModule | SW-REQ-025

Diagram Type: **Function Flow**

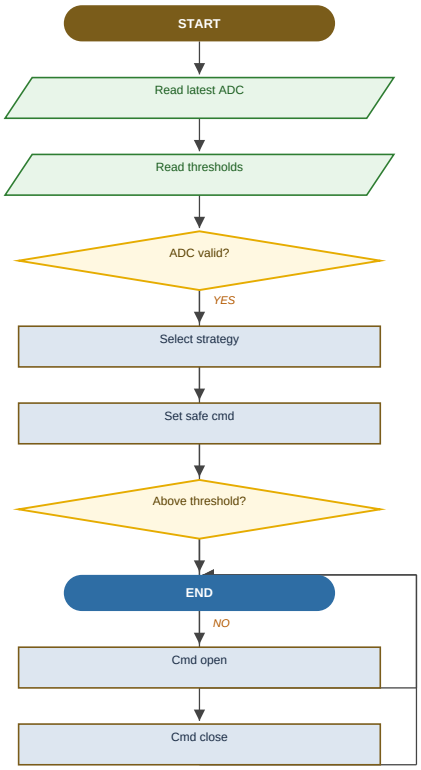
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-037	**Flap Control Computation** (Service) | Module: ControlModule | SW-REQ-026

Diagram Type: **Function Flow**

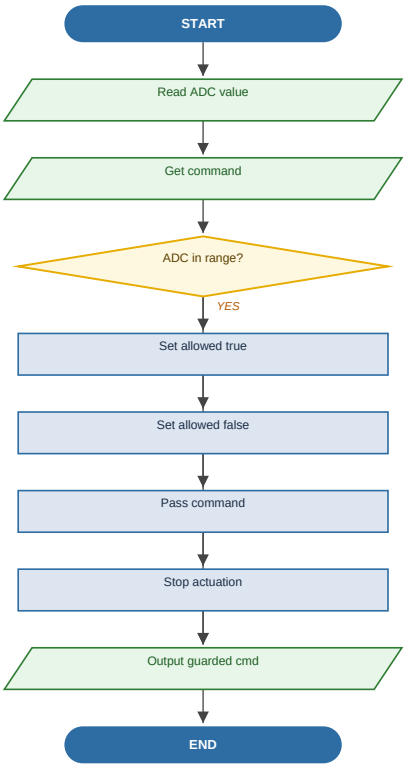
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-038	**Actuation Range Guard** (Algorithm) | Module: SafetyModule | SW-REQ-027

Diagram Type: **Function Flow**

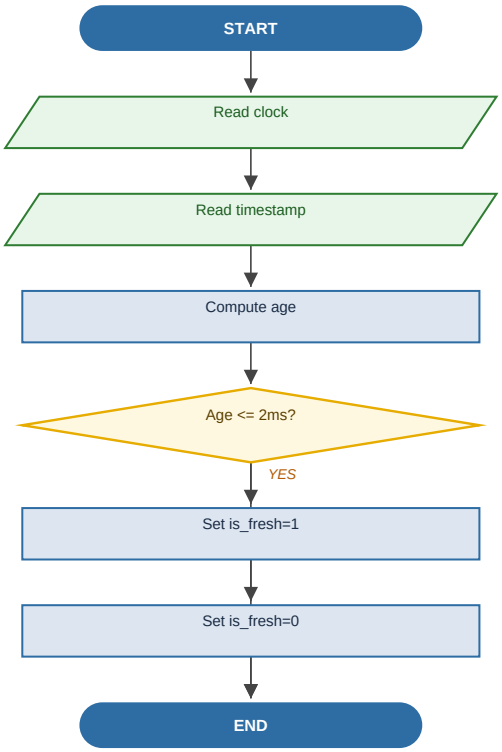
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-039	**Real-Time ADC Freshness Check** (Algorithm) | Module: SensingModule | SW-REQ-028

Diagram Type: **Function Flow**

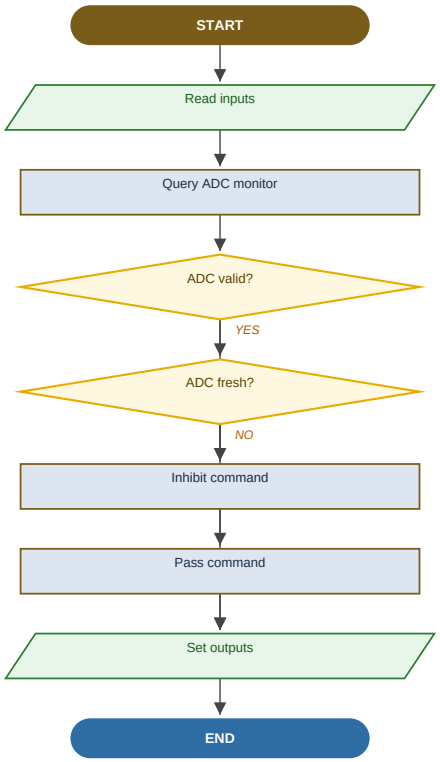
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-040	**Open-Loop Inhibit Gate** (Service) | Module: SafetyModule | SW-REQ-029

Diagram Type: **Function Flow**

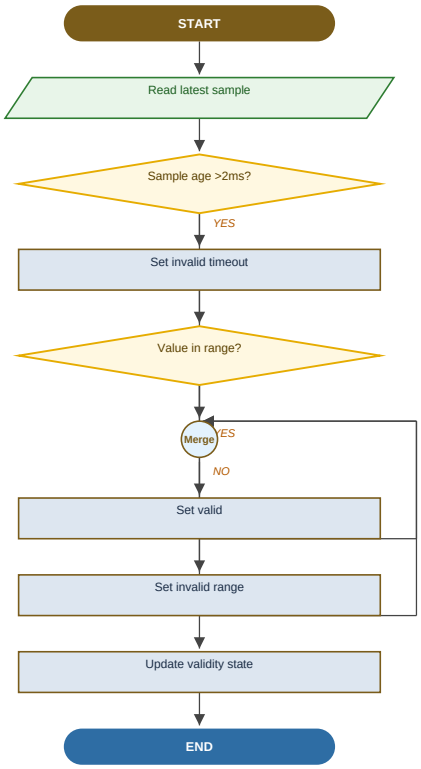
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-041	**ADC Validity Monitor** (Service) | Module: SensingModule | SW-REQ-030

Diagram Type: **Function Flow**

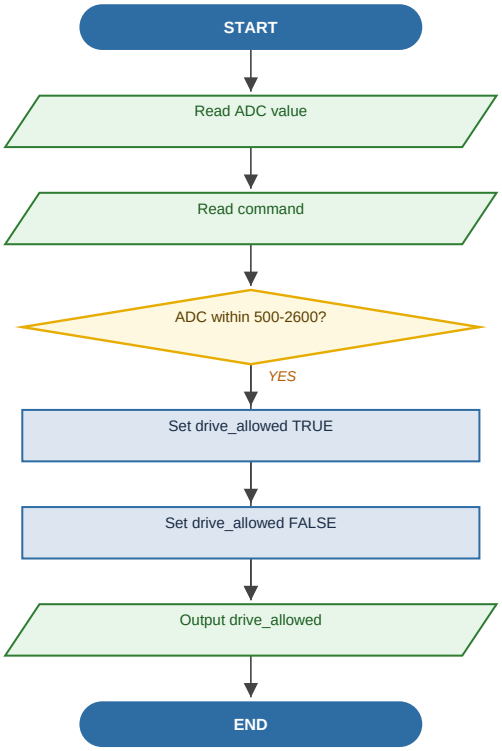
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-042	**Drive Permission Check** (Algorithm) | Module: SafetyModule | SW-REQ-031

Diagram Type: **Function Flow**

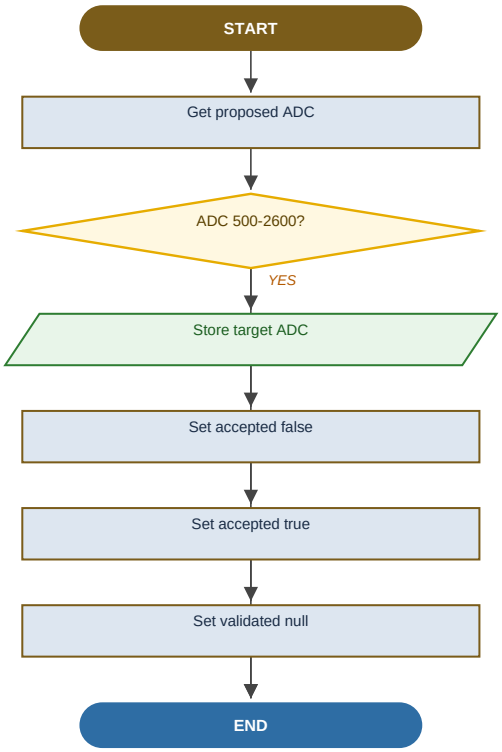
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-043	**Target ADC Validator** (Service) | Module: ConfigModule | SW-REQ-032

Diagram Type: **Function Flow**

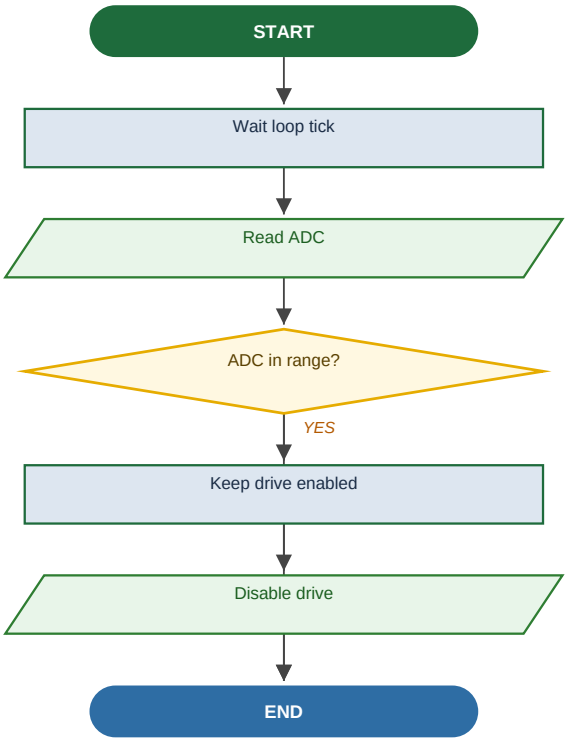
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-044	**ActuatorSafetyMonitor** (Class) | Module: ControlLoopModule | SW-REQ-033

Diagram Type: **Function Flow**

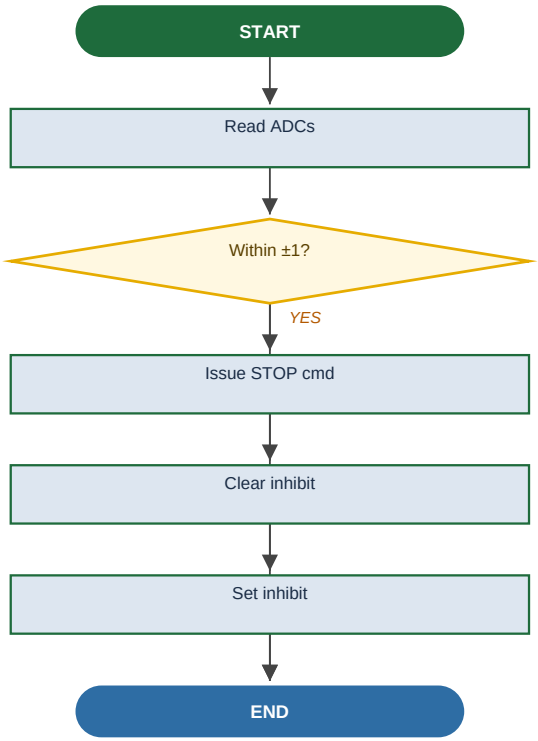
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-045	**ActuatorTargetStopController** (Class) | Module: ControlLoopModule | SW-REQ-034

Diagram Type: **Function Flow**

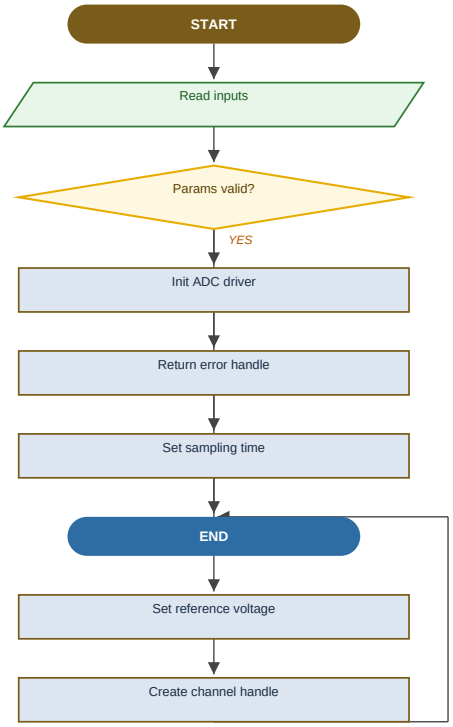
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-046	**SupplyVoltageADCChannelConfig** (Service) | Module: HardwareAbstractionLayer | SW-REQ-035

Diagram Type: **Function Flow**

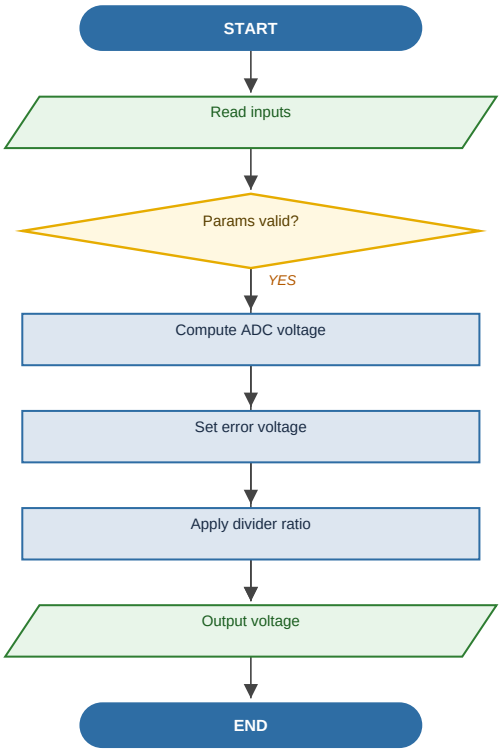
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-047	**SupplyVoltageScaler** (Algorithm) | Module: VoltageMonitorModule | SW-REQ-036

Diagram Type: **Function Flow**

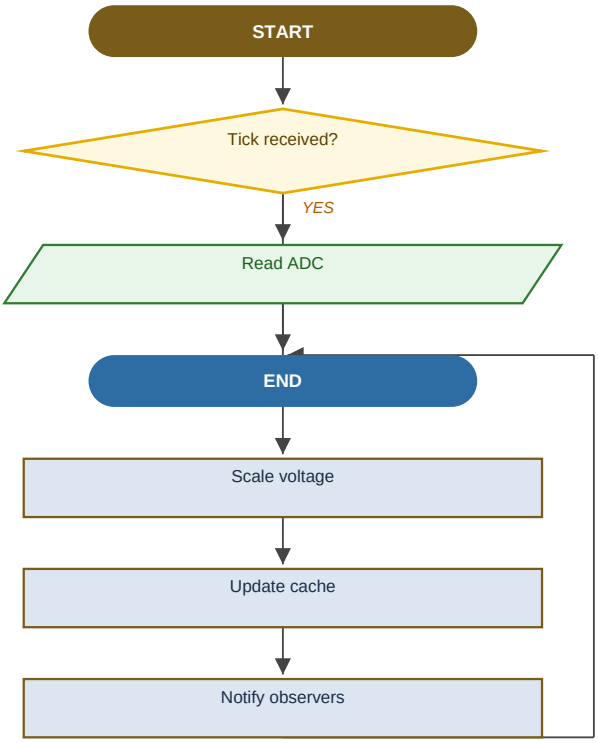
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-048	**SupplyVoltageSampler** (Service) | Module: VoltageMonitorModule | SW-REQ-037

Diagram Type: **Function Flow**

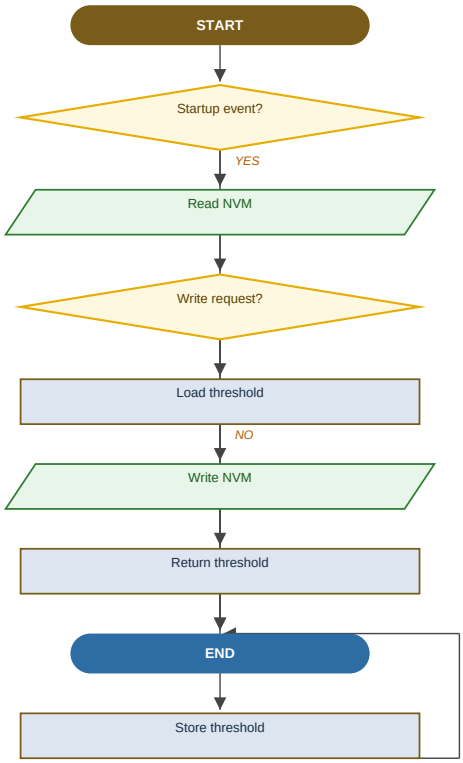
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-049	**OvervoltageThresholdStore** (Service) | Module: ConfigModule | SW-REQ-038

Diagram Type: **Function Flow**

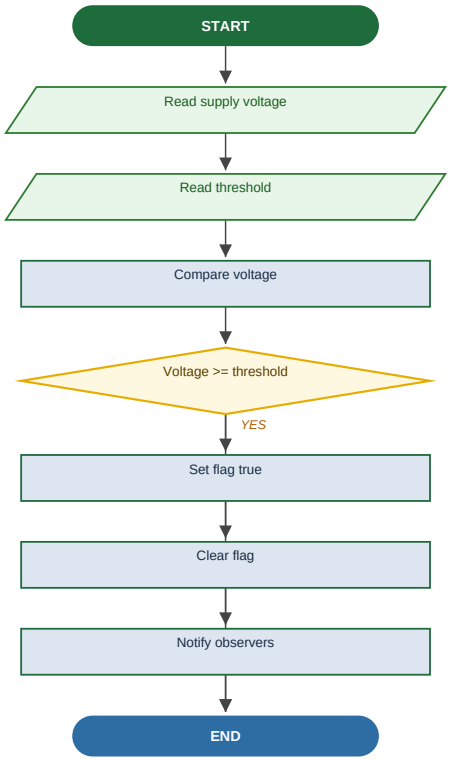
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-050	**OvervoltageDetector** (Class) | Module: VoltageMonitorModule | SW-REQ-039

Diagram Type: **Function Flow**

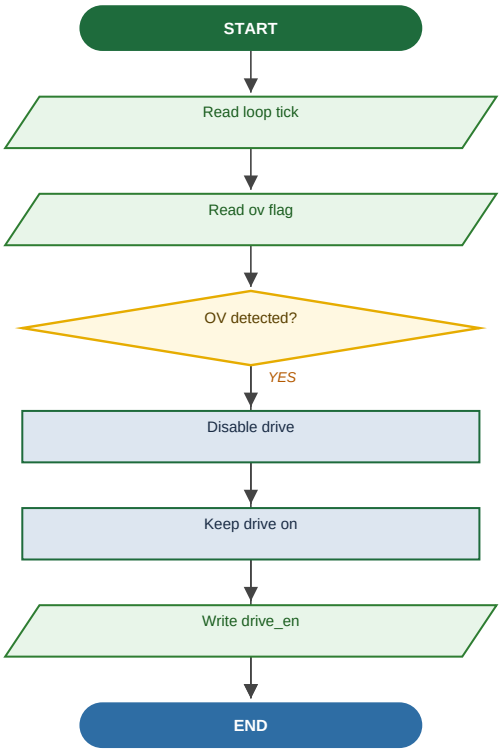
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-051	**OvervoltageDriveDisabler** (Class) | Module: ControlLoopModule | SW-REQ-040

Diagram Type: **Function Flow**

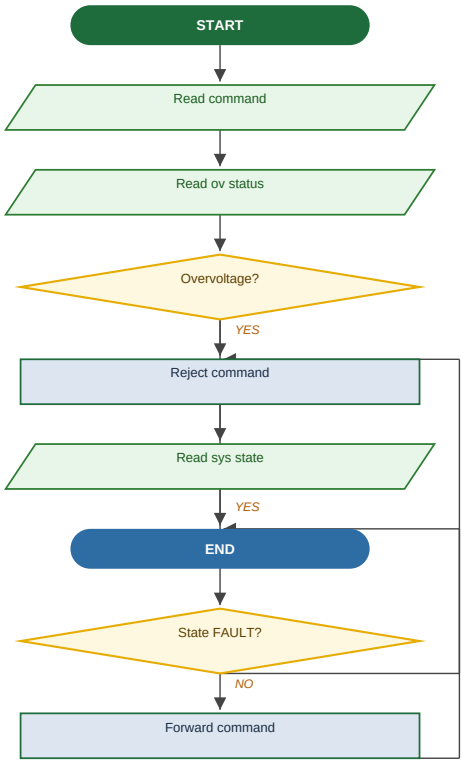
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-052	**ActuatorCommandGate** (Class) | Module: ControlModule | SW-REQ-041

Diagram Type: **Function Flow**

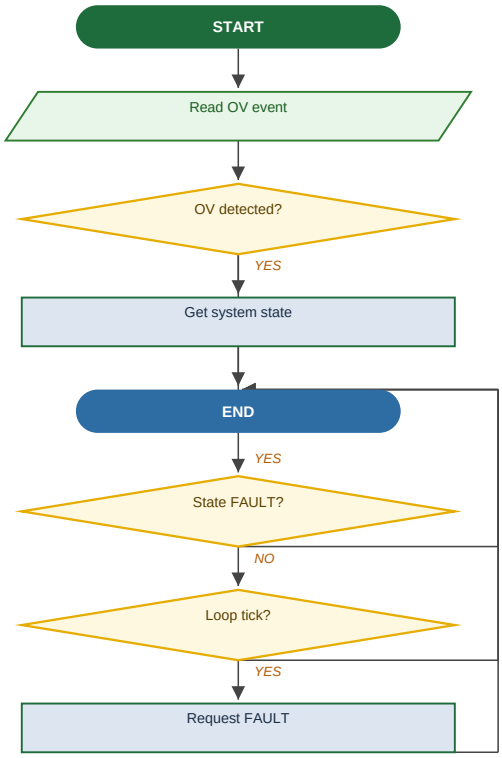
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-053	**FaultTransitionHandler** (Class) | Module: FaultManagementModule | SW-REQ-042

Diagram Type: **Function Flow**

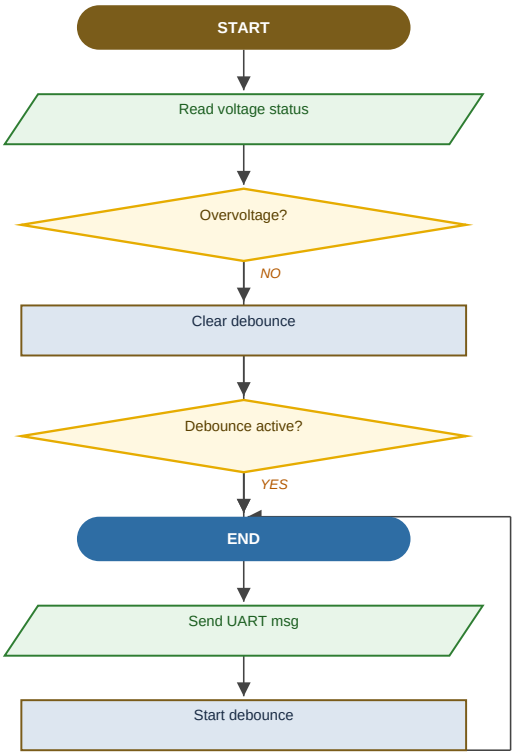
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-054	**OvervoltageUartNotifier** (Service) | Module: DiagnosticsModule | SW-REQ-043

Diagram Type: **Function Flow**

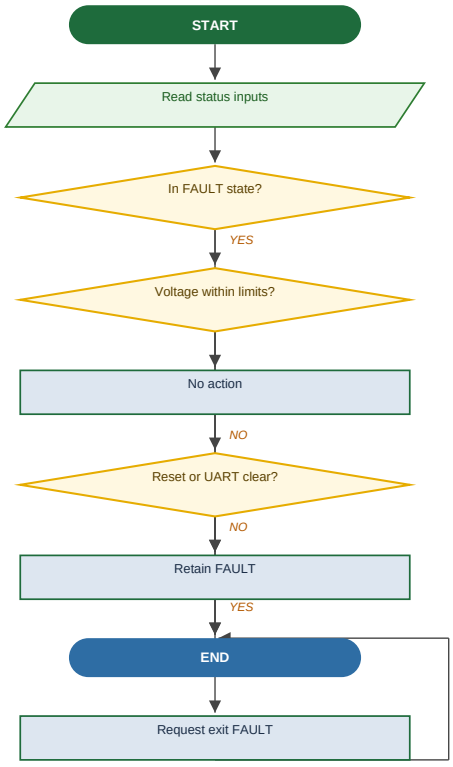
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-055	**FaultStateRetentionLogic** (Class) | Module: FaultManagementModule | SW-REQ-044

Diagram Type: **Function Flow**

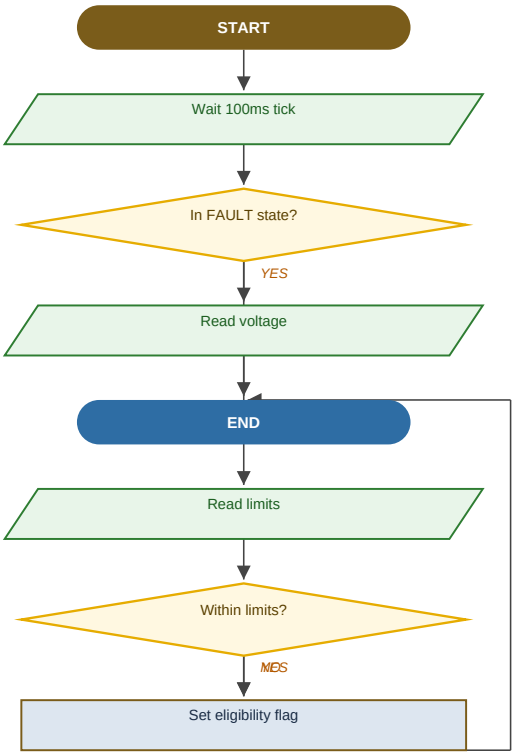
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-056	**FaultExitVoltageChecker** (Service) | Module: FaultManagementModule | SW-REQ-045

Diagram Type: **Function Flow**

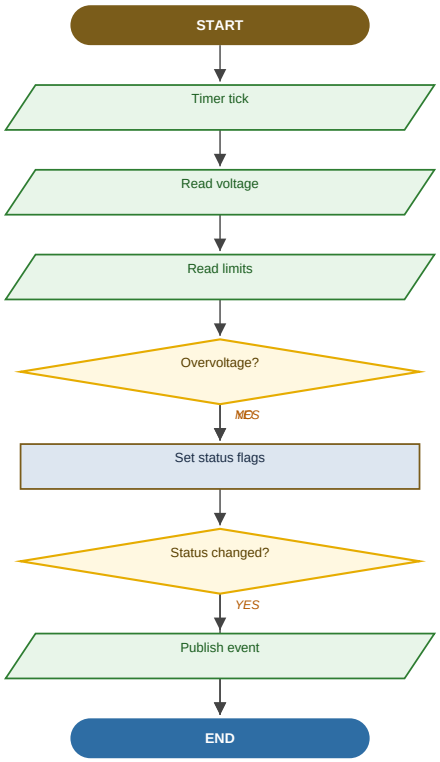
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-057	**OvervoltageMonitor** (Service) | Module: PowerSafetyModule | SW-REQ-046

Diagram Type: **Function Flow**

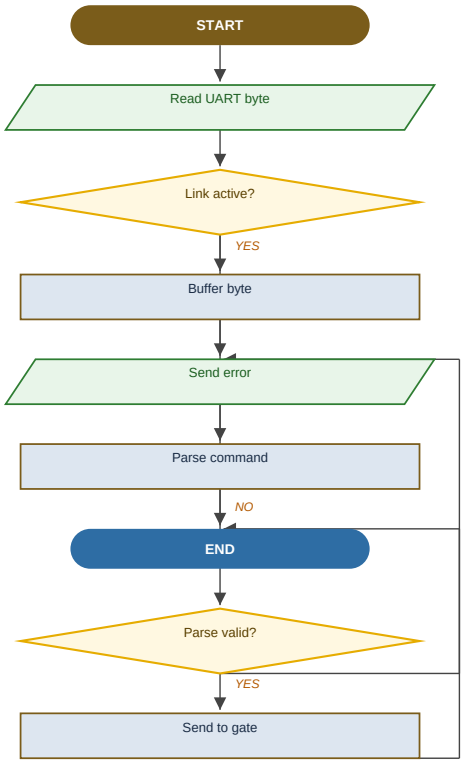
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-058	**UartActuatorCommandInterface** (Service) | Module: UartModule | SW-REQ-047

Diagram Type: **Function Flow**

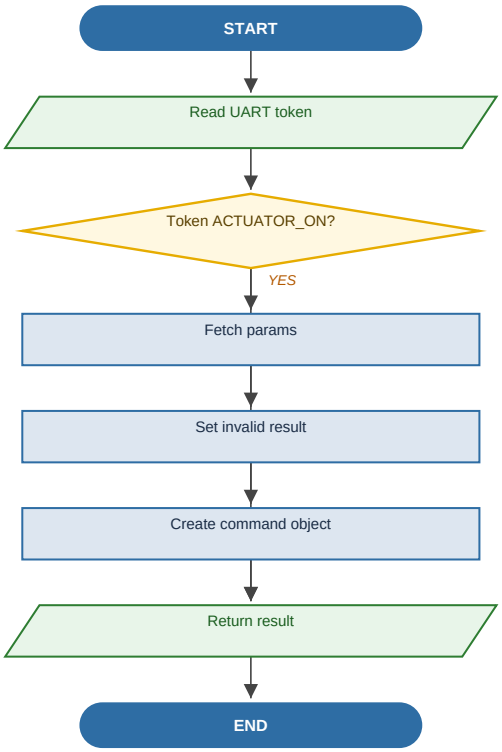
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-059	**ActuatorOnCommandParser** (Algorithm) | Module: UartModule | SW-REQ-048

Diagram Type: **Function Flow**

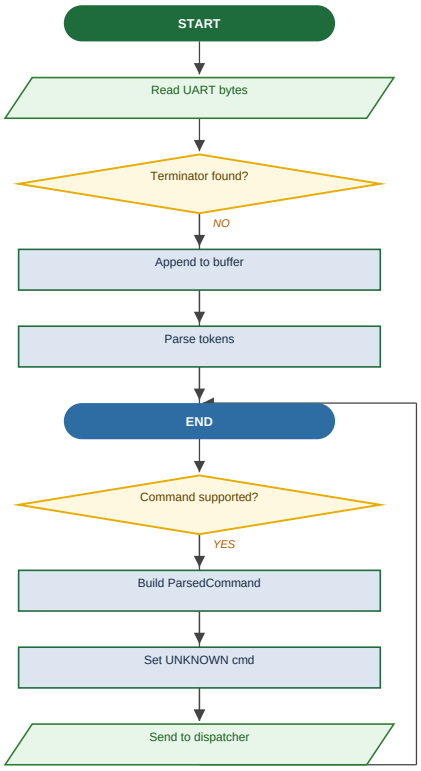
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-060	**UART Command Parser** (Class) | Module: UartCommModule | SW-REQ-049

Diagram Type: **Function Flow**

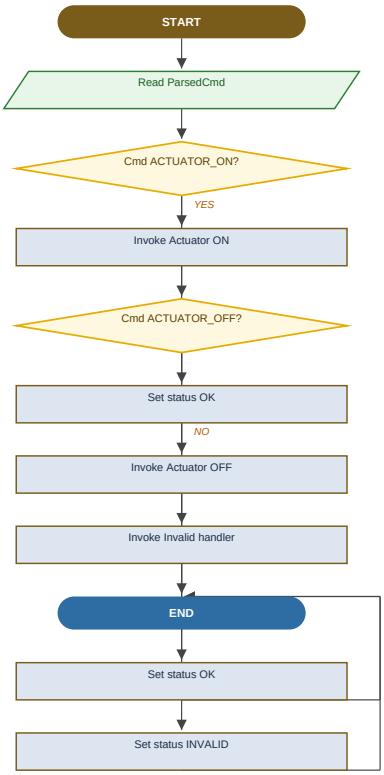
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-061	**Command Dispatcher** (Service) | Module: CommandModule | SW-REQ-050

Diagram Type: **Function Flow**

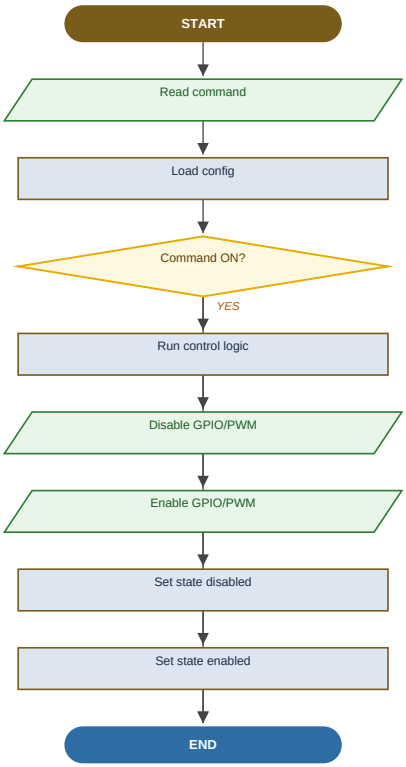
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-062	**Actuator Controller** (Service) | Module: ActuatorModule | SW-REQ-051

Diagram Type: **Function Flow**

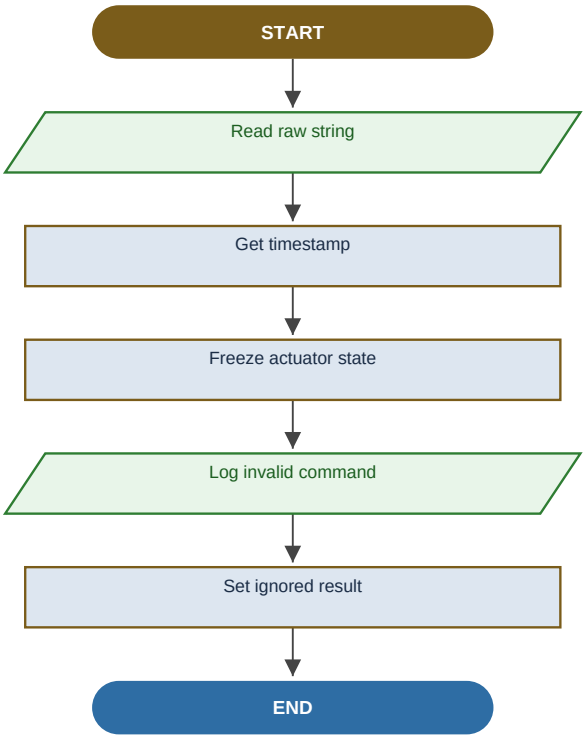
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-063	**Invalid Command Handler** (Service) | Module: CommandModule | SW-REQ-052

Diagram Type: **Function Flow**

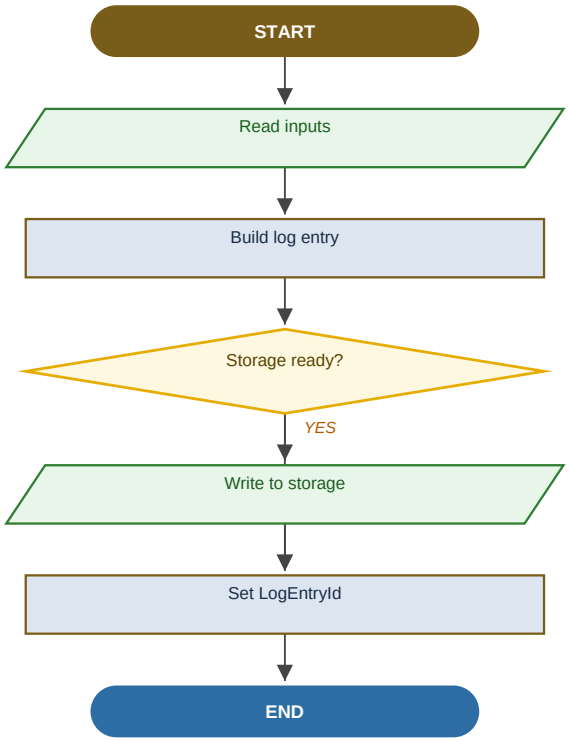
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-064	**Invalid Command Logger** (Service) | Module: LoggingModule | SW-REQ-053

Diagram Type: **Function Flow**

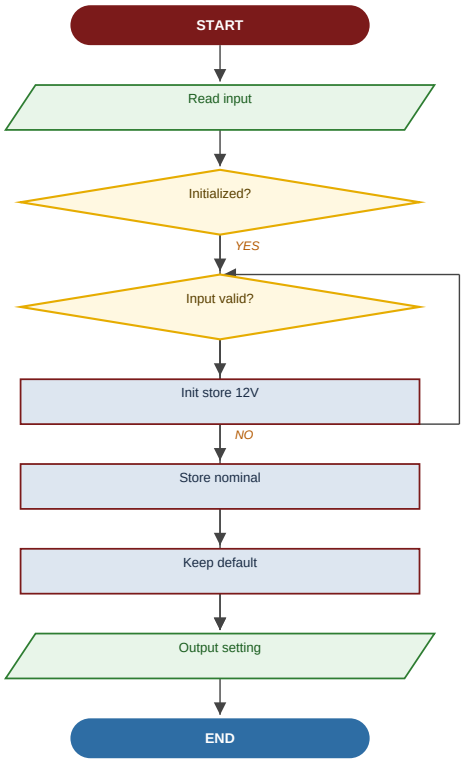
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-065	**Voltage Configuration Store** (Database) | Module: ConfigModule | SW-REQ-054

Diagram Type: **Data Flow**

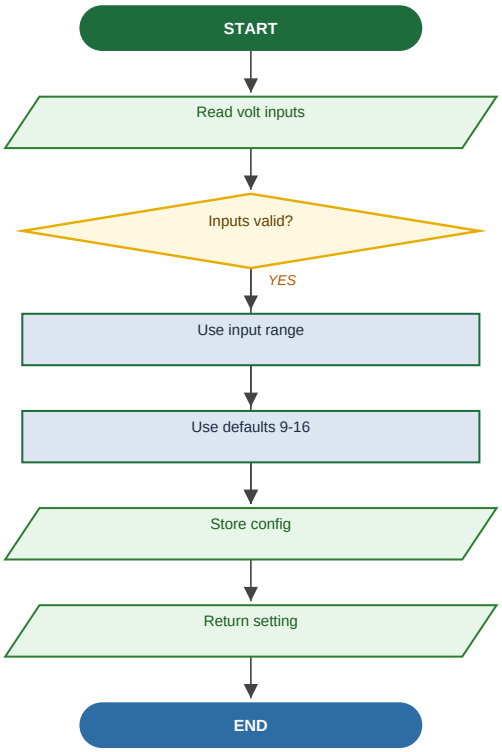
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-066	**Voltage Range Configuration** (Class) | Module: ConfigModule | SW-REQ-055

Diagram Type: **Function Flow**

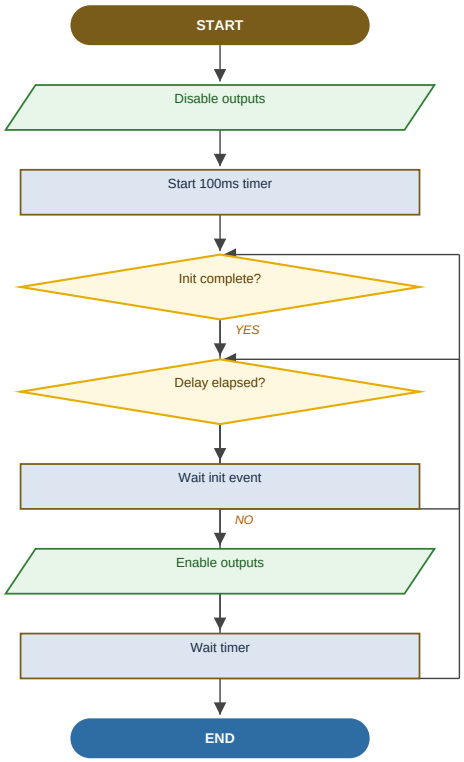
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-067	**Startup Safety Manager** (Service) | Module: StartupModule | SW-REQ-056

Diagram Type: **Function Flow**

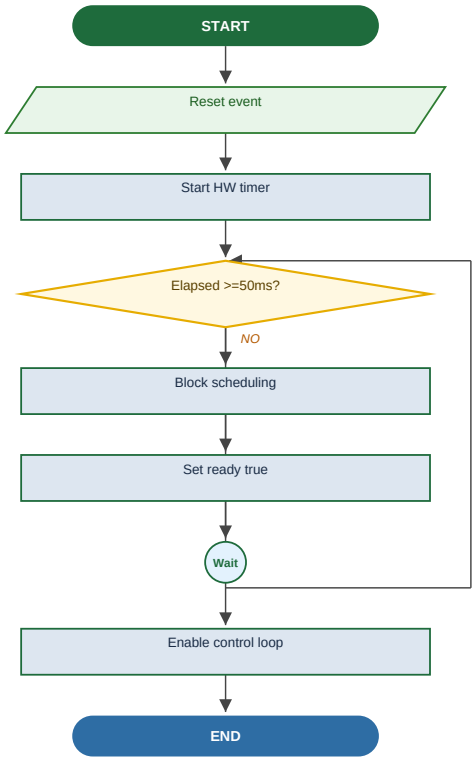
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-068	**StartupDelayManager** (Class) | Module: StartupModule | SW-REQ-057

Diagram Type: **State Flow**

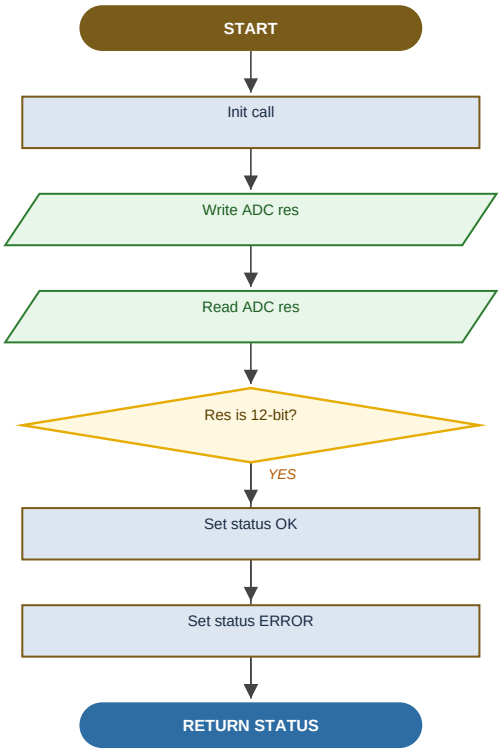
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-069	**ADCResolutionConfigurator** (Service) | Module: ADCModule | SW-REQ-058

Diagram Type: **Function Flow**

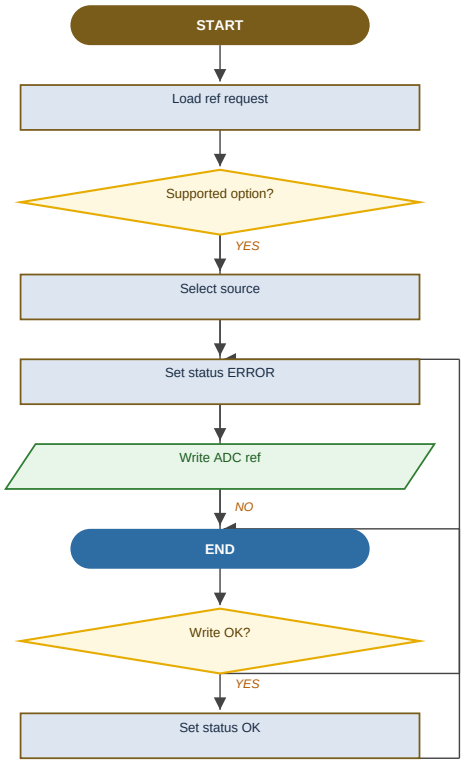
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-070	**ADCReferenceConfigurator** (Service) | Module: ADCModule | SW-REQ-059

Diagram Type: **Function Flow**

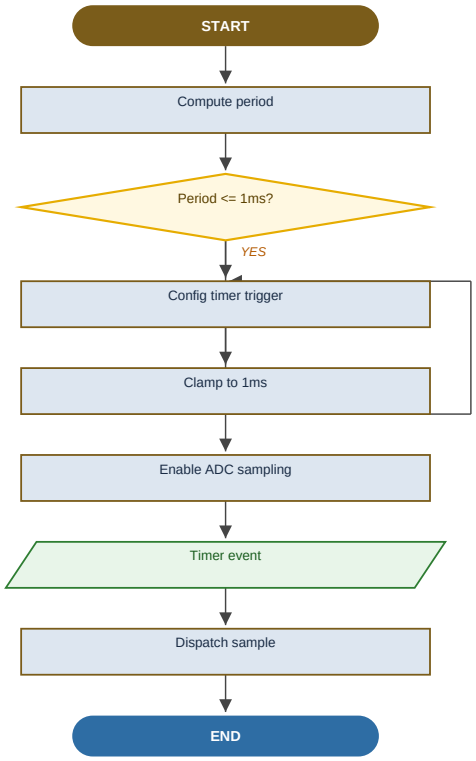
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-071	**ADCSamplingScheduler** (Service) | Module: ADCModule | SW-REQ-060

Diagram Type: **Function Flow**

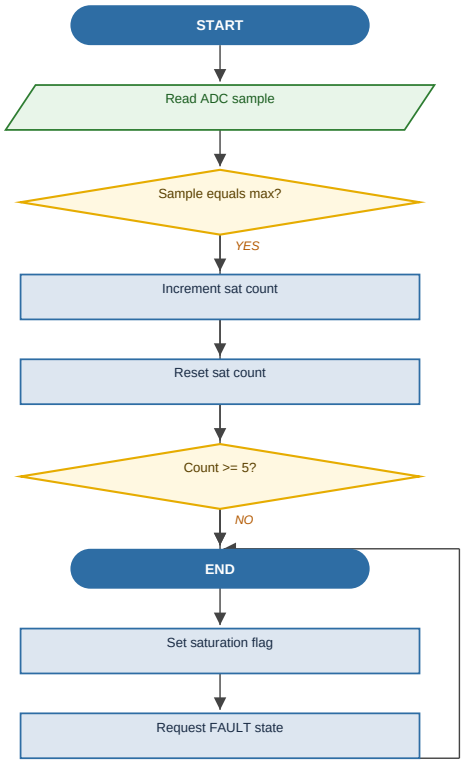
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-072	**ADCSaturationDetector** (Algorithm) | Module: ADCModule | SW-REQ-061

Diagram Type: **Function Flow**

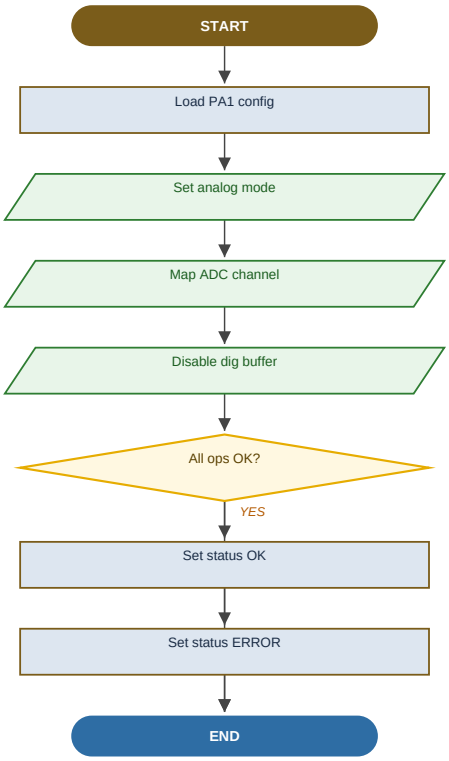
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-073	**ADCPinInitializer** (Service) | Module: BoardSupportModule | SW-REQ-062

Diagram Type: **Function Flow**

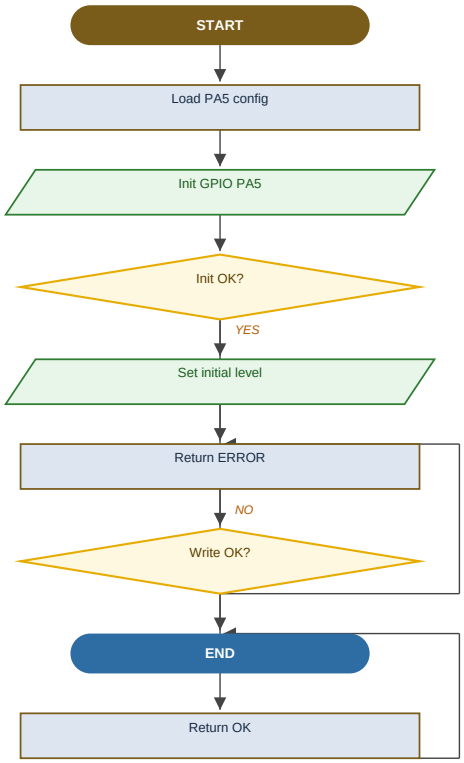
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-074	**ActuatorGPIOInitializer** (Service) | Module: BoardSupportModule | SW-REQ-063

Diagram Type: **Function Flow**

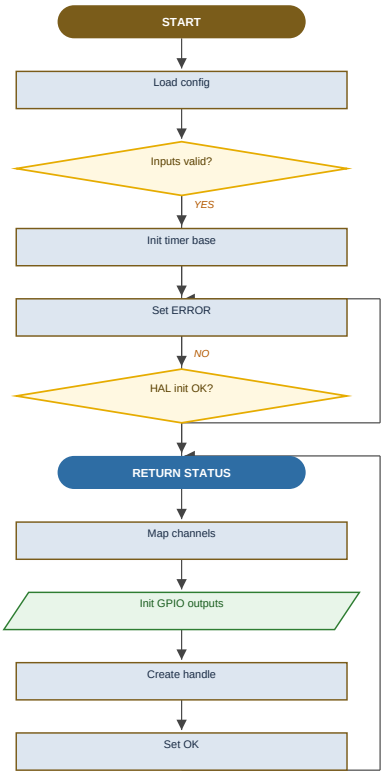
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-075	**PWMTimerInitializer** (Service) | Module: ActuatorControlModule | SW-REQ-064

Diagram Type: **Function Flow**

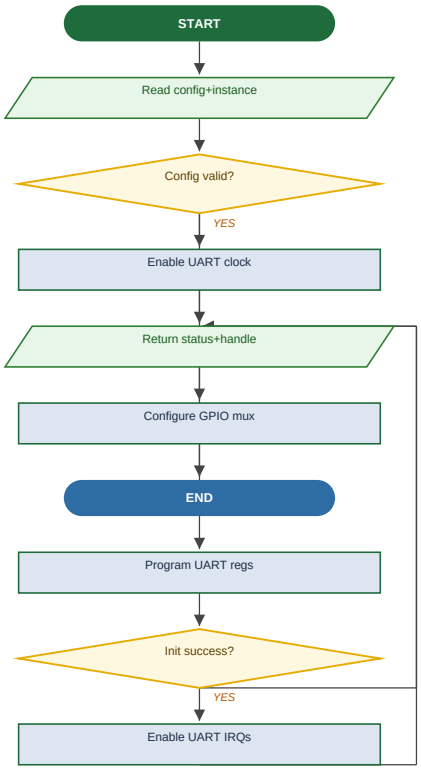
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-076	**UART Peripheral Initializer** (Class) | Module: HAL | SW-REQ-065

Diagram Type: **Function Flow**

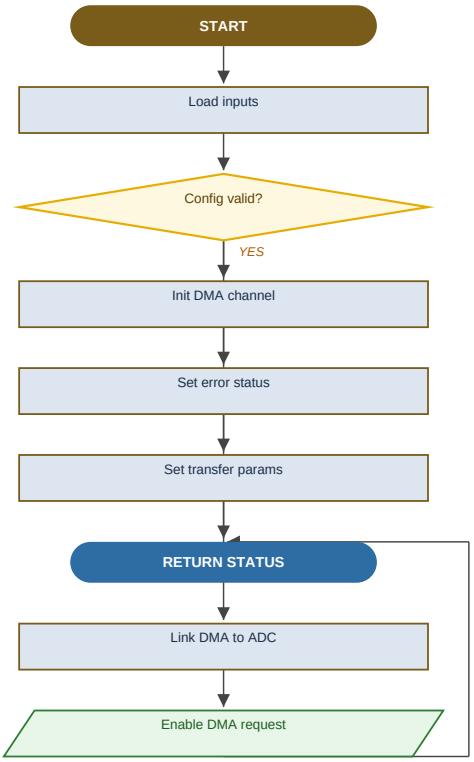
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-077	**ADC DMA Configuration Service** (Service) | Module: Driver | SW-REQ-066

Diagram Type: **Function Flow**

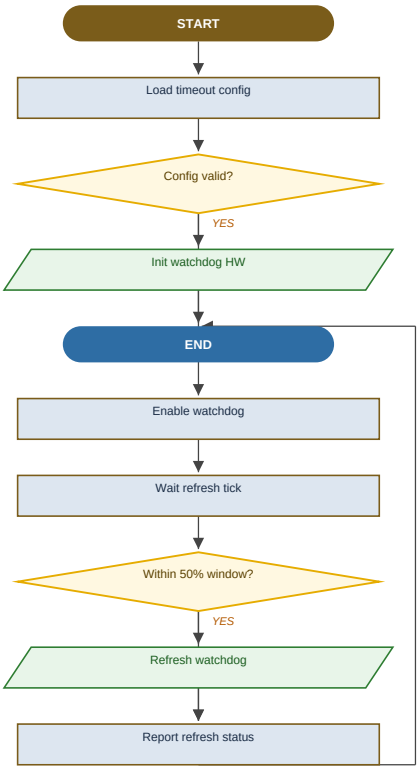
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-078	**Independent Watchdog Service** (Service) | Module: Middleware | SW-REQ-067

Diagram Type: **Function Flow**

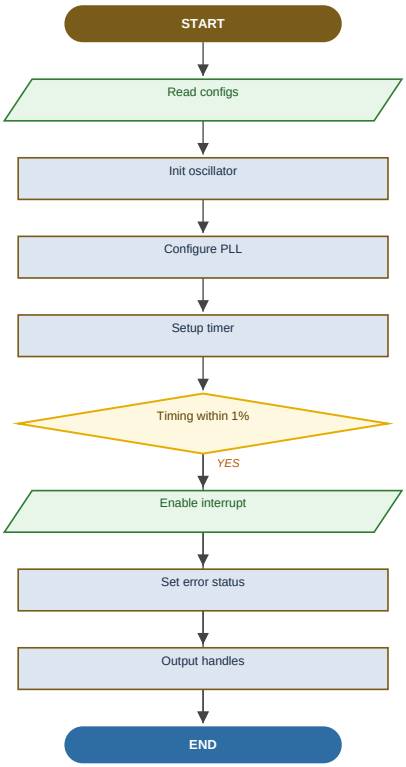
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-079	**Clock and Timer Configuration** (Service) | Module: HAL | SW-REQ-068

Diagram Type: **Function Flow**

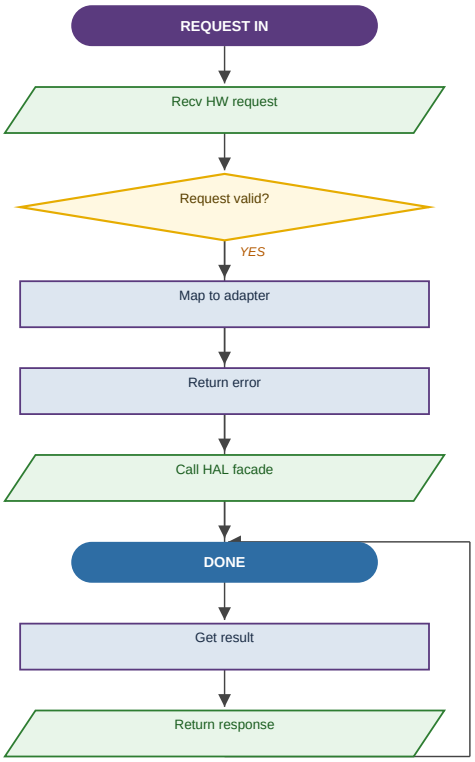
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-080	**Layered Architecture Enforcement Interface** (Interface) | Module: Application | SW-REQ-069

Diagram Type: **Request Response**

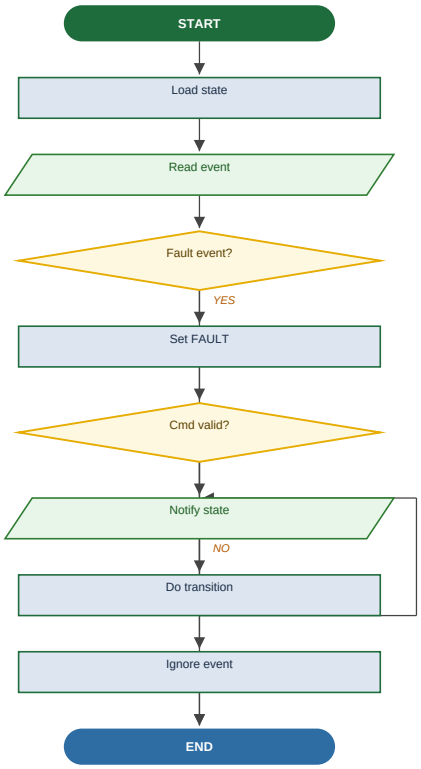
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-081	**System State Machine** (Class) | Module: Application | SW-REQ-070

Diagram Type: **State Flow**

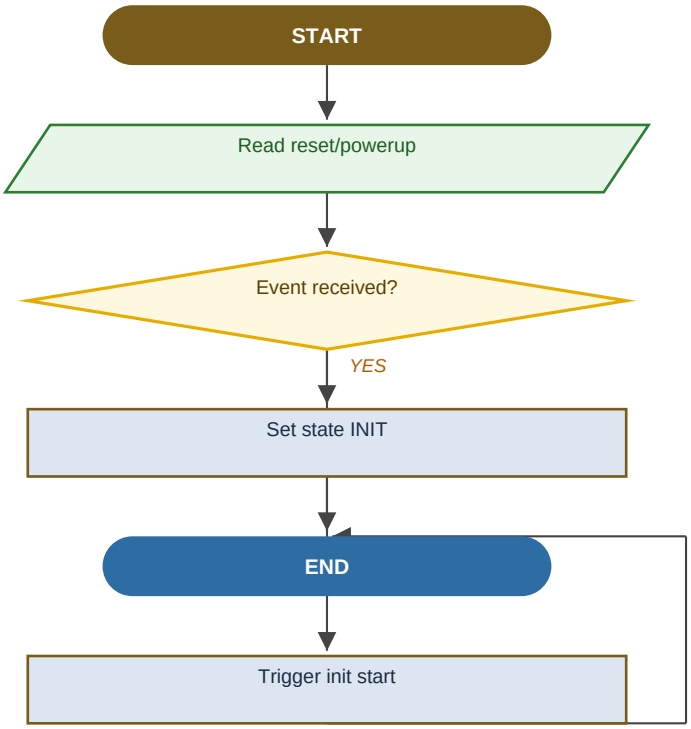
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-082	**Reset Initialization Handler** (Service) | Module: Application | SW-REQ-071

Diagram Type: **Function Flow**

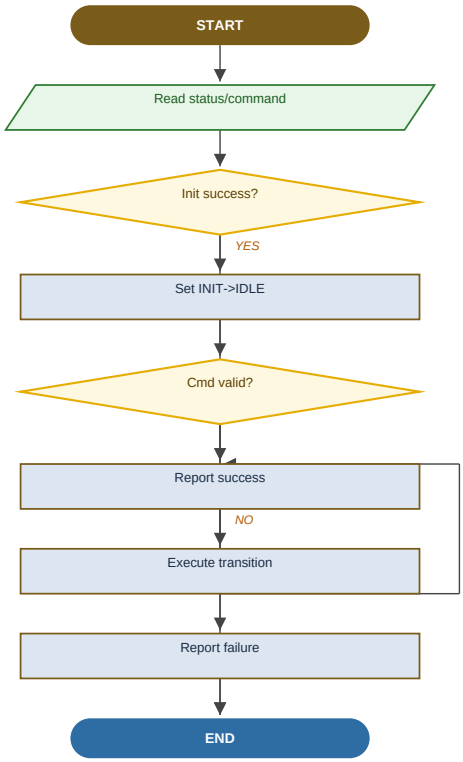
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-083	**Nominal Transition Controller** (Service) | Module: Application | SW-REQ-072

Diagram Type: **Function Flow**

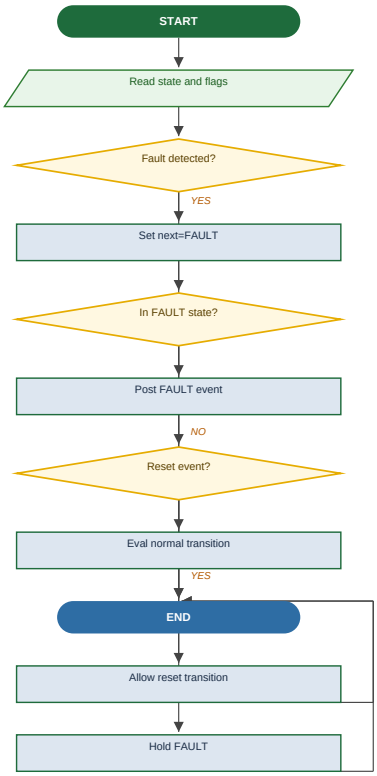
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-084	**FaultStateManager** (Class) | Module: StateManagementModule | SW-REQ-073

Diagram Type: **State Flow**

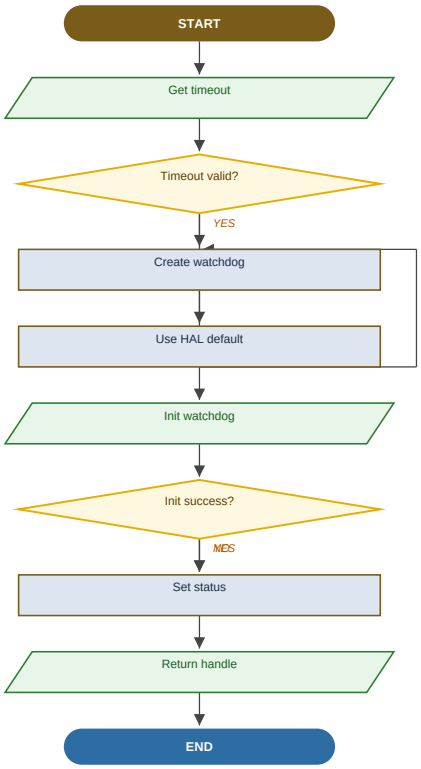
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-085	**HardwareWatchdogInitializer** (Service) | Module: WatchdogModule | SW-REQ-074

Diagram Type: **Function Flow**

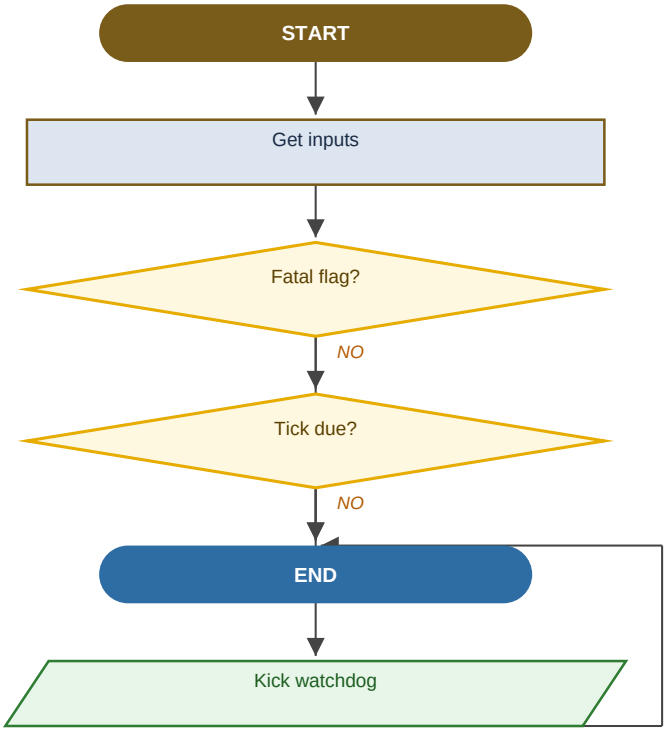
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-086	**WatchdogServicer** (Service) | Module: WatchdogModule | SW-REQ-075

Diagram Type: **Function Flow**

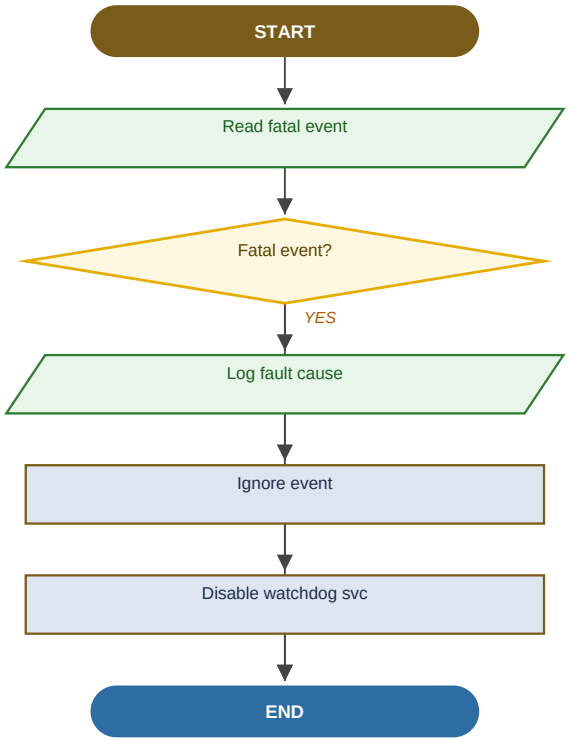
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-087	**FatalMalfunctionResetHandler** (Service) | Module: FaultHandlingModule | SW-REQ-076

Diagram Type: **Function Flow**

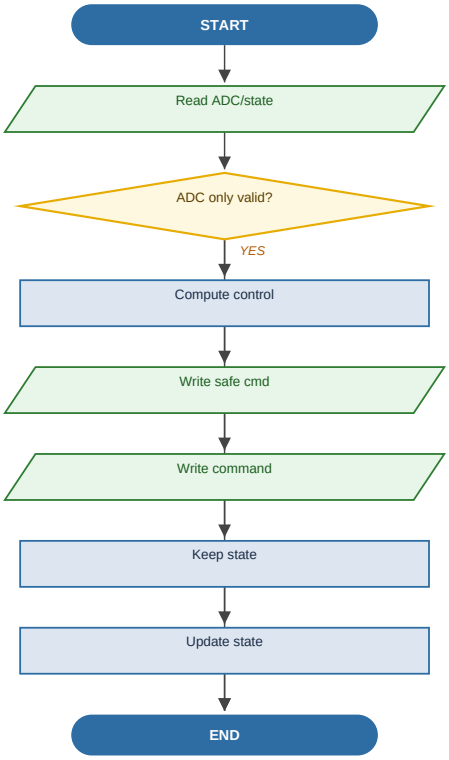
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-088	**ADCOnlyControlAlgorithm** (Algorithm) | Module: ControlModule | SW-REQ-077

Diagram Type: **Function Flow**

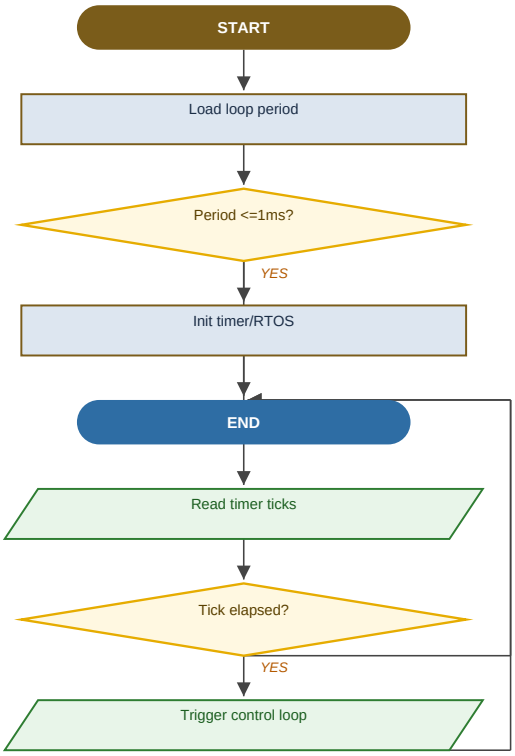
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-089	**ControlLoopScheduler** (Service) | Module: ControlModule | SW-REQ-078

Diagram Type: **Function Flow**

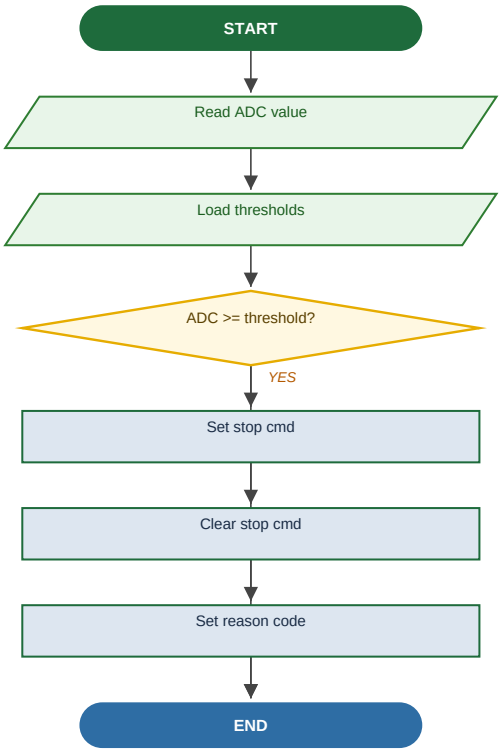
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-090	**ThresholdStopEvaluator** (Class) | Module: SafetyControlModule | SW-REQ-079

Diagram Type: **Function Flow**

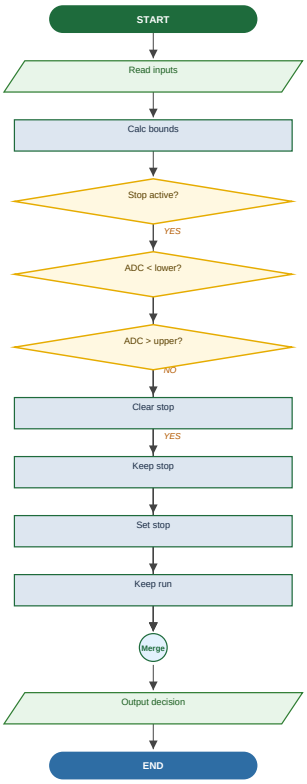
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-091	**HysteresisBandManager** (Class) | Module: SafetyControlModule | SW-REQ-080

Diagram Type: **Function Flow**

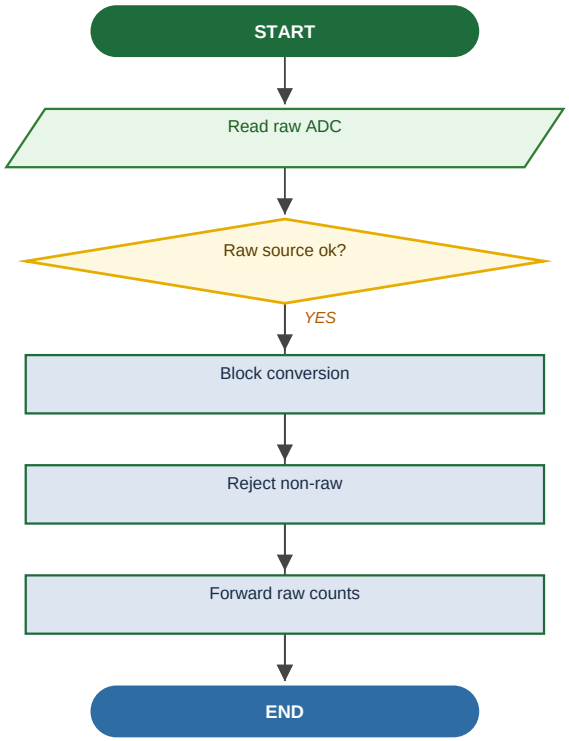
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-092	**RawAdcDataPolicy** (Class) | Module: ControlModule | SW-REQ-081

Diagram Type: **Function Flow**

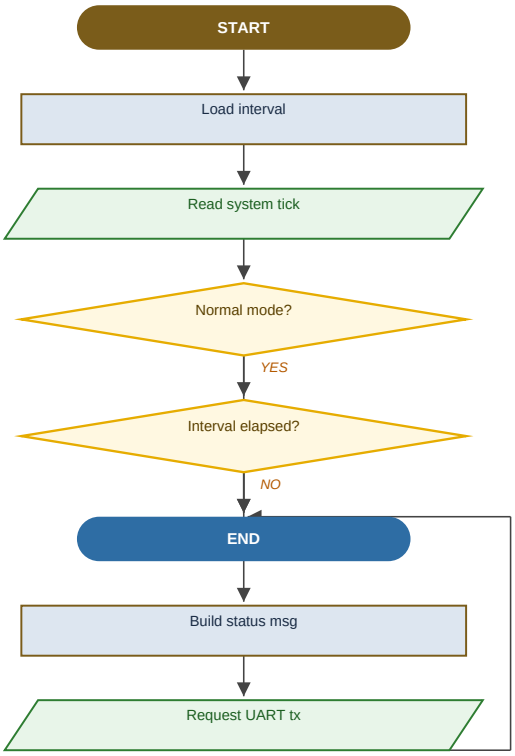
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-093	**UartStatusScheduler** (Service) | Module: CommsModule | SW-REQ-082

Diagram Type: **Function Flow**

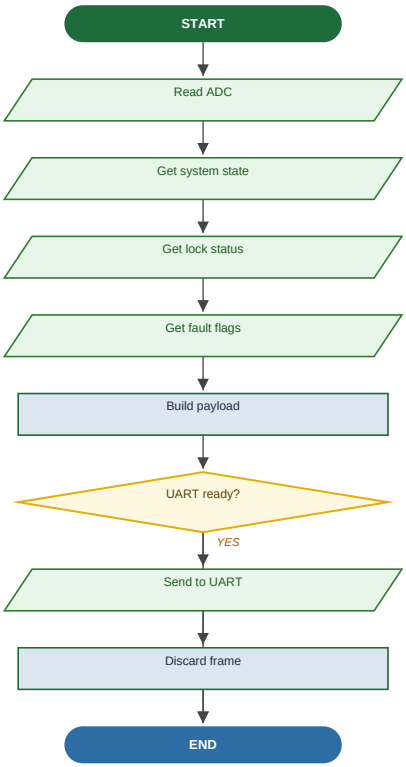
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-094	**UartStatusMessageBuilder** (Class) | Module: CommsModule | SW-REQ-082

Diagram Type: **Function Flow**

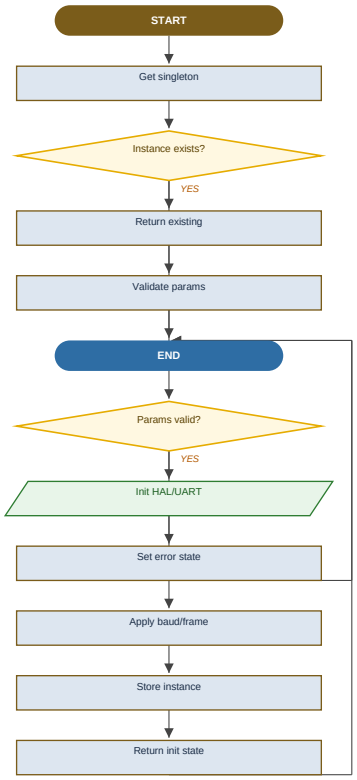
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-095	**UartConfigService** (Service) | Module: CommsModule | SW-REQ-083

Diagram Type: **Function Flow**

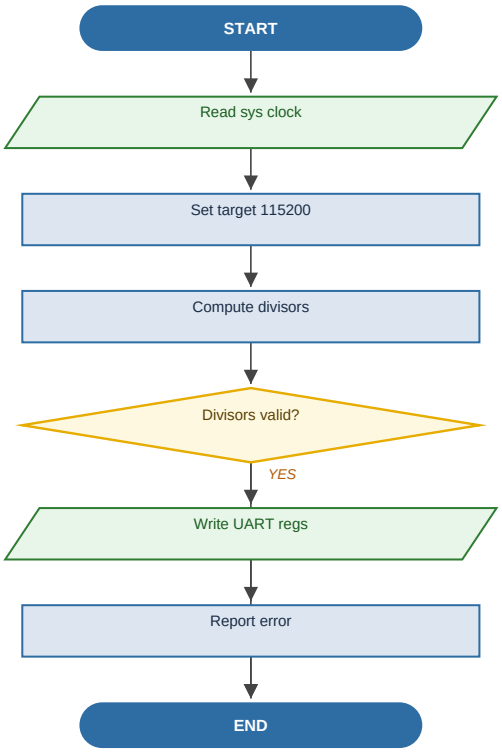
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-096	**UartBaudRateConfig** (Algorithm) | Module: CommsModule | SW-REQ-083

Diagram Type: **Function Flow**

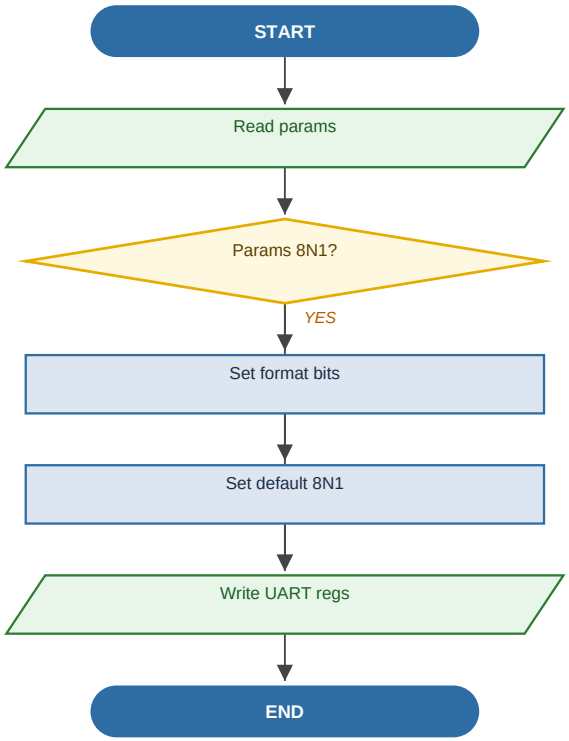
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-097	**UartFrameFormatConfig** (Algorithm) | Module: CommsModule | SW-REQ-084

Diagram Type: **Function Flow**

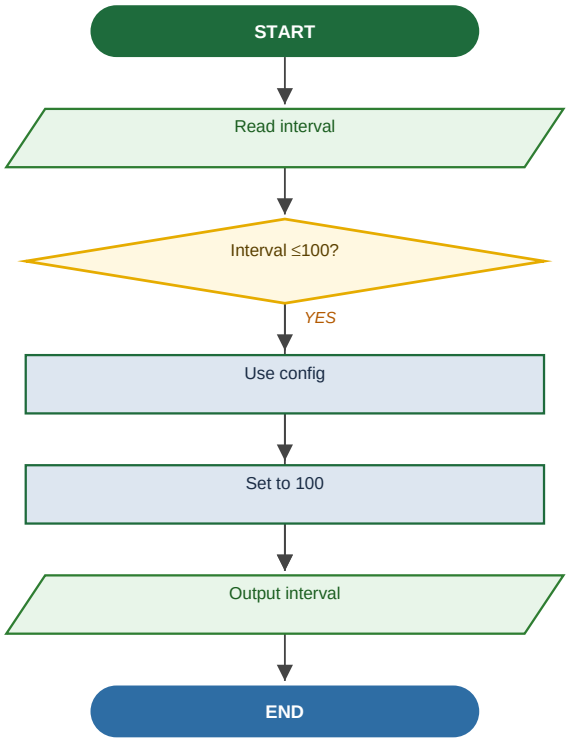
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-098	**UartTransmitIntervalValidator** (Class) | Module: CommsModule | SW-REQ-085

Diagram Type: **Function Flow**

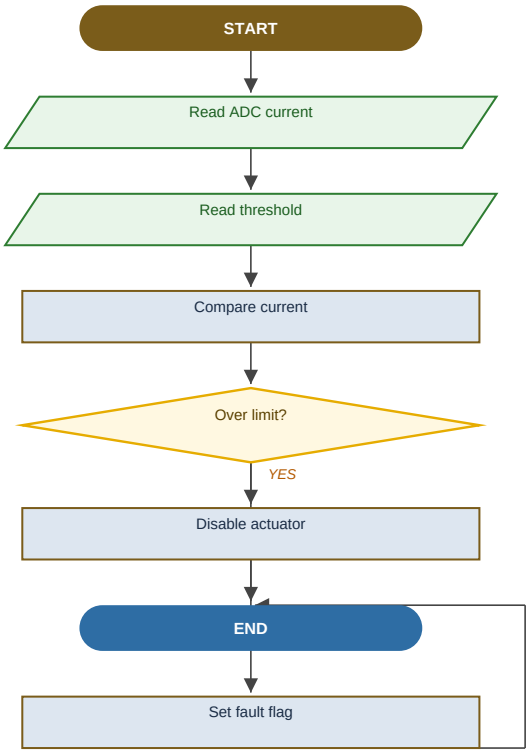
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-099	**OvercurrentMonitor** (Service) | Module: SafetyModule | SW-REQ-086

Diagram Type: **Function Flow**

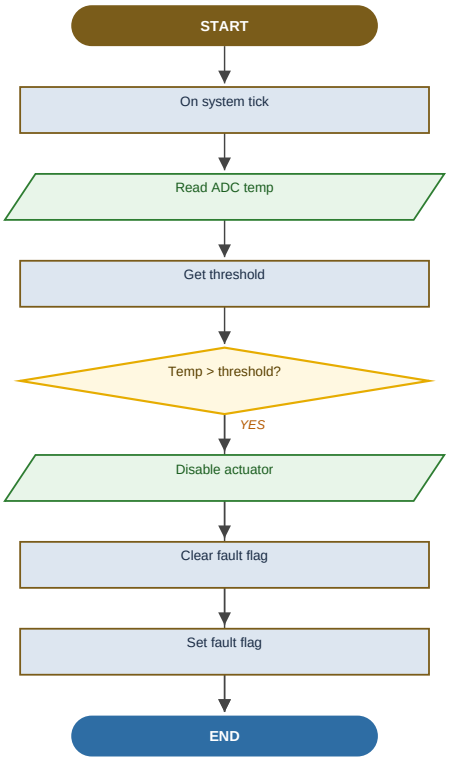
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-100	**ThermalMonitor** (Service) | Module: SafetyModule | SW-REQ-087

Diagram Type: **Function Flow**

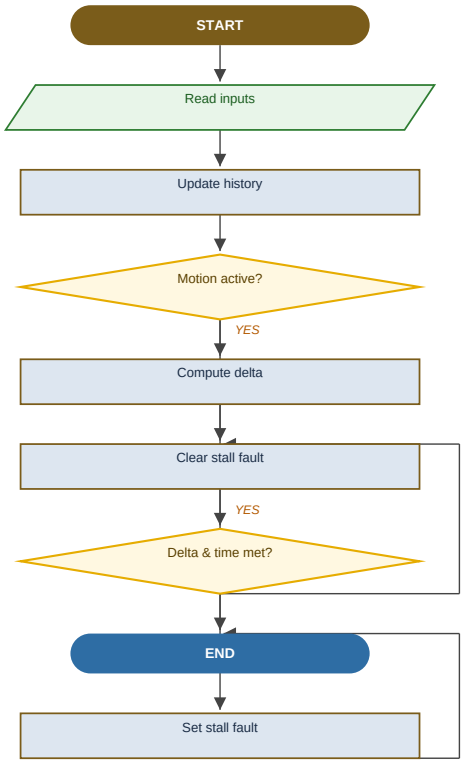
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-101	**StallDetectionService** (Service) | Module: SafetyModule | SW-REQ-088

Diagram Type: **Function Flow**

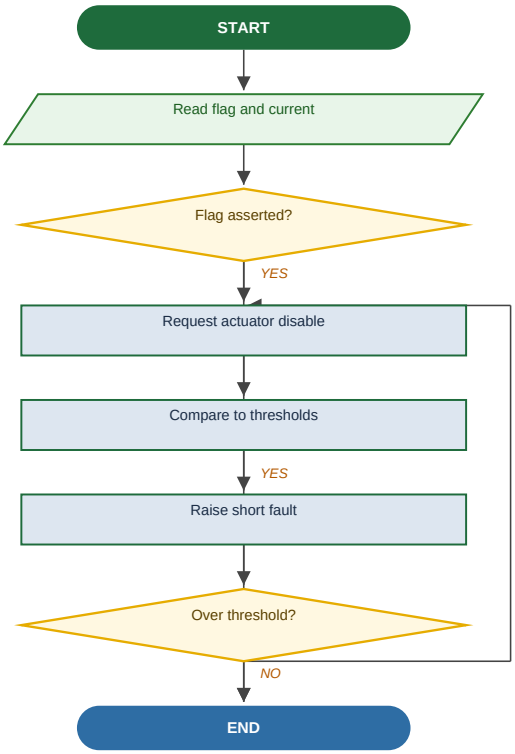
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-102	**ShortCircuitDetector** (Class) | Module: FaultMonitoringModule | SW-REQ-089

Diagram Type: **Function Flow**

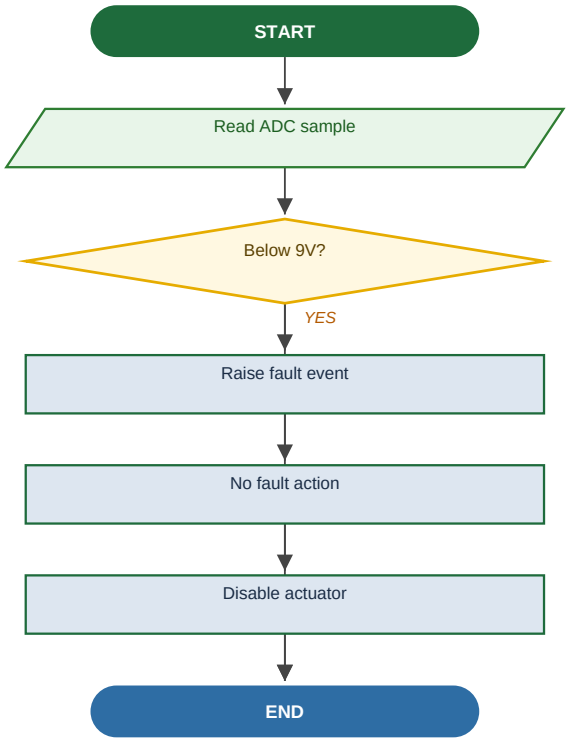
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-103	**UndervoltageMonitor** (Class) | Module: PowerMonitoringModule | SW-REQ-090

Diagram Type: **Function Flow**

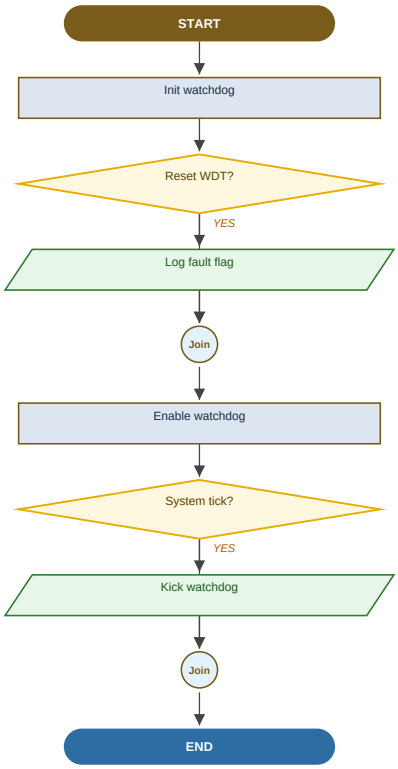
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-104	**WatchdogService** (Service) | Module: ReliabilityModule | SW-REQ-091

Diagram Type: **Function Flow**

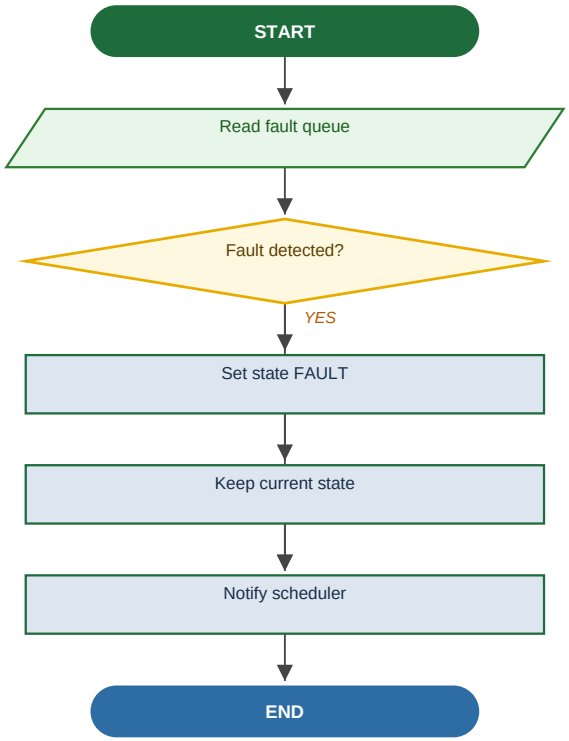
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-105	**FaultStateTransitionHandler** (Class) | Module: StateMachineModule | SW-REQ-092

Diagram Type: **State Flow**

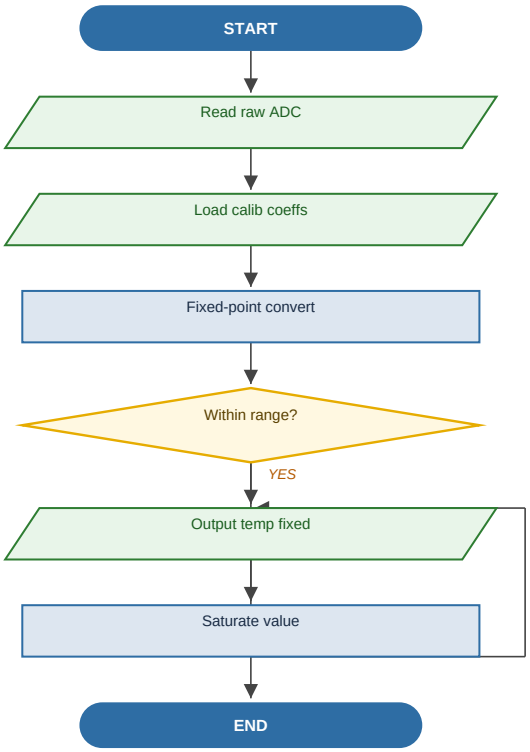
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-106	**TemperatureProcessing** (Algorithm) | Module: SensorProcessingModule | SW-REQ-093

Diagram Type: **Function Flow**

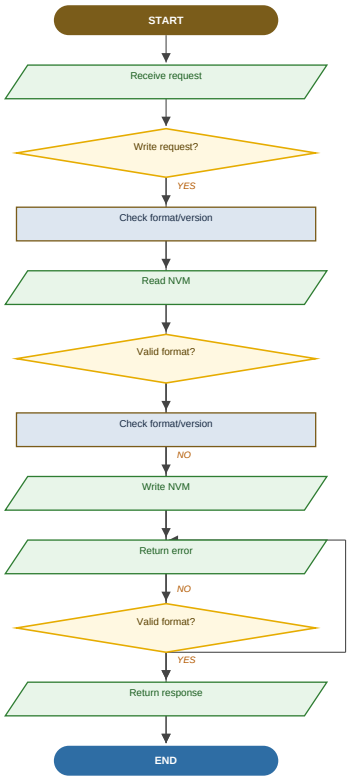
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-107	**NvmConfigStorage** (Service) | Module: DataPersistenceModule | SW-REQ-094

Diagram Type: **Function Flow**

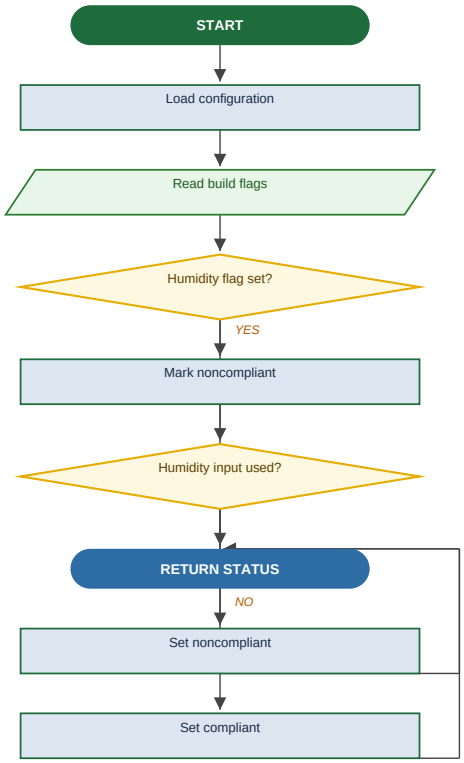
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-108	**HumidityIndependencePolicy** (Class) | Module: SystemConstraintsModule | SW-REQ-095

Diagram Type: **Function Flow**

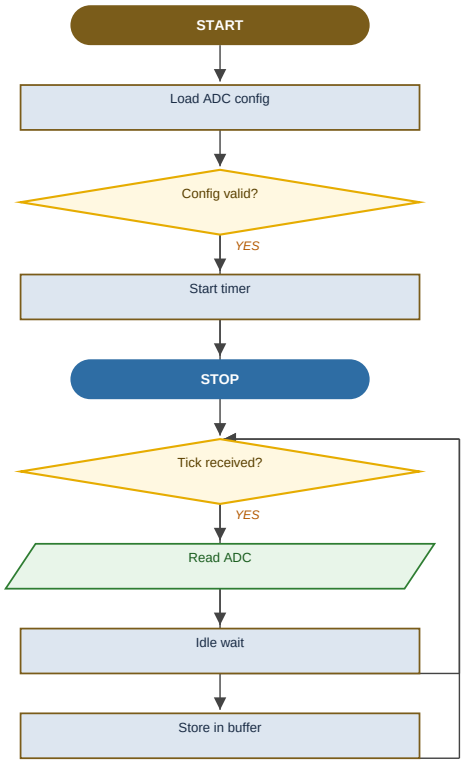
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-109	**AdcAcquisitionService** (Service) | Module: AcquisitionModule | SW-REQ-096

Diagram Type: **Function Flow**

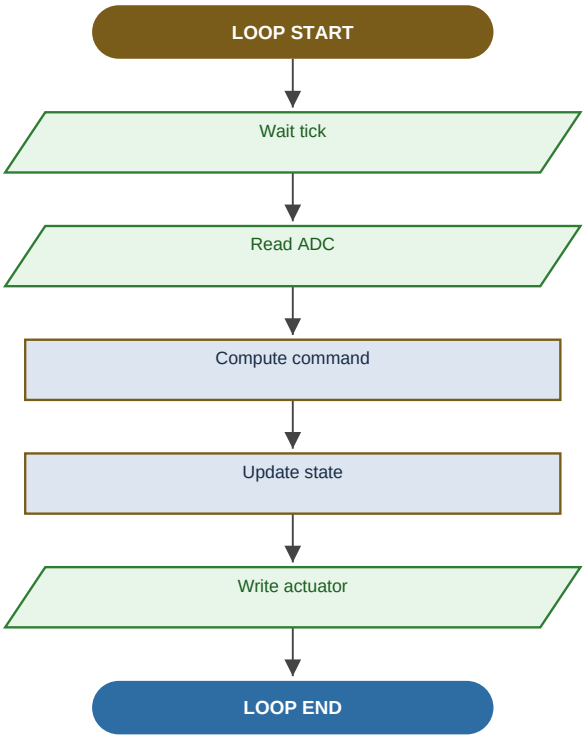
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-110	**ControlLoopScheduler** (Service) | Module: ControlModule | SW-REQ-097

Diagram Type: **Function Flow**

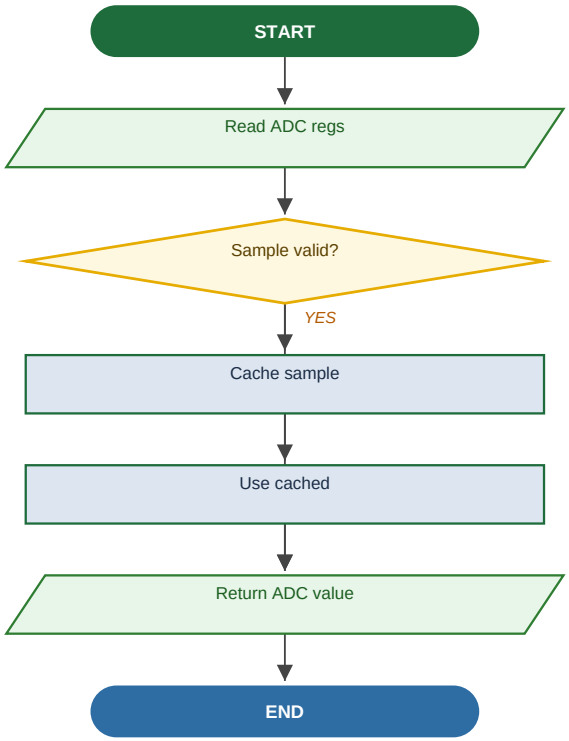
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-111	**ADCReader** (Class) | Module: SensorModule | SW-REQ-097

Diagram Type: **Function Flow**

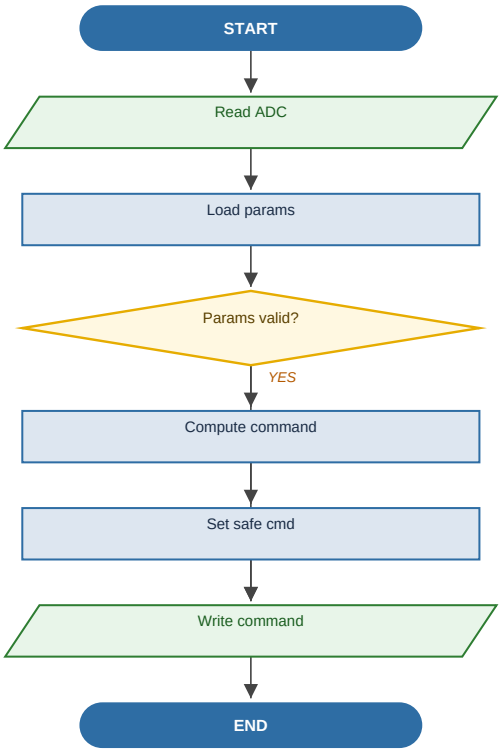
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-112	**CommandCalculator** (Algorithm) | Module: ControlModule | SW-REQ-097

Diagram Type: **Function Flow**

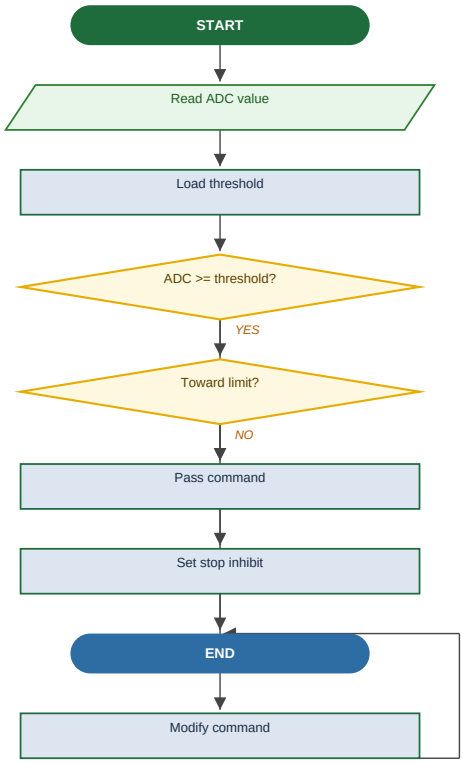
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-113	**ShippingLimitGuard** (Class) | Module: SafetyModule | SW-REQ-098

Diagram Type: **Function Flow**

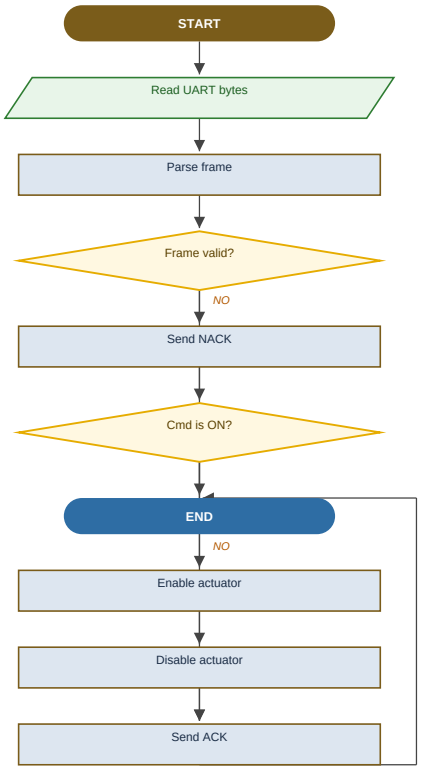
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-114	**UARTCommandHandler** (Service) | Module: CommunicationModule | SW-REQ-099

Diagram Type: **Function Flow**

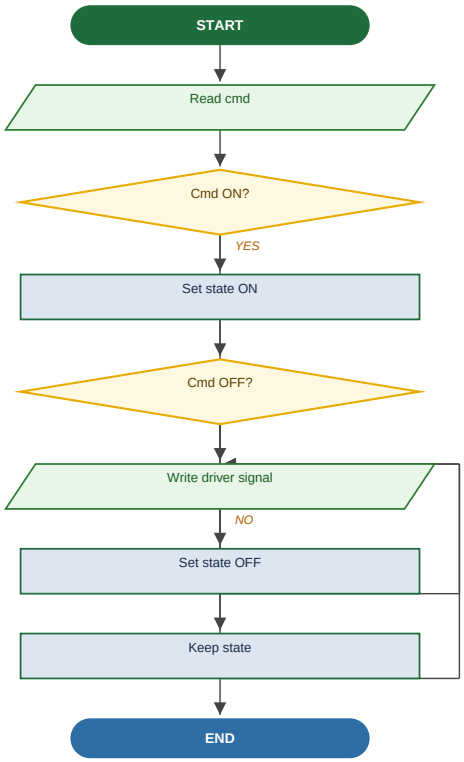
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-115	**ActuatorEnableController** (Class) | Module: ControlModule | SW-REQ-099

Diagram Type: **State Flow**

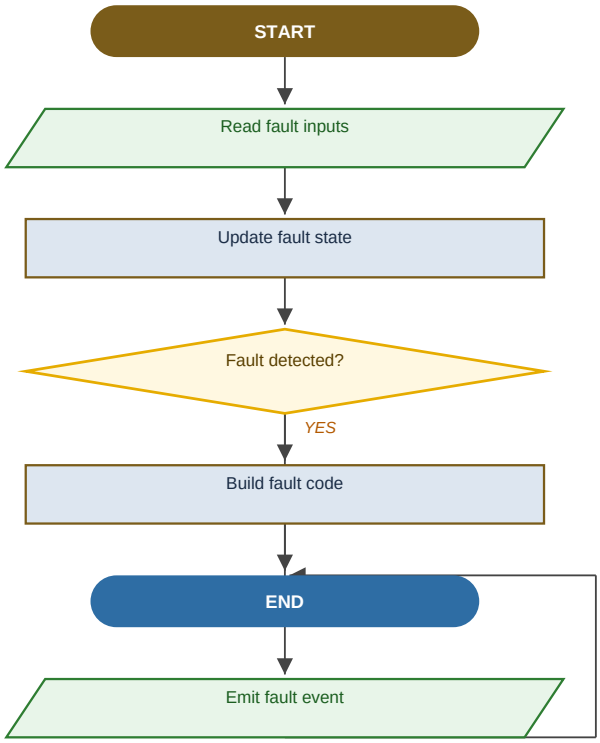
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-116	**FaultMonitor** (Service) | Module: FaultModule | SW-REQ-100

Diagram Type: **Function Flow**

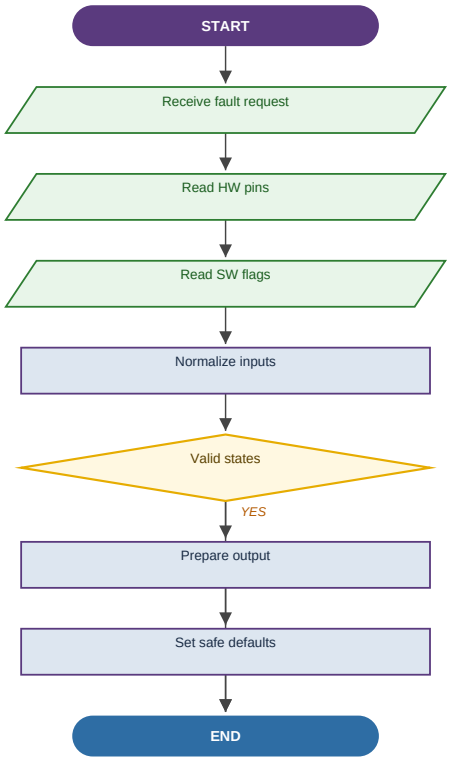
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-117	**FaultInputProvider** (Interface) | Module: FaultModule | SW-REQ-100

Diagram Type: **Request Response**

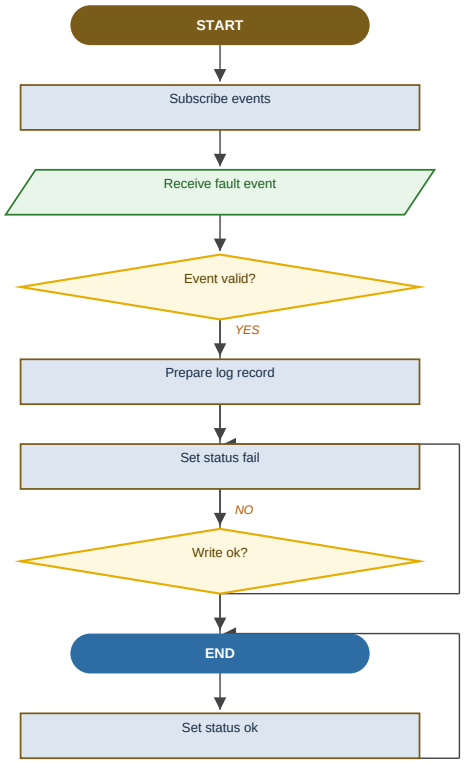
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-118	**FaultLogger** (Service) | Module: FaultModule | SW-REQ-101

Diagram Type: **Function Flow**

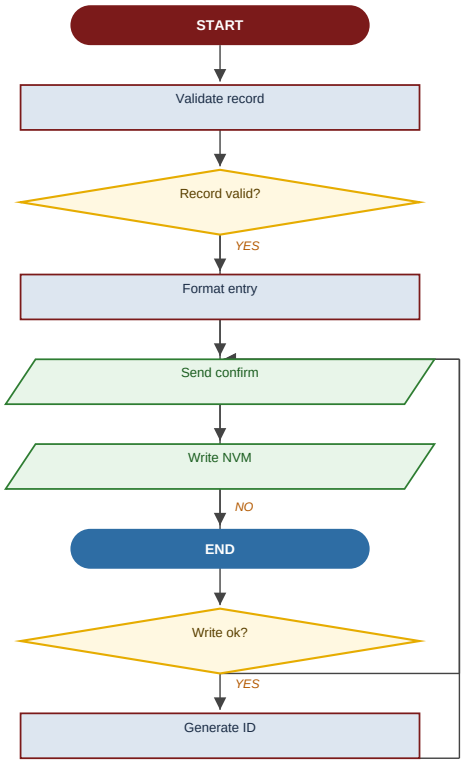
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-119	**FaultLogStore** (Database) | Module: DataLayer | SW-REQ-101

Diagram Type: **Data Flow**

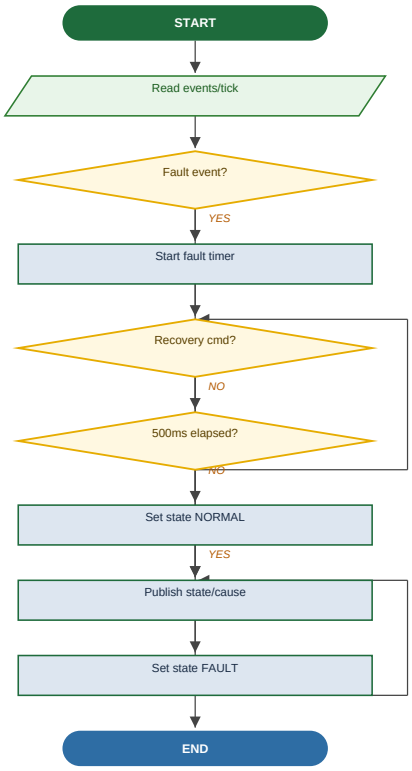
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-120	**SystemStateMachine** (Class) | Module: StateModule | SW-REQ-102

Diagram Type: **State Flow**

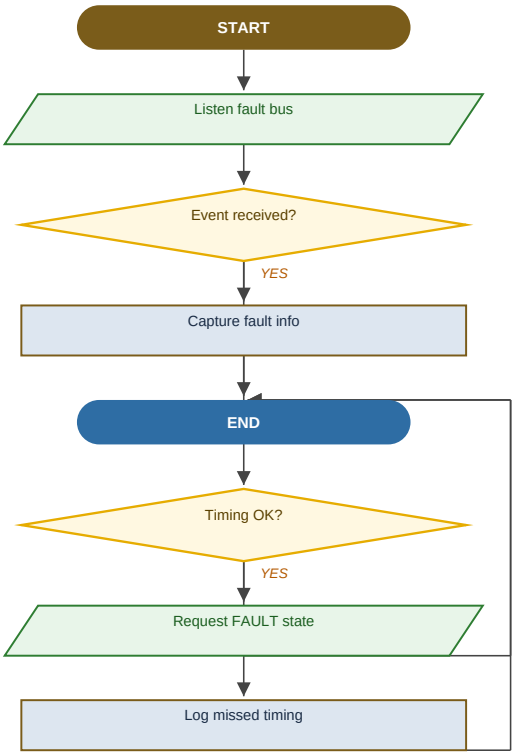
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-121	**FaultToStateTransitionService** (Service) | Module: StateModule | SW-REQ-102

Diagram Type: **Function Flow**

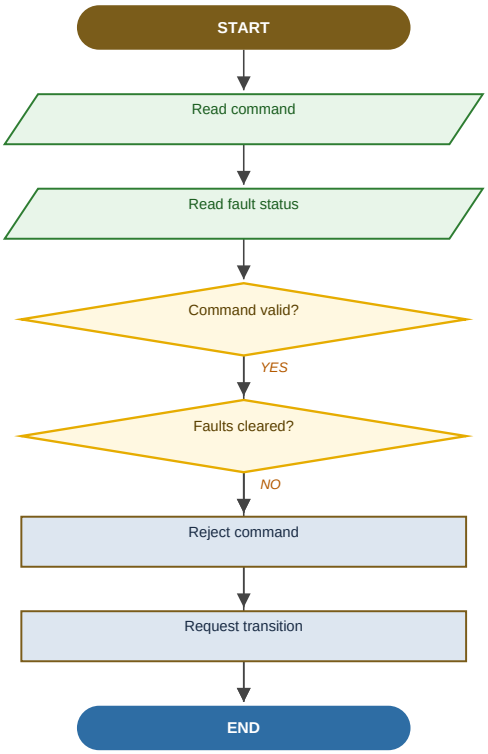
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-122	**RecoveryCommandHandler** (Service) | Module: StateModule | SW-REQ-103

Diagram Type: **Function Flow**

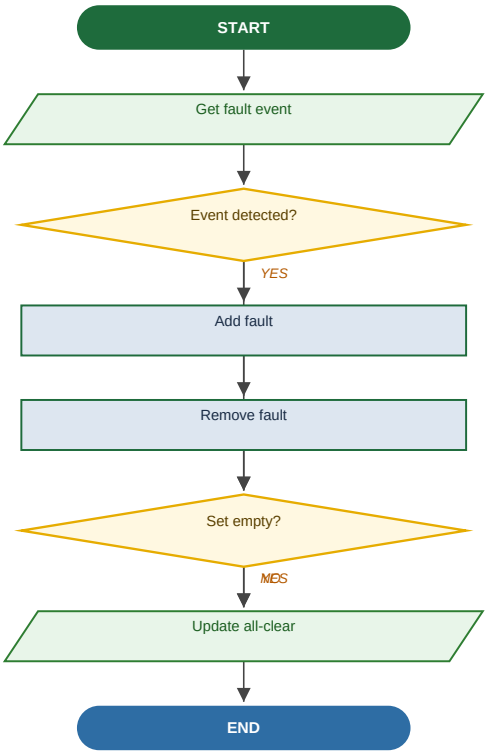
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-123	**FaultStatusTracker** (Class) | Module: FaultModule | SW-REQ-103

Diagram Type: **Function Flow**

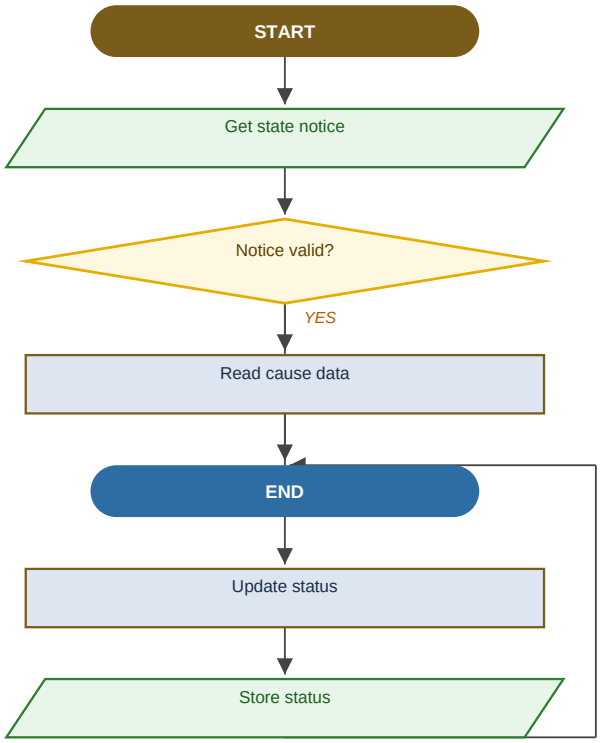
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-124	**StateStatusProvider** (Service) | Module: StateModule | SW-REQ-104

Diagram Type: **Function Flow**

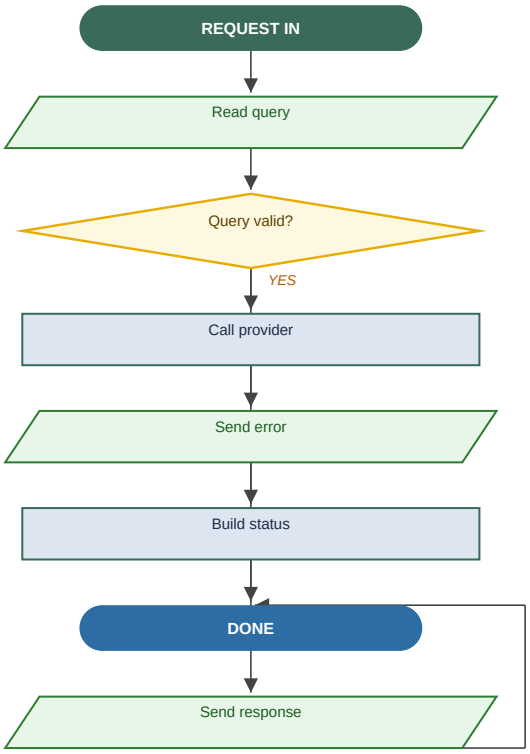
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-125	**StateStatusAPI** (API Endpoint) | Module: InterfaceModule | SW-REQ-104

Diagram Type: **Request Response**

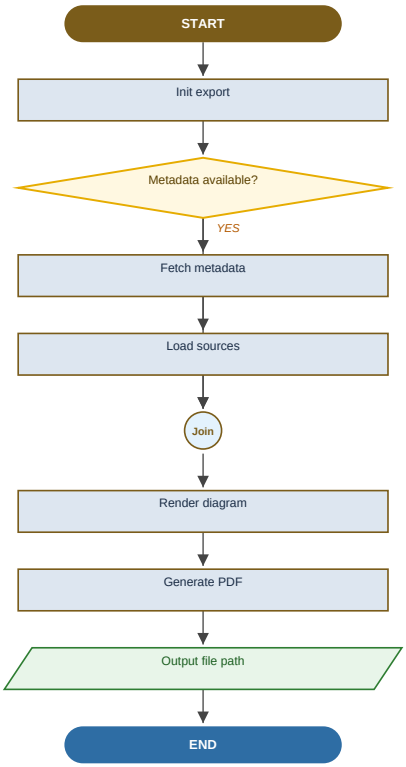
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-126	**ArchitectureDiagramExporter** (Service) | Module: ArtifactGenerationModule | SW-REQ-105

Diagram Type: **Function Flow**

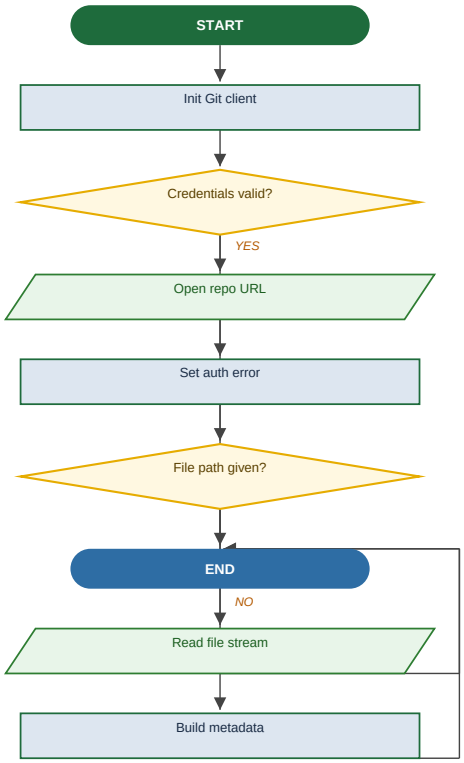
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-127	**DocumentationRepositoryAdapter** (Class) | Module: DocRepositoryModule | SW-REQ-105

Diagram Type: **Function Flow**

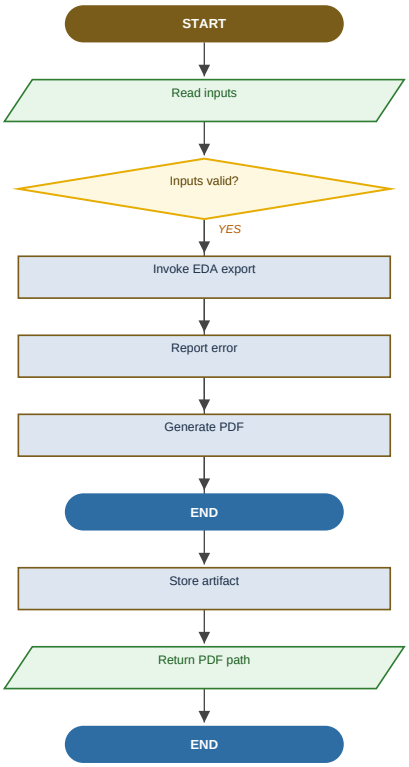
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-128	**SchematicPdfExporter** (Service) | Module: ArtifactGenerationModule | SW-REQ-106

Diagram Type: **Function Flow**

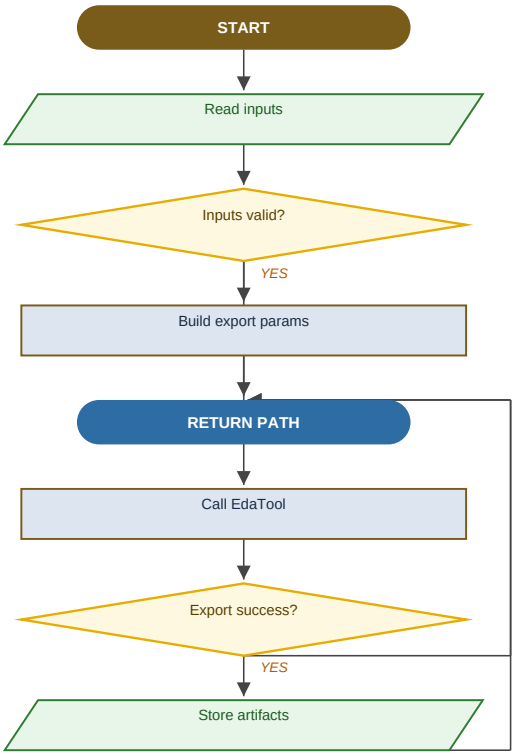
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-129	**PcbGerberExporter** (Service) | Module: ArtifactGenerationModule | SW-REQ-107

Diagram Type: **Function Flow**

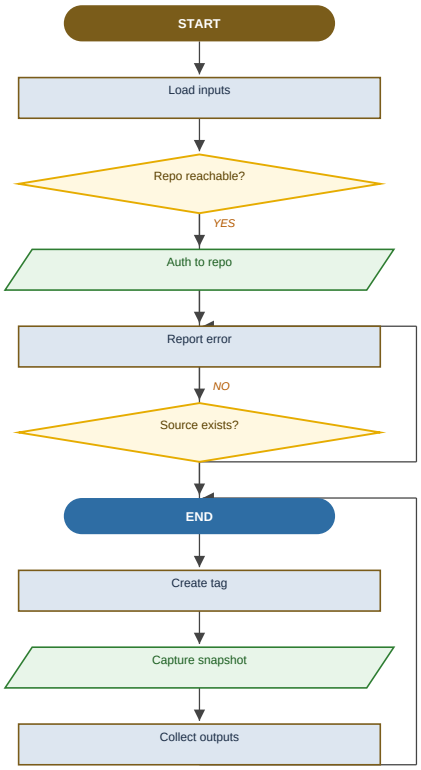
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-130	**FirmwareReleaseTagger** (Service) | Module: VCSModule | SW-REQ-108

Diagram Type: **Function Flow**

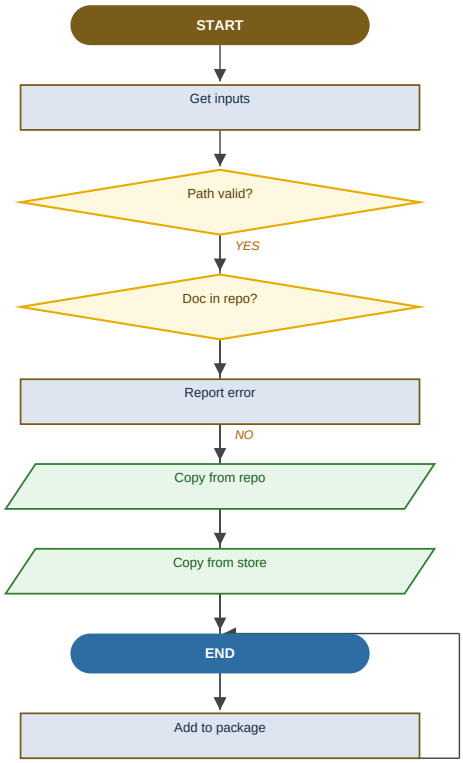
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-131	**CalibrationDocIncluder** (Service) | Module: ReleasePackagingModule | SW-REQ-109

Diagram Type: **Function Flow**

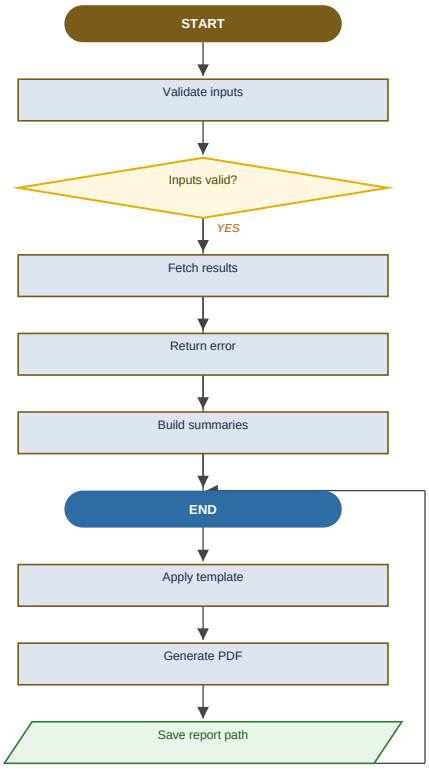
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-132	**TestReportGenerator** (Service) | Module: TestReportingModule | SW-REQ-110

Diagram Type: **Function Flow**

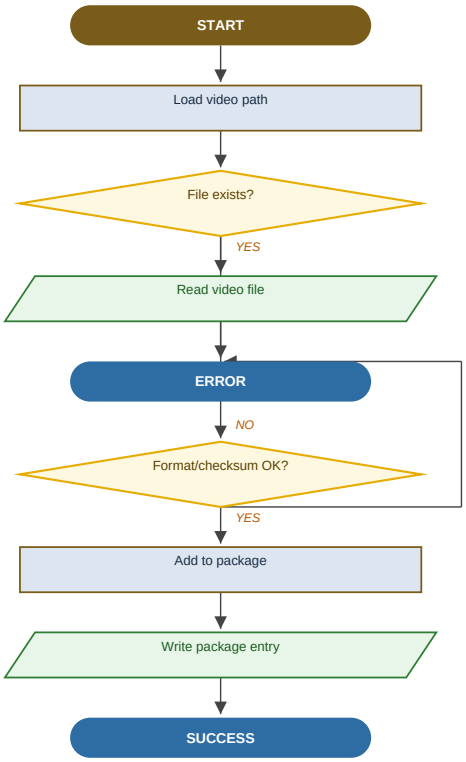
Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



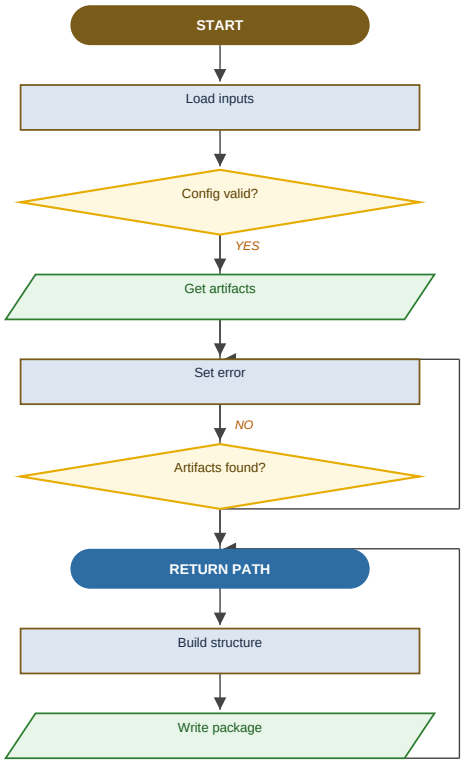
DD-COMP-133	**DemonstrationVideoIncluder** (Service) | Module: ReleasePackagingModule | SW-REQ-111

Diagram Type: **Function Flow**

Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector



DD-COMP-134	**ReleasePackageBuilder** (Service) | Module: ReleasePackagingModule | SW-REQ-106

Diagram Type: **Function Flow**

Shape	Meaning
Rounded Rect	Start / End
Rectangle	Process
Diamond	Decision
Parallelogram	I/O (Register/UART)
Circle	Connector